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Aim of the module

The aim of this module is to create understanding of on basic concepts of Combinatorics and graph theory for distance learner on counting principles, advanced counting, elements of graph theory.

Module description:

Combinatorics is a branch of Mathematics that deals with the existence, enumeration, analysis, and optimization of discrete structures and states. Combinatorics has become an important branch of Mathematics since the middle of the twentieth century because of the major impact that computers have had and continue to have, in our society. Large scale problems that were previously difficult to solve are being solved with the advent of computers.

But computers do not function independently; they need to be programmed to carry out these functions. The basis for these computer programs often are combinatorial algorithms for the solutions of problems. Analysis of these algorithms for efficiency and effectiveness with regard to running time and storage requirements requires clever combinatorial thinking and enumeration. In this course, students will be introduced fundamental principles of counting, recurrence relations, networks and flows, and graph theory and its applications and.

Module Objectives:

On completion of this module, successful students will be able to:

- ✓ know basic concepts of discrete mathematics,
- ✓ understand the principles of counting, recurrence relations and generating functions,
- ✓ understand the basic concepts of graph and their types
- ✓ know the basic algorithms on graphs
- ✓ use the methods and principles of Combinatorics
- ✓ apply Combinatorics in counting problems
- ✓ solve simple counting problems
- ✓ construct graphs with given degree patterns
- ✓ apply graph theory to solve network oriented problems.

Chapter One

Elementary Counting Principle

Introduction

Suppose that a password on a computer system consists of six, seven, or eight characters. Each of these characters must be a digit or a letter of the alphabet. Each password must contain at least one digit. How many such passwords are there? The techniques needed to answer this question and a wide variety of other counting problems will be introduced in this section. Counting problems arise throughout mathematics and computer science. For example, we must count the successful outcomes of experiments and all the possible outcomes of these experiments to determine probabilities of discrete events. We need to count the number of operations used by an algorithm to study its time complexity.

We will introduce the basic techniques of counting in this section. These methods serve as the foundation for almost all counting techniques.

Chapter Objectives

On completion of this chapter, students will be able to:

- Understand the basic counting principles.
- Understand permutations and combinations.
- Apply basic counting principles to solve counting problems.
- Apply counting techniques to solve problems related with permutations and combinations.
- Understand the inclusion – exclusion principle.
- Understand the pigeonhole principle.

1.1 Basic Counting Principles

Objectives: After completing this sub topic, students will be able to:

- ✓ State addition and multiplication principles.
- ✓ Solve counting problems using addition and multiplication principles.

The two basic counting principles.

- I. Addition Principle
- II. Multiplication Principle

I. Addition Principle:

If $E_1, E_2, \dots, E_k$ are mutually exclusive events and E_1 can happen n_1 ways, E_2 can happen n_2 ways, $\dots$, E_k can happen n_k ways, then E_1 or E_2 , or $\dots$, or E_k can happen $n_1 + n_2 + \dots + n_k$ ways.

In terms of set addition principle is stated as: Suppose that a set S is partitioned into pairwise disjoint parts $S_1, S_2, \dots, S_m$. The number of objects in S can be determined by finding the number of objects in each of the parts, and adding the numbers so obtained:

$$|S| = |S_1| + |S_2| + \dots + |S_m|.$$

If the sets $S_1, S_2, \dots, S_m$ are allowed to overlap, then a more profound principle, the inclusion-exclusion principle in the next section, can be used to count the number of objects in S .

Example 1: A student wishes to take either a mathematics course or a biology course, but not both. If there are four mathematics courses and three biology courses for which the student has the necessary prerequisites, then the student can choose a course to take in $4 + 3 = 7$ ways.

Example 2: In how many ways can we get a sum of 4 or of 8 when two distinguishable dice are rolled?

Solution: Let denote the possible outcomes obtained from rolling the dice by order pair (a, b) where a and b are numbers from 1 to 6 on both dice with $a+b = 4$ or $a+b = 8$. Then the possible outcomes to get sum of 4 are $(1, 3)$, $(2, 2)$ and $(3, 1)$, implies that there are $e_1 = 3$ ways to obtain the sum 4 (E_1).

And also we obtain the sum 8 (E_2) from the outcomes $(2, 6)$, $(3, 5)$, $(4, 4)$, $(5, 3)$ and $(6, 2)$.

Here $e_2 = 5$ ways. Therefore there are $3+5 = 8$ ways to get the sum of 4 or 8.

Example 3: A student can choose a computer project from one of three lists. The three lists contain 23, 15 and 19 possible projects, respectively. No project is on more than one list. How many possible projects are there to choose from?

Solution: The student can choose a project by selecting a project from the first list, the second list, or the third list. Because no project is on more than one list, by the addition principle there are $23 + 15 + 19 = 57$ ways to choose a project.

Example 4: In how many ways can we draw a heart or a spade from an ordinary deck of playing cards? A heart or an ace? An ace or a king? A card numbered 2 through 10? A numbered card or a king?

Solution. Since there are 13 hearts and 13 spades we may draw a heart or a spade in

$$13 + 13 = 26 \text{ ways.}$$

We may draw a heart or an ace in $13 + 3 = 16$ ways, since there are only 3 aces that are not hearts.

We may draw an ace or a king in $4 + 4 = 8$ ways.

These are 9 cards numbered 2 through 10 in each of 4 suits, clubs, diamonds, hearts, or spades.

So we may choose a numbered card in 36 ways.

Thus, we may choose a numbered card or a king in $36 + 4 = 40$ ways.

Example 5: How many ways can we get a sum of 8 when two indistinguishable dice are rolled?

An even sum?

Solution: Had the dice been distinguishable, we should obtain a sum of 8 by the outcomes (2, 6), (3, 5), (4, 4), (5, 3) and (6, 2), but since the dice are similar, the outcomes (2, 6) and (6, 2) and, as well, (3, 5) and (5, 3) cannot be differentiated and thus we obtain the sum of 8 with the roll of two similar dice in only 3 ways. Likewise, we can get an even sum in $1 + 2 + 3 + 3 + 2 + 1 = 12$ ways.

Example 6: If there are 14 boys and 12 girls in a class, find the number of ways of selecting one student as class representative.

Solution: Using sum rule, there are $14 + 12 = 26$ ways of selecting one student (either a boy or a girl) as class representative.

II. Multiplication Principle

If events $E_1, E_2, \dots, E_k$ can happen $n_1, n_2, \dots, n_k$ ways respectively, then the sequence of events E_1 first, followed by $E_2, \dots$, followed by E_k can happen $n_1 \cdot n_2 \dots n_k$ ways.

NB. The Multiplication Principle applies when a procedure is made up of separate tasks.

Example 1: How many two letter words start with one of the 5 vowels?

Solution: There are 26 two letter words starting with A, another 26 starting with E, and so on.

We will have 5 groups of 26. So we add 26 to itself 5 times. Of course it would be easier to just multiply $5 \cdot 26$.

Example 2: If 2 distinguishable dice are rolled, in how many ways can they fall? If 5 distinguishable dice are rolled, how many possible outcomes are there?

Solution: The first die can fall (event E_1) in 6 ways and the second can fall (event E_2) in 6 ways. There are $6 \cdot 6 = 6^2 = 36$ outcomes are rolled dice when 2 dice are rolled. And there are $6 \cdot 6 \cdot 6 \cdot 6 \cdot 6 = 6^5$ possible outcome when 5 dice are rolled.

Example 3: There are 32 microcomputers in a computer center. Each microcomputer has 24 ports. How many different ports to a microcomputer in the center are there?

Solution: The procedure of choosing a port consists of two tasks, first picking a microcomputer and then picking a port on this microcomputer. Because there are 32 ways to choose the microcomputer and 24 ways to choose the port no matter which microcomputer has been selected, the product rule shows that there are $32 \cdot 24 = 768$ ports.

Example 4: How many different license plates can be made if each plate contains a sequence of three uppercase English letters followed by three digits (and no sequences of letters are prohibited, even if they are obscene)?

Solution: There are 26 choices for each of the three uppercase English letters and ten choices for each of the three digits. Hence, by the product rule there are a total of $26 \cdot 26 \cdot 26 \cdot 10 \cdot 10 \cdot 10 = 17,576,000$ possible license plates.

Example 5: The number of ways a man, woman, boy, and girl can be selected from five men, six women, two boys, and four girls is $5 \times 6 \times 2 \times 4 = 240$.

Example 6: How many two-digit numbers have distinct and nonzero digits? A two-digit number ab can be regarded as an ordered pair (a, b) , where a is the tens digit and b is the units digit. Neither of these digits is allowed to be 0 in the problem, and the two digits are to be different. There are nine choices for a , namely 1, 2, ..., 9. Once a is chosen, there are eight choices for b . If $a = 1$, these eight choices are 2, 3, ..., 9, if $a = 2$, the eight choices are 1, 3, ..., 9, and so on. What is important for application of the multiplication principle is that the number of choices is always 8. The answer to the questions is, by the multiplication principle, $9 \times 8 = 72$. We can arrive at the answer 72 in another way. There are 90 two-digit numbers, 10, 11, 12, ..., 99. Of these numbers, nine have a 0, (namely, 10, 20, ..., 90) and nine have identical digits (namely, 11, 22, ..., 99). Thus the number of two-digit numbers with distinct and nonzero digits equals $90 - 9 - 9 = 72$.

Example 7: Each user on a computer system has a password, which is six to eight characters long, where each character is an upper case letter or a digit. Each password must contain at least one digit. How many possible passwords are there?

Solution: Let P be the total number of possible passwords, and let P_6 , P_7 and P_8 denote the number of possible passwords of length 6, 7, and 8 respectively.

By the sum rule, $P = P_6 + P_7 + P_8$

We will now find P_6 , P_7 , and P_8 . Finding P_6 directly is difficult.

To find P_6 it is easier to find the number of strings of upper case letters and digits that are six characters long, including those with no digits, and subtract from this the number of strings with no digits.

By the product rule, the number of strings of six characters is 366 and the number of strings with no digits is 266.

$$\begin{aligned}\text{Hence, } P_6 &= 366 - 266 = 2, 176, 782, 336 - 308, 915, 776 \\ &= 1, 867, 866, 560.\end{aligned}$$

Similarly, it can be shown that

$$\begin{aligned}P_7 &= 367 - 267 = 78, 364, 164, 096 - 8, 031, 810, 176 \\ &= 70, 332, 353, 920.\end{aligned}$$

$$\begin{aligned}\text{and } P_8 &= 368 - 268 = 2, 821, 109, 907, 456 - 208, 827, 064, 576 \\ &= 2, 612, 282, 842, 880.\end{aligned}$$

Consequently,

$$\begin{aligned}P &= P_6 + P_7 + P_8 \\ &= 2, 684, 483, 063, 360.\end{aligned}$$

Example 8:

- How many 2 digit or 3-digit numbers can be formed using the digits 1, 3, 4, 5, 6, 8 and 9 if no repetition is allowed ?
- How many numbers can be formed using the digits 1, 3, 4, 5, 6, 8 and 9 if no repetition are allowed?

Solution:

- There are 7.6.5, three-digit numbers possible. Likewise, we can apply the product rule to see that these are 7.6 possible 2-digit numbers.
Hence, these are $7.6 + 7.6.5$ possible two-digit or three-digit numbers.
- The number of digits are not specified in this problem so we can form one-digit numbers, two-digit numbers, or three digit numbers, etc.

But since no repetitions are allowed and we have only the 7 integers to work with, the maximum number of digits would have to be 7.

Applying the product rule, we see that we may form 7 one-digit numbers, $7 \cdot 6 = 42$ two digit numbers $7 \cdot 6 \cdot 5$ three digit numbers, $7 \cdot 6 \cdot 5 \cdot 4$ four digit numbers, $7 \cdot 6 \cdot 5 \cdot 4 \cdot 3$ five digit numbers, $7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2$ six-digit numbers, and $7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1$ seven-digit numbers.

The events of forming one-digit numbers, two digit numbers, three digit numbers, etc., are mutually exclusive events so,

We apply the sum rule to see that there are

$$7 + 7 \cdot 6 + 7 \cdot 6 \cdot 5 + 7 \cdot 6 \cdot 5 \cdot 4 + 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 + 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 + 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 13,741$$

different numbers we can form under the restrictions of this problem.

Check list

To check your mastery the topic put (✓) if you can perform it.; otherwise put (X) in front of each statement.

Important points	I Can	I Can't
1. State addition principle.		
2. State multiplication principle.		
3. Solve problems given this subtopics related to addition and multiplication principles.		

Is your response positive for the above questions? If your answer is “I Can”, go to the next section. If your answer is “I Can’t”, go back and read once again.

Exercise 1.1

1.

To reach city D from city A , one has to pass through city B and then city C

If there are 2 ways to travel from A to B , 5 ways from B to C , and 3

Ways to travel from C to D , how many ways are there to travel from A to D ?

2. Suppose repetitions are not permitted.

- Find the number of three-digit numbers that can be formed from the six digits 2, 3, 5, 6, 7, and 9.
- How many of them are less than 400?
- How many of them are even?

3. A Mathematics class contains 8 male students and 6 female students. Find the number n of ways that the class can elect:
 - a) 1 class representative
 - b) 2 class representatives, 1 male and 1 female
 - c) 1 president and 1 vice president.
4. How many license plates consisting of 2 nonrepeating letters (A to Z) followed by 3 even digits (0 to 8) are possible?
5. How many 6-digit numbers have odd digits in the odd places and even digits in the even places?
6. A certain Internet provider requires that its e-mail addresses consist of 6 characters (letters A to Z or digits 0 to 9). Further, the first character must be a letter, and the last character must be a digit. How many different e-mail accounts are possible from this provider?
7. How many 5-digit numbers neither start with zero nor end in zero?
8. How many 6-digit numbers (leading zeros are allowed) have the last two digits the same as the first two digits (in the same order)?

1.2 Permutations and Combinations

Many counting problems can be solved by finding the number of ways to arrange a specified number of distinct elements of a set of a particular size, where the order of these elements matters. Many other counting problems can be solved by finding the number of ways to select a particular number of elements from a set of a particular size, where the order of the elements selected does not matter. For example, in how many ways can we select three students from a group of five students to stand in line for a picture? How many different committees of three students can be formed from a group of four students? In this section we will develop methods to answer questions such as these.

Objectives: After completing this sub topic, students will be able to:

- ✓ Define permutation and combination.
- ✓ Evaluate the given number of permutations and combinations without repetition and with repetitions allowed.
- ✓ Solve problems related to number of permutations and combinations.

Many counting problems can be thought of as a selection of some objects from a fixed set of objects. In some cases, the order in which the objects are selected is important, and in others it is not. As we shall see, permutations handle the former case, and combinations handle the latter.

Definition: A permutation of r objects from a set of size n is an ordered list of r of the n objects. An r -permutation is an ordered arrangement of r elements of a set.

Example 1: Consider the set of letters A, B, C, D . Then:

- i) $BDCA, DCBA$, and $ACDB$ are permutations of the four letters (taken all at a time).
- ii) BAD, ACB, DBC are permutations of the four letters taken three at a time.
- iii) AD, BC, CA are permutations of the four letters taken two at a time.

Example 2: If $S = \{A, B, C, D, E\}$ is the set of $n = 5$ objects under consideration and $r = 3$, then all of the possible permutations of 3 elements from S are listed below.

ABC ABD ABE ACD ACE ADE BCD BCE BDE CDE
ACB ADB AEB ADC AEC AED BDC BEC BED CED
BAC BAD BAE CAD CAE DAE CBD CBE DBE DCE
BCA BDA BEA CDA CEA DEA CDB CEB DEB DEC
CAB DAB EAB DAC EAC EAD DBC EBC EBD ECD
CBA DBA EBA DCA ECA EDA DCB ECB EDB EDC

They are 60 in number.

Example 3: In how many ways can we arrange three students from a group of five students to stand in line for a picture? In how many ways can we arrange all five of these students in a line for a picture?

Solution: There are five ways to take the first student to stand at the start of the line. Once this student has been taken, there are four ways to take the second student in the line. After the first and second students have been taken, there are three ways to take the third student in the line. There are $5 \cdot 4 \cdot 3 = 60$ ways to arrange three students from a group of five students to stand in line for a picture.

To arrange all five students in a line for a picture, we take the first student in five ways, the second in four ways, the third in three ways, the fourth in two ways, and the fifth in one way. Consequently, there are $5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 120$ ways to arrange all five students in a line for a picture.

Example 4: Let $S = \{a, b, c\}$. The 2-permutations of S are the ordered arrangements ab, ac, ba, bc, ca, cb . Consequently, there are 6 2-permutations of this set with three elements.

Theorem 1. If n is a positive integer and r is an integer with $1 \leq r \leq n$, then there are

$$P(n, r) = n(n-1)(n-2) \cdots (n-r+1)$$

r -permutations of a set with n distinct elements.

Corollary 1. If n and r are integers with $0 \leq r \leq n$, then

$$P(n, r) = \frac{n!}{(n-r)!}$$

Example 5: How many ways are there to select a first-prize winner, a second-prize winner, and a third-prize winner from 100 different people who have entered a contest?

Solution: Because it matters which person wins which prize, the number of ways to pick the three prize winners is the number of ordered selections of three elements from a set of 100 elements, that is, the number of 3 permutations of a set of 100 elements. Consequently, the answer is $P(100, 3) = 100 \cdot 99 \cdot 98 = 970,200$.

Example 6: Suppose that there are eight runners in a race. The winner receives a gold medal, the second place finisher receives a silver medal, and the third-place finisher receives a bronze medal. How many different ways are there to award these medals, if all possible outcomes of the race can occur and there are no ties?

Solution: The number of different ways to award the medals is the number of 3-permutations of a set with eight elements. Hence, there are $P(8, 3) = 8 \cdot 7 \cdot 6 = 336$ possible ways to award the medals.

Example 7: The number of four-letter "words" that can be formed by using each of the letters a, b, c, d, e at most once is $P(5, 4)$, and this equals $5!/(5-4)! = 120$. The number of five-letter words equals $P(5, 5)$, which is also 120.

Example 8: What is the number of ways to order the 26 letters of the alphabet so that no two of the vowels a, e, i, o, u occur consecutively?

Solution: Here we think of two main tasks to be accomplished. The first task is to decide how to order the consonants among themselves. There are 21 consonants, and so $21!$ permutations of the consonants. Since we cannot have two consecutive vowels in our final arrangement, the vowels must be in 5 of the 22 spaces before, between, and after the consonants. Our second task is to put the vowels in these places. There are 22 places for the a , then 21 for the e , 20 for the i , 19 for the o , and 18 for the u . That is, the second task can be accomplished in

$$P(22, 5) = \frac{22!}{17!}$$

ways. By the multiplication principle, we determine that the number of ordered arrangements of the letters of the alphabet with no two vowels consecutive is

$$21! \times \frac{22!}{17!}.$$

Example 9: Find n if $P(n, 2) = 72$.

Solution: $P(n, 2) = n(n-1) = n^2 - n$. Thus, we get $n^2 - n = 72$ or $n^2 - n - 72 = 0$ or

$$(n-9)(n+8) = 0 \implies n = 9 \text{ and } n = -8$$

Since n must be positive, the only answer is $n = 9$.

Definition: The number of circular r -permutations of a set of n elements is given by

$$\frac{P(n, r)}{r} = \frac{n!}{r \cdot (n-r)!}.$$

If $r = n$, then the number of such permutation is given by

$$(n-1)!$$

Example 10: Find the number of 3- permutations of the letters p, q, r, s and t in circle.

Solution: $n = 5$ and $r = 3$, the using the above formula,

$$\frac{n!}{r(n-r)!} = \frac{5!}{3(5-3)!} = 20.$$

Example:

Simplify $\frac{(n+1)!}{n!}$

Solution:

$$\frac{(n+1)!}{n!} = \frac{(n+1) n(n-1) \dots 3 \cdot 2 \cdot 1}{n(n-1) \dots 3 \cdot 2 \cdot 1} = n+1$$

or

$$\frac{(n+1)!}{n!} = \frac{(n+1)n!}{n!} = (n+1)$$

Example 11: Prove that $2 P(n, n-2) = P(n, n)$

Solution:

$$= 2 P(n, n-2) = 2 \frac{n!}{(n-(n-2))!}$$

$$\begin{aligned}
 &= 2 \cdot \frac{n!}{n - n + 2!} = \frac{2 \cdot n!}{2!} \\
 &= n! = P(n, n).
 \end{aligned}$$

Example 12: In how many ways can the letters of the English alphabet be arranged so that there are exactly ten letters between a and z?

Solution. There are $P(24, 10)$ arrangements of the letters of the alphabet (excluding a and z) taken ten at a time, and hence $2 \cdot P(24, 10)$ strings of 12 letters, each beginning and ending with an a and z (either letter coming first in a string).

For each of these strings, there are $15!$ ways to arrange the 14 remaining letters and the string. So there are altogether

$$2 \cdot P(24, 10) \cdot 15! \text{ arrangements of the desired type.}$$

Example 13: A man, a woman, a boy, a girl, a dog, and a cat are walking down a long and winding road one after the other.

- a) In how many ways can this happen?
- b) In how many ways can this happen if the dog comes first?
- c) In how many ways can this happen if the dog immediately follows the boy?
- d) In how many ways can this happen if the dog (and only the dog) is between the man and the boy?

Solution: a) There are $6! = 720$ ways for six creatures to form a line.

b) If the dog comes first, the others can form $= 5!$ lines behind.

c) If the dog immediately follows the boy, then the dog-boy pair should be thought of as a single object to be put into a line with four others.

There are $5! = 120$ such lines.

d) If the man, dog, and boy appear in this order, then thinking of man-dog-boy as a

e) single object to be put into a line with three others, we see that there are $4!$ possible lines. Similarly, there are $4!$ lines in which the boy, dog, and man appear in this order.

So, by the addition rule, there are $4! + 4! = 48$ lines in which the dog (and only the dog) is between the man and the boy.

Example 14: Find the number of positive integers greater than a million that can be formed with the digits 2, 3, 0, 3, 4, 2 and 3.

Solution: The numbers greater than a million must be of 7 digits.

In the given set of digits, 2 appear twice, 3 appear thrice and all others are distinct.

Thus, the total number of seven digit numbers that can be formed with given digits is

$$\frac{7!}{2!3!} = 420.$$

The set of these 420 positive integers, include some numbers which begin with 0.

Clearly, these numbers are less than a million and they must not be counted in our answer.

The number of such numbers is given by the permutations of 6 non-zero digits and is equal to

$$\frac{6!}{2!3!} = 60.$$

Therefore, the number of positive integers greater than a million that can be formed with given digits is equal to $420 - 60 = 360$.

Example 15: In how many different ways can 5 men and 5 women sit around a table, if

i) there is no restriction

ii) no two women sit together?

Solution: i) The problem is related to circular permutation of 10 objects (5 men and 5 women). If there is no restriction then the number of permutations is $(10 - 1)! = 9! = 362880$.

Notice here the difference in arrangement between clockwise and anticlockwise.

ii) there is a restriction that no two women are allowed to sit side by side. To meet this restriction each woman should occupy a sit between two men.

The number of ways five men can sit around a table is $4! = 24$.

Once these five men have sat on alternate chairs, the five women can occupy the 5 empty chairs in $5!$ ways.

Thus, total number of ways, in this case is $24 * 5! = 24 * 120 = 2880$.

Definition: A combination of r elements from a set of size n is a subset of size r . An r -combination of elements of a set is an unordered selection of r elements from the set of n elements.

Example 1: If $S = \{a, b, c, d, e\}$ is the set of $n = 5$ objects under consideration and $r = 3$, then all of the possible combinations of 3 elements from S are listed below.

$$\begin{aligned} &\{a, b, c\} \{a, b, d\} \{a, b, e\} \{a, c, d\} \{a, c, e\} \\ &\{a, d, e\} \{b, c, d\} \{b, c, e\} \{b, d, e\} \{c, d, e\} \end{aligned}$$

Example 2: Let S be the set $\{1, 2, 3, 4\}$. Then $\{1, 3, 4\}$ is a 3-combination from S . Note that $\{4, 1, 3\}$ is the same 3-combination as $\{1, 3, 4\}$, because the order in which the elements of a set are listed does not matter.

Example 3: The 2-combinations of $\{a, b, c, d\}$ are the six subsets $\{a, b\}, \{a, c\}, \{a, d\}, \{b, c\}, \{b, d\}$, and $\{c, d\}$.

Theorem 2: The number of r -combinations of a set with n elements, where n is a nonnegative integer and r is an integer with $0 \leq r \leq n$, is given by

$$C(n, r) = \frac{n!}{r!(n-r)!}$$

Note: The number $C(n, r)$ is also denoted by $\binom{n}{r}$

Example 4: How many poker hands of five cards can be dealt from a standard deck of 52 cards? Also, how many ways are there to select 47 cards from a standard deck of 52 cards?

Solution: Because the order in which the five cards are dealt from a deck of 52 cards does not matter, there are $C(52, 5) = \frac{52!}{5!(52-5)!} = \frac{52!}{5!47!} = 2,598,960$

different hands of five cards that can be dealt.

Example 5: A farmer buys 3 cows, 2 pigs, and 4 hens from a man who has 6 cows, 5 pigs, and 8 hens. Find the number of choices that the farmer has.

Solution: The farmer can choose the cows in $C(6, 3)$ ways, the pigs in $C(5, 2)$ ways, and the hens in $C(8, 4)$ ways.

Thus the number of choices is

$$\binom{6}{3} \binom{5}{2} \binom{8}{4} = 20 \cdot 10 \cdot 70 = 14\,000 \quad \text{by Multiplication principle.}$$

Corollary 2: Let n and r be nonnegative integers with $r \leq n$. Then

$$C(n, r) = C(n, n - r).$$

1.3 The Binomial Theorem

Theorem 3: The binomial theorem

Let x and y be variables, and let n be a non-negative integer. Then

$$(x + y)^n = \sum_{r=0}^n \binom{n}{r} x^{n-r} y^r$$

Example 1: Expand $(x + y)^4$ using binomial theorem

Solution: From the binomial theorem it follows that

$$(x + y)^4 = x^4 + 4x^3y + 6x^2y^2 + 4xy^3 + y^4.$$

Example 2:

Find the binomial expansion of $(2a - 3b)^4$.

SOLUTION:

Here $x = 2a$, $y = -3b$, and $n = 4$. Using the binomial theorem,

$$\begin{aligned}(2a - 3b)^4 &= \binom{4}{0} (2a)^4 (-3b)^0 + \binom{4}{1} (2a)^3 (-3b)^1 + \binom{4}{2} (2a)^2 (-3b)^2 \\&\quad + \binom{4}{3} (2a)^1 (-3b)^3 + \binom{4}{4} (2a)^0 (-3b)^4 \\&= (2a)^4 + 4(2a)^3(-3b) + 6(2a)^2(-3b)^2 + 4(2a)(-3b)^3 + (-3b)^4 \\&= 16a^4 - 96a^3b + 216a^2b^2 - 216ab^3 + 81b^4 \quad \blacksquare\end{aligned}$$

Example 3: What is the coefficient of x^5y^4 in the expansion of $(x + y)^9$?

Solution: From the binomial theorem it follows that this coefficient is $\binom{25}{13} = \frac{9!}{4!(9-4)!} = 126$.

Example 4: What is the coefficient of x^3y^2 in the expansion of $(2x - 3y)^5$?

Solution: First, note that this expression equals $(2x + (-3y))^5$.

By the binomial theorem, the term containing x^3y^2 is

$$\binom{5}{2} (2x)^{5-2} (-3y)^2 = 10 \cdot 2^3 (-3)^2 x^3 y^2 = 720x^3y^2.$$

Therefore the required coefficient is 720.

Corollary 3: Let n be a nonnegative integer. Then

$$\sum_{r=0}^n \binom{n}{r} = 2^n$$

Proof:

By the binomial theorem,

$$(x + y)^n = \sum_{r=0}^n \binom{n}{r} x^{n-r} y^r$$

Let $x = y = 1$. Then

$$2^n = (1 + 1)^n = \sum_{r=0}^n \binom{n}{r} 1^{n-r} 1^r$$

That is,

$$\sum_{r=0}^n \binom{n}{r} = 2^n$$

Pascal's triangle identity

Theorem 4: Pascal's identity

Let n and k be positive integers with $n \geq k$. Then

$$\binom{n}{r} = \binom{n-1}{r-1} + \binom{n-1}{r}$$

Pascal's triangle identity

$$\begin{array}{ccccccc}
 & & \binom{0}{0} & & \dots & n=0 & \\
 & & & & & & \\
 & \binom{1}{0} & & \binom{1}{1} & & \dots & n=1 \\
 & & & & & & \\
 & \binom{2}{0} & & \binom{2}{1} & & \binom{2}{2} & \dots & n=2 \\
 & & & & & & \\
 \binom{3}{0} & & \binom{3}{1} & & \binom{3}{2} & & \binom{3}{3} & \dots & n=3
 \end{array}$$

Theorem 5: The multinomial theorem:

Let $a_1, a_2, \dots, a_m \in \mathbb{R}$ and $n \in \mathbb{Z}$ with $n \geq 1$. Then

$$(a_1 + a_2 + \dots + a_m)^n = \sum_{\substack{0 \leq k_1, k_2, \dots, k_m \leq n \\ k_1 + k_2 + \dots + k_m = n}} \binom{n}{k_1, k_2, \dots, k_m} a_1^{k_1} a_2^{k_2} \dots a_m^{k_m}.$$

Here, the sum is indexed over all ordered m -tuples of integers $k_1, k_2, \dots, k_m$ such that $0 \leq k_1, k_2, \dots, k_m \leq n$ and $k_1 + k_2 + \dots + k_m = n$.

Example 1:

Use the Multinomial Theorem to expand $(x + y + z)^4$.

Solution: Here, the sum in our expansion is indexed over the triples

(k_1, k_2, k_3) such that $0 \leq k_1, k_2, k_3 \leq 4$ and $k_1 + k_2 + k_3 = 4$. That is, the triples

$$\begin{array}{cccccc}
 (4, 0, 0), & (3, 1, 0), & (3, 0, 1), & (2, 2, 0), & (2, 1, 1), \\
 (2, 0, 2), & (1, 3, 0), & (1, 2, 1), & (1, 1, 2), & (1, 0, 3), \\
 (0, 4, 0), & (0, 3, 1), & (0, 2, 2), & (0, 1, 3), & (0, 0, 4)
 \end{array}$$

index our sum. In each case, the triple (k_1, k_2, k_3) corresponds to the term $\binom{4}{k_1, k_2, k_3} x^{k_1} y^{k_2} z^{k_3}$. That is, the triple determines both the exponents

on x , y , and z and the bottom row of the multinomial coefficient. For example, the triple $(3, 0, 1)$ corresponds to the term $\binom{4}{3, 0, 1} x^3 y^0 z^1 = 4x^3 z$. Thus, in total, we get

$$\begin{aligned} (x + y + z)^4 &= \sum_{\substack{0 \leq k_1, k_2, k_3 \leq 4 \\ k_1 + k_2 + k_3 = 4}} \binom{4}{k_1, k_2, k_3} x^{k_1} y^{k_2} z^{k_3} \\ &= x^4 + 4x^3 y + 4x^3 z + 6x^2 y^2 + 12x^2 yz \\ &\quad + 6x^2 z^2 + 4xy^3 + 12xy^2 z + 12xyz^2 + 4xz^3 \\ &\quad + y^4 + 4y^3 z + 6y^2 z^2 + 4yz^3 + z^4. \end{aligned}$$

Example 2: What is the coefficient of $u^2 w^3 x^4 y^2$ in the expansion of $(u + v + 2w + x + 3y + z)^{11}$?

Solution:

The expansion includes the summand

$$\begin{aligned} \binom{11}{2, 0, 3, 4, 2, 0} u^2 v^0 (2w)^3 x^4 (3y)^2 z^0 &= 69300 u^2 \cdot 8w^3 x^4 \cdot 9y^2 \\ &= 4989600 u^2 w^3 x^4 y^2. \end{aligned}$$

Hence, the desired coefficient is 4989600.

Generalized Permutations and Combinations

In many counting problems, elements may be used repeatedly. For instance, a letter or digit may be used more than once on a license plate. When a dozen donuts are selected, each variety can be chosen repeatedly. This contrasts with the counting problems discussed earlier in the chapter where we considered only permutations and combinations in which each item could be used at most once. In this section we will show how to solve counting problems where elements may be used more than once.

Also, some counting problems involve indistinguishable elements. For instance, to count the number of ways the letters of the word *SUCCESS* can be rearranged, the placement of identical letters must be considered. This contrasts with the counting problems discussed earlier where all elements were considered distinguishable. In this section we will describe how to solve counting problems in which some elements are indistinguishable.

Counting permutations when repetition of elements is allowed can easily be done using the product rule, look example below.

Example: How many strings of length r can be formed from the uppercase letters of the English alphabet?

Solution: By the product rule, because there are 26 uppercase English letters, and because each letter can be used repeatedly, we see that there are 26^r strings of uppercase English letters of length r .

Theorem 6: The number of r -permutations of a set of n objects with repetition allowed is given by

$$U(n, r) = n^r$$

Proof: There are n ways to select an element of the set for each of the r positions in the r -permutation when repetition is allowed, because for each choice all n objects are available. Hence, by the multiplication principle there are n^r r -permutations when repetition is allowed.

Combinations with Repetition

Theorem 7: There are $C(n + r - 1, r) = C(n + r - 1, n - 1)$ r -combinations from a set with n elements when repetition of elements is allowed.

Example 1: Suppose that a cookie shop has four different kinds of cookies. How many different ways can six cookies be chosen? Assume that only the type of cookie, and not the individual cookies or the order in which they are chosen, matters.

Solution: The number of ways to choose six cookies is the number of 6-combinations of a set with four elements.

Using Theorem 7 this equals $C(4 + 6 - 1, 6) = C(9, 6) = 84$

Therefore, there are 84 different ways to choose the six cookies.

Theorem 7 can also be used to find the number of solutions of certain linear equations where the variables are integers subject to constraints.

Example 2: How many solutions does the equation $x_1 + x_2 + x_3 = 11$ have, where x_1, x_2 , and x_3 are nonnegative integers?

Solution: Using theorem 7 it follows that there are $C(3 + 11 - 1, 11) = C(13, 11) = C(13, 2) = 78$. Therefore there are 78 non – negative integer solution to the equation of the form (x_1, x_2, x_3) .

Theorem 8: The number of different permutations of n objects, where there are n_1 indistinguishable objects of type 1, n_2 indistinguishable objects of type 2, $\dots$, and n_k indistinguishable objects of type k , is given by

$$P(n; n_1, n_2, \dots, n_k) = \frac{n!}{n_1! n_2! \dots n_k!}$$

Example 1: Find the number of words that can be formed using the letters of the word “BENZENE.”

Solution:

$$P(7; 3, 2) = \frac{7!}{3!2!} = \frac{7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}{3 \cdot 2 \cdot 1 \cdot 2 \cdot 1} = 420$$

Example 2: The number of permutations of the letters in the word MISSISSIPPI is

$$P(11; 1, 4, 4, 2) = \frac{11!}{1!4!4!2!} = 34650$$

1.4 The Inclusion – Exclusion principle

Objectives: At the end of this subtopic the students will be able to

- ✓ State the inclusion – exclusion principle.
- ✓ Solve problems related to inclusion – exclusion principle.

The Inclusion – Exclusion principle

Let A and B be any finite sets. Then

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

For any finite sets A, B, C we have

$$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(A \cap C) - n(B \cap C) + n(A \cap B \cap C)$$

Example: Find the number of mathematics students at a college taking at least one of the languages French, German, and Russian, given the following data:

- 65 study French, 20 study French and German,
- 45 study German, 25 study French and Russian,
- 42 study Russian, 15 study German and Russian,
- and 8 study all three languages.

We want to find $n(F \cup G \cup R)$ where F , G , and R denote the sets of students studying French, German, and Russian, respectively.

By the Inclusion–Exclusion Principle,

$$\begin{aligned}n(F \cup G \cup R) &= n(F) + n(G) + n(R) - n(F \cap G) - n(F \cap R) - n(G \cap R) + n(F \cap G \cap R) \\&= 65 + 45 + 42 - 20 - 25 - 15 + 8 = 100\end{aligned}$$

This implies, 100 students study at least one of the three languages.

1.5 The Pigeonhole Principle

Objectives: At the end of this subtopic the students will be able to

- ✓ State the pigeonhole principle.
- ✓ Solve problems related to pigeonhole principle.

The Pigeonhole Principle

If $k + 1$ or more objects are placed in to k boxes, then there is at least one box containing two or more of the objects.

Example 1: Suppose a department contains 13 professors, then two of the professors were born in the same month. Because there are 12 months in a year.

Example 2: There are n married couples. How many of the $2n$ people must be selected to guarantee that a married couple has been selected?

Solution. To apply the pigeonhole principle in this case, think of n boxes, one corresponding to each of the n couples. If we select $n + 1$ people and put each of them in the box corresponding to the couple to which they belong, then some box contains two people; that is, we have selected a married couple. Two of the ways to select n people without getting a married couple are to select all the husbands or all the wives.

Therefore, $n + 1$ is the smallest number that will guarantee a married couple has been selected.

Example 3: Find the minimum number of elements that one needs to take from the set $S = \{1, 2, 3, \dots, 9\}$ to be sure that two of the numbers add up to 10.

Here the pigeonholes are the five sets $\{1, 9\}$, $\{2, 8\}$, $\{3, 7\}$, $\{4, 6\}$, $\{5\}$. Thus any choice of six elements of S will guarantee that two of the numbers add up to ten.

Generalized Pigeonhole Principle:

If N objects are placed into k boxes, then there is at least one box containing $\left\lceil \frac{N}{k} \right\rceil$ or more objects.

Where $\left\lceil \frac{N}{k} \right\rceil$ is the greatest integer greater than or equal to $\frac{N}{k}$.

Example 1: Find the minimum number of students in a class to be sure that three of them are born in the same month.

Solution: the number of months on one year is $k = 12$. Then the required minimum number of students in a class is equal to $2(12) + 1 = 24 + 1 = 25$.

Check list

To check your mastery the topic put ($\checkmark$) if you can perform it.; otherwise put (X) in front of each statement.

Important points	I Can	I Can't
1. Define permutation and combination.		
2. List r – permutations and r – combinations of n distinct objects.		
3. Evaluate the number of r – permutations and r – combinations of n distinct objects.		
4. Solve problems related to permutations and combinations.		
5. State inclusion – exclusion principle.		
6. State pigeonhole principle.		
7. Solve problems related to inclusion – exclusion principle and pigeonhole principle.		

Is your response positive for the above questions? If your answer is “I Can”, go to the next section. If your answer is “I Can’t”, go back and read once again.

Exercise 1.2

- List all the permutations of $\{a, b, c\}$.
- How many different permutations are there of the set $\{a, b, c, d, e, f, g\}$
- How many permutations of $\{a, b, c, d, e, f, g\}$ end with a ?
- Let $S = \{1, 2, 3, 4, 5\}$
 - List all the 3-permutations of S .
 - List all the 3-combinations of S .
- Find the value of each of these quantities.

a) $P(6, 3)$ **b)** $P(6, 5)$

6. Find the value of each of these quantities.

a) $C(5, 1)$ **b)** $C(5, 3)$

7. Find the number of 5-permutations of a set with nine elements.

8. A class contains 10 students with 6 men and 4 women. Find the number of ways to:

a) Select a 4 member committee from the students.

b) Select a 4 member committee with 2 men and 2 women.

9. Find n if

a) $P(n, 2) = 110$.

b) $P(n, n) = 5040$.

c) $P(n, 4) = 12P(n, 2)$.

d) $C(n, 2) = 45$.

e) $C(n, 3) = P(n, 2)$.

f) $C(n, 5) = C(n, 2)$.

10. How many integers greater than 5400 have both of the following properties?

(a) The digits are distinct.

(b) The digits 2 and 7 do not occur.

11. In how many ways can four men and eight women be seated at a round table if there are to be two women between consecutive men around the table?

12. In how many ways can six men and six women be seated at a round table if the men and women are to sit in alternate seats?

13. In how many ways can 15 people be seated at a round table if B refuses to sit next to A?
What if B only refuses to sit on A's right?

14. Consider all 5 letter "words" made from the letters a through h . (Recall, words are just strings of letters, not necessarily actual English words.)

(a) How many of these words are there total?

(b) How many of these words contain no repeated letters?

(c) How many of these words start with the sub-word "aha"?

(d) How many of these words either start with "aha" or end with "bah" or both?

(e) How many of the words containing no repeats also do not contain the sub-word "bad"?

15. Let $S = \{1, 2, 3, 4, 5, 6\}$

- (a) How many subsets are there total?
 - (b) How many subsets have $\{2, 3, 5\}$ as a subset?
 - (c) How many subsets contain at least one odd number?
 - (d) How many subsets contain exactly one even number?
16. Let $S = \{1, 2, 3, 4, 5, 6\}$
- a) How many subsets are there of cardinality 4?
 - b) How many subsets of cardinality 4 have $\{2, 3, 5\}$ as a subset?
 - c) How many subsets of cardinality 4 contain at least one odd number?
 - d) How many subsets of cardinality 4 contain exactly one even number?
17. Find the expansion of $(x + y)^6$ using binomial theorem.
18. Find the coefficient of x^5y^8 in $(x + y)^{13}$.
19. What is the coefficient of x^7 in $(1 + x)^{11}$?
20. What is the coefficient of x^9 in $(2 - x)^{19}$?
21. Find the middle term in the binomial expansion of each.

a) $\left(x + \frac{1}{x}\right)^4$ b) $\left(2x + \frac{2}{x}\right)^8$ c) $\left(x^2 + \frac{1}{x^2}\right)^{10}$

22. What is the coefficient of $x^4y^6z^2w^4$ in the expansion of $(w - z + y - x)^{20}$?
23. What is the coefficient of $x^4y^2z^5w^3$ in the expansion of $(x + y - 2z + w)^{14}$?

Review Exercise

1. In how many different orders can five runners finish a race if no ties are allowed?
2. How many possibilities are there for the win, place, and show (first, second, and third) positions in a horse race with 12 horses if all orders of finish are possible?
3. Find the number of automobile license plates where
 - a) Each plate contains 2 different letters followed by 3 different digits.
 - b) The first digit cannot be 0.
4. A class has 10 male students and 8 female students. Find the number of ways the class can elect
 - a) a class representative
 - b) 2 class representatives, one male and one female
 - c) a class president and vice president.
5. Suppose a code consists of five characters, two letters followed by three digits. Find the number of: (a) codes; (b) codes with distinct letter; (c) codes with the same letters.

6. A class contains 9 men and 3 women. Find the number of ways a teacher can select a committee of 4 from the class where there is:
- no restrictions
 - 2 men and 2 women
 - exactly one woman
 - at least one woman.
7. There are six different candidates for governor of a state. In how many different orders can the names of the candidates be printed on a ballot?
8. How many bit strings of length 10 contain
- exactly four 1s?
 - at most four 1s?
 - at least four 1s?
 - an equal number of 0s and 1s?
9. A group contains n men and n women. How many ways are there to arrange these people in a row if the men and women alternate?
10. In how many ways can a set of two positive integers less than 100 be chosen?
11. In how many ways can a set of five letters be selected from the English alphabet?
12. How many subsets with an odd number of elements does a set with 10 elements have?
13. How many subsets with more than two elements does a set with 100 elements have?
14. A coin is flipped eight times where each flip comes up either heads or tails. How many possible outcomes
- are there in total?
 - contain exactly three heads?
 - contain at least three heads?
 - contain the same number of heads and tails?
15. A coin is flipped 10 times where each flip comes up either heads or tails. How many possible outcomes
- are there in total?
 - contain exactly two heads?
 - contain at most three tails?
 - contain the same number of heads and tails?
16. Simplify

$$(a) \frac{n!}{(n-1)!}; \quad (b) \frac{(n+2)!}{n!}.$$

17.

If we select any group of 1000 students on campus, show that at least three of them must have the same birthday.

18. Find the minimum number of students needed to guarantee that five of them belong to the same class (Freshman, Sophomore, Junior, Senior). Ans 17
19. Find the minimum number of students needed to guarantee that 3 of them:
- a) have last names which begin with the same first letter;
 - b) were born on the same day of a month (with 31 days).
20. Find the number of permutations that can be formed from all the letters of each word:
- a) QUEUE b) COMMITTEE c) PROPOSITION d) BASEBALL.
21. Suppose you have sets A and B with $|A| = 10$ and $|B| = 15$.
- a) What is the largest possible value for $|A \cap B|$?
 - b) What is the smallest possible value for $|A \cap B|$?
 - c) What are the possible values for $|A \cup B|$?
22. If $|A| = 8$ and $|B| = 5$, what is $|A \cup B| + |A \cap B|$?

Chapter Two

Recurrence Relations

Chapter Objectives

On completion of this chapter, students will be able to

- Understand a recurrence relation for a sequence.
- Form a recurrence relation which represent a given problem.
- Understand a recurrence relation with constant coefficients of degree k .
- Know the methods solving linear recurrence relation with constant coefficients.
- Understand generating function of infinite sequence.

Introduction

Some of the counting problems that cannot be solved using the techniques discussed in chapter 1 can be solved by finding recurrence relations involving the terms of a sequence. In this section we will study a variety of counting problems that can be modeled using recurrence relations. In this section we will study methods for finding explicit formulae for the terms of sequences that satisfy certain types of recurrence relations.

2.1 Definition and Examples of recurrence relations

Objectives: On completion of subtopic, students will be able to

- ✓ Define recurrence relation for a sequence.
- ✓ Derive recurrence relation which represent the given problem.
- ✓ Define linear recurrence relation with constant coefficients.

Definition: A recurrence relation for a sequence $\{a_n\}_{n=0}^{\infty}$ is an equation that express a_n in terms of one or more of previous term $a_0, a_1, \dots, a_{n-1}$ for all $n \geq n_0$, where n_0 is a non – negative integer.

Example 1: Consider the following sequence which begins with the number 3 and for which each of the following terms is found by multiplying the previous term by 2:

$$3, 6, 12, 24, 48, \dots$$

It can be defined recursively by

$$a_0 = 3, a_n = 2a_{n-1} \text{ for } n \geq 1 \text{ or } a_0 = 3, a_{n+1} = 2a_n \text{ for } n \geq 0$$

The second definition may be obtained from the first by setting $n = n + 1$.

Clearly, the formula $a_n = 3(2^n)$ gives us the n th term of the sequence without calculating any previous term.

The following remarks about the above example are in order.

- 1) The equation $a_n = 2a_{n-1}$ or, equivalently, $a_{n+1} = 2a_n$, where one term of the sequence is defined in terms of previous terms of the sequence, is called a recurrence relation.
- 2) The equation $a_0 = 3$, which gives a specific value to one of the terms, is called an initial condition.
- 3) The function $a_n = 3(2^n)$, which gives a formula for a_n as a function of n , not of previous terms, is called a solution of the recurrence relation.
- 4) There may be many sequences which satisfy a given recurrence relation.

For example, each of the following is a solution of the recurrence relation

$$a_n = 2a_{n-1}.$$

1, 2, 4, 8, 16, ... and 7, 14, 28, 56, 112, ...

All such solutions form the so called general solution of the recurrence relation.

- 5) On the other hand, there may be only a unique solution to a recurrence relation which also satisfies given initial conditions. For example, the initial condition $a_0 = 3$ uniquely yields the solution 3, 6, 12, 24, ... of the recurrence relation $a_n = 2a_{n-1}$.

Example 2: Find a_6 in the sequence defined by $a_n = 2a_{n-1} - a_{n-2}$ with $a_0 = 3$ and $a_1 = 4$.

Solution. We know that $a_6 = 2a_5 - a_4$. So to find a_6 we need to find a_5 and a_4 . Well

$$a_5 = 2a_4 - a_3 \quad \text{and} \quad a_4 = 2a_3 - a_2,$$

so if we can only find a_3 and a_2 we would be set. Of course

$$a_3 = 2a_2 - a_1 \quad \text{and} \quad a_2 = 2a_1 - a_0,$$

so we only need to find a_1 and a_0 . But we are given these. Thus

$$a_0 = 3$$

$$a_1 = 4$$

$$\begin{aligned}
 a_2 &= 2 \cdot 4 - 3 = 5 \\
 a_3 &= 2 \cdot 5 - 4 = 6 \\
 a_4 &= 2 \cdot 6 - 5 = 7 \\
 a_5 &= 2 \cdot 7 - 6 = 8 \\
 a_6 &= 2 \cdot 8 - 7 = 9.
 \end{aligned}$$

Example 3: Find a recurrence relation and initial conditions for 1, 5, 17, 53, 161, 485 . . .

Solution: Notice that these are growing by a factor of 3. i.e $1 \cdot 3 = 3$, $5 \cdot 3 = 15$, $17 \cdot 3 = 51$ and so on. It appears that we always end up with 2 less than the next term.

So $a_n = 3a_{n-1} + 2$ is our recurrence relation and the initial condition is $a_0 = 1$.

Example 4: A person invests 10,000 birr at 12% interest compounded annually. How much will be there at the end of 15 years.

Solution: Let A_n represents the amount at the end of n years.

So at the end of $n - 1$ years, the amount is A_{n-1} .

Since the amount after n years equals the amount after $n - 1$ years plus interest for the n th year.

Thus the sequence $\{A_n\}$ satisfies the recurrence relation

$$A_n = A_{n-1} + (0.12) A_{n-1} = (1.12) A_{n-1}, n \geq 1.$$

With initial condition $A_0 = 10,000$.

The recurrence relation with the initial condition allow us to compute the value of A_n for any n .

For example, $A_1 = (1.12) A_0$

$$A_2 = (1.12) A_1 = (1.12)^2 A_0$$

$$A_3 = (1.12) A_2 = (1.12)^3 A_0$$

$$A_n = (1.12)^n A_0$$

which is an explicit formula and the required amount can be derived from the formula by putting $n = 15$. So,

$$A_{15} = (1.12)^{15}(10000).$$

Note: A sequence is solution of the recurrence relation if its terms satisfy the recurrence relation.

Example 4: Determine whether the sequences

$$a) \quad a_n = 3n, n = 0, 1, 2, \dots$$

$$b) \quad a_n = 2^n, n = 0, 1, 2, \dots$$

$$c) \ a_n = 5, \ n = 0, 1, 2, \dots$$

is the solution of the recurrence relation $a_n = 2a_{n-1} - a_{n-2}$, $n \geq 2$.

Solution. a) $a_n = 3n$, $n = 0, 1, 2, \dots$. From this formula we have

$$a_{n-1} = 3(n-1) \text{ and } a_{n-2} = 3(n-2)$$

By substituting these values on the recurrence relation

$$a_n = 2[3(n-1)] - 3(n-2) = 6n - 6 - 3n + 6 = 3n. \text{ which is true.}$$

(b) and (c) are left as exercise for the reader.

Example 5: Check that $a_n = 2^n + 1$ is a solution to the recurrence relation $a_n = 2a_{n-1} - 1$ with $a_1 = 3$.

Solution. First, it is easy to check the initial condition: a_1 should be $2^1 + 1$ according to our closed formula. Indeed, $2^1 + 1 = 3$, which is what we want. To check that our proposed solution satisfies the recurrence relation, by substituting it in.

$$\begin{aligned} 2a_{n-1} - 1 &= 2(2^{n-1} + 1) - 1 \\ &= 2^n + 2 - 1 \\ &= 2^n + 1 \\ &= a_n. \end{aligned}$$

Thus, it is true.

Example 6: Find a recurrence relation and give initial conditions for the number of bit strings of length n that do not have two consecutive 0s. How many such bit strings are there of length five?

Solution: Let a_n denote the number of bit strings of length n that do not have two consecutive 0s. To obtain a recurrence relation for a_n , note that by the sum rule, the number of bit strings of length n that do not have two consecutive 0s equals the number of such bit strings ending with a 0 plus the number of such bit strings ending with a 1. We will assume that $n \geq 3$, so that the bit string has at least three bits.

The bit strings of length n ending with 1 that do not have two consecutive 0s are precisely the bit strings of length $n-1$ with no two consecutive 0s with a 1 added at the end. Consequently, there are a_{n-1} such bit strings. Bit strings of length n ending with a 0 that do not have two consecutive 0s must have 1 as their $(n-1)$ bit; otherwise they would end with a pair of 0s. It

follows that the bit strings of length n ending with a 0 that have no two consecutive 0s are precisely the bit strings of length $n - 2$ with no two consecutive 0s with 10 added at the end. Consequently, there are a_{n-2} such bit strings.

We conclude that, $a_n = a_{n-1} + a_{n-2}$ for $n \geq 3$. The initial conditions are $a_1 = 2$, $a_2 = 3$,

To obtain a_5 , we use the recurrence relation three times to find that

$$a_3 = a_2 + a_1 = 3 + 2 = 5,$$

$$a_4 = a_3 + a_2 = 5 + 3 = 8,$$

$$a_5 = a_4 + a_3 = 8 + 5 = 13.$$

Linear Recurrence Relations

Definition: A linear recurrence relation of degree k with constant coefficients is a recurrence relation of the form

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k} + f(n)$$

where $c_1, c_2, \dots, c_k$ are real numbers, $c_k \neq 0$ and $f(n)$ is a function of n .

The recurrence relation in the definition is linear because the right-hand side is a sum of previous terms of the sequence.

Example 1: Determine whether the following recurrence relations are linear recurrence relation with constant or not.

Definition: A linear homogeneous recurrence relation of degree k with constant coefficients is a recurrence relation of the form

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k}$$

where $c_1, c_2, \dots, c_k$ are real numbers, $c_k \neq 0$ and the coefficients of the terms of the sequence are all constants, rather than functions that depend on n . The degree is k because a_n is expressed in terms of the previous k terms of the sequence and $f(n) = 0$ for all $n \geq 0$.

Example 1: The following are examples of a linear homogeneous recurrence relations of degree with the indicated degree k .

a) $a_n = 2a_{n-1} + a_{n-2}; n \geq 2, k = 2$

b) $a_n = 5a_{n-1}; n \geq 1, k = 1$

c) $c_n = 4a_{n-1} - 2a_{n-2} + a_{n-3}; n \geq 3, k = 3$

Example 2: The recurrence relation $a_n = a_{n-1}^2 + a_{n-2}$ is not linear. The recurrence relation $h_n = 2h_{n-1} + 1$ is not homogeneous. The recurrence relation $a_n = na_{n-1} + a_{n-2}$ does not have constant coefficients.

Linear recurrence relations are studied for two reasons. First, they often occur in modeling of problems. Second, they can be systematically solved.

Check list

To check your mastery the topic put (✓) if you can perform it.; otherwise put (X) in front of each statement.

Important points	I Can	I Can't
1. Define recurrence relation for a sequence.		
2. Derive recurrence relation for a sequence and from a given problem.		
3. Check whether a given sequence is solution for a given recurrence relation.		
4. Find terms of a sequence from the given recurrence relation with initial conditions.		
5. Identify whether the given recurrence relation is linear recurrence relation with constant coefficient.		

Is your response positive for the above questions? If your answer is “I Can”, go to the next section. If your answer is “I Can’t”, go back and read once again.

Exercise 2.1

- Determine which of these linear homogeneous recurrence relations with constant coefficients are. Also, find the degree of those that are.

a) $a_n = 3a_{n-1} + 4a_{n-2} + 5a_{n-3}$

b) $a_n = 2na_{n-1} + a_{n-2}$ c) $a_n = a_{n-1} + a_{n-4}$

d) $a_n = a_{n-1} + 2$ e) $a_n = a_{n-1}^2 + a_{n-2}$

f) $a_n = a_{n-2}$ g) $a_n = a_{n-1} + n$

- Determine which of these linear homogeneous recurrence relations with constant coefficients are. Also, find the degree of those that are.

- a) $a_n = 3a_{n-2}$ b) $a_n = 3$
 c) $a_n = a_{n-1}^2$ d) $a_n = a_{n-1} + 2a_{n-3}$
 e) $a_n = a_{n-1}/n$
 f) $a_n = a_{n-1} + a_{n-2} + n + 3$
 g) $a_n = 4a_{n-2} + 5a_{n-4} + 9a_{n-7}$

3. Show that the following sequences are solutions of the given recurrence relations.

- a) $a_n = 3(2^n)$; $a_n = 2a_{n-1}$ for $n \geq 1$, $a_0 = 3$
 b) $a_n = 3 \cdot 2^n - 2 \cdot 3^n$; $a_n = 5a_{n-1} - 6a_{n-2}$ for $n \geq 2$, $a_0 = 1$, $a_1 = 0$
 c) $a_n = 6 \cdot 2^n - 2 \cdot n2^n$; $a_n = 4a_{n-1} - 4a_{n-2}$ for $n \geq 2$, $a_0 = 6$, $a_1 = 8$

2.2 Solving Linear Homogeneous Recurrence Relations with Constant Coefficients

Objectives: On completion of subtopic, students will be able to

- ✓ Solve the given linear recurrence relation with constant coefficients using different methods.
- ✓ Define generating function.
- ✓ Find the generating function for a given sequence.

i) Characteristic roots method

The basic approach for solving linear homogeneous recurrence relations is to look for solutions of the form $a_n = r^n$, where r is a constant. Note that $a_n = r^n$ is a solution of the recurrence relation $a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k}$ if and only if

$$r^n = c_1 r^{n-1} + c_2 r^{n-2} + \dots + c_k r^{n-k}.$$

When both sides of this equation are divided by r^{n-k} and the right-hand side is subtracted from the left, we obtain the equation $r^k - c_1 r^{k-1} - c_2 r^{k-2} - \dots - c_k = 0$. Consequently, the sequence $\{a_n\}$ with $a_n = r^n$ is a solution if and only if r is a solution of this last equation. We call this the characteristic equation of the recurrence relation. The solutions of this equation are called the characteristic roots of the recurrence relation. As we will see, these characteristic roots can be used to give an explicit formula for all the solutions of the recurrence relation.

Theorem 1: Let c_1 and c_2 be real numbers. Suppose that $r^2 - c_1 r - c_2 = 0$ has two distinct roots r_1 and r_2 . Then the sequence $\{a_n\}$ is a solution of the recurrence relation $a_n = c_1 a_{n-1} +$

$c_2 a_{n-2}$ if and only if $a_n = \alpha_1 r_1^n + \alpha_2 r_2^n$ for $n = 0, 1, 2, \dots$, where α_1 and α_2 are constants which are obtained from initial conditions.

Example 1: Solve the recurrence relation $a_n = a_{n-1} + 2a_{n-2}$; $n \geq 2$ with $a_0 = 2, a_1 = 7$.

Solution: The characteristic equation of the recurrence relation is $r^2 - r - 2 = 0$. This implies $(r - 2)(r + 1) = 0$. Its roots are $r = 2$ and $r = -1$.

Hence, the sequence $\{a_n\}$ is a solution to the recurrence relation if and only if $a_n = \alpha_1 2^n + \alpha_2 (-1)^n$, for some constants α_1 and α_2 .

From the initial conditions, it follows that $a_0 = \alpha_1 2^0 + \alpha_2 (-1)^0 = 2$ and

$$a_1 = \alpha_1 2^1 + \alpha_2 (-1)^1 = 7 \text{ by rewriting we have}$$

$$\alpha_1 + \alpha_2 = 2 \text{ and}$$

$$2\alpha_1 - \alpha_2 = 7$$

Solving these system of equations shows that $\alpha_1 = 3$ and $\alpha_2 = -1$. Hence, the solution to the recurrence relation and initial conditions is the sequence $a_n = 3 \cdot 2^n - (-1)^n, n \geq 0$

Example 2: Solve the recurrence relation $a_n = 7a_{n-1} - 10a_{n-2}$ with $a_0 = 2$ and $a_1 = 3$.

Solution. Rewrite the recurrence relation $a_n - 7a_{n-1} + 10a_{n-2} = 0$.

Now form the characteristic equation:

$$x^2 - 7x + 10 = 0$$

and solve for x :

$$(x - 2)(x - 5) = 0$$

so $x = 2$ and $x = 5$ are the characteristic roots. We therefore know that the solution to the recurrence relation will have the form

$$a_n = a2^n + b5^n.$$

To find a and b , plug in $n = 0$ and $n = 1$ to get a system of two equations with two unknowns:

$$2 = a2^0 + b5^0 = a + b$$

$$3 = a2^1 + b5^1 = 2a + 5b$$

Solving this system gives $a = \frac{7}{3}$ and $b = -\frac{1}{3}$ so the solution to the recurrence relation is

$$a_n = \frac{7}{3}2^n - \frac{1}{3}5^n.$$

Theorem 2: Let c_1 and c_2 be real numbers with $c_2 \neq 0$. Suppose that $r^2 - c_1r - c_2 = 0$ has only one root r . Then the sequence $\{a_n\}$ is a solution of the recurrence relation $a_n = c_1a_{n-1} + c_2a_{n-2}$ if and only if $a_n = \alpha_1r^n + \alpha_2nr^n$ for $n = 0, 1, 2, \dots$ where α_1 and α_2 are constants.

Example 1: Solve the recurrence relation $a_n = 6a_{n-1} - 9a_{n-2}$ with initial conditions $a_0 = 1, a_1 = 6$.

Solution: The characteristic equation of the recurrence relation is $r^2 - 6r + 9 = 0$ and by solving the equation, it has only one root $r = 3$. Hence, the solution to this recurrence relation is $a_n = \alpha_13^n + \alpha_2n3^n$ for some constants α_1 and α_2 . Using the initial conditions,

$$\alpha_0 = 1 = \alpha_1$$

$$a_1 = 6 = \alpha_1 \cdot 3 + \alpha_2 \cdot 3$$

it follows that $\alpha_1 = 1$ and $\alpha_2 = 1$.

Therefore, the solution to this recurrence relation and the initial conditions is $a_n = 3^n + n3^n$.

Theorem 3: Let $c_1, c_2, \dots, c_k$ be real numbers. Suppose that the characteristic equation $r^k - c_1r^{k-1} - c_2r^{k-2} - \dots - c_k = 0$ has k distinct roots $r_1, r_2, \dots, r_k$. Then a sequence a_n is a solution of the recurrence relation $a_n = c_1a_{n-1} + c_2a_{n-2} + \dots + c_ka_{n-k}$ if and only if $a_n = \alpha_1r_1^n + \alpha_2r_2^n + \dots + \alpha_kr_k^n$ for $n = 0, 1, 2, \dots$, where $\alpha_1, \alpha_2, \dots, \alpha_k$ are constants.

Example 1: Find the solution to the recurrence relation $a_n = 6a_{n-1} - 11a_{n-2} + 6a_{n-3}; n \geq 3$. with the initial conditions $a_0 = 2, a_1 = 5$ and $a_2 = 15$.

Solution: The characteristic polynomial of this recurrence relation is $r^3 - 6r^2 + 11r - 6 = 0$. $r^3 - 6r^2 + 11r - 6 = (r - 1)(r - 2)(r - 3) = 0$. This implies the characteristic roots are $r_1 = 1, r_2 = 2$, and $r_3 = 3$. Hence, the solutions to this recurrence relation are of the form $a_n = \alpha_11^n + \alpha_22^n + \alpha_33^n$. To find the constants α_1, α_2 , and α_3 , use the initial conditions. This gives

$$a_0 = 2 = \alpha_1 + \alpha_2 + \alpha_3,$$

$$a_1 = 5 = \alpha_1 + \alpha_2 \cdot 2 + \alpha_3 \cdot 3,$$

$$a_2 = 15 = \alpha_1 + \alpha_2 \cdot 4 + \alpha_3 \cdot 9.$$

When these three simultaneous equations are solved for α_1 , α_2 , and α_3 , we find that $\alpha_1 = 1$, $\alpha_2 = -1$, and $\alpha_3 = 2$. Hence, the unique solution to this recurrence relation and the given initial conditions is the sequence $a_n = 1 - 2^n + 2 \cdot 3^n$.

Example 2: Suppose that the roots of the characteristic equation of a linear homogeneous recurrence relation are 2, 2, 2, 5, 5, and 9 (that is, there are three roots, the root 2 with multiplicity three, the root 5 with multiplicity two, and the root 9 with multiplicity one). What is the form of the general solution?

Solution: The general form of the solution is

$$a_n = \alpha_1 2^n + \alpha_2 n 2^n + \alpha_3 n^2 2^n + \beta_1 5^n + \beta_2 n 5^n + \sigma 9^n.$$

Example 3: Find the solution to the recurrence relation $a_n = -3a_{n-1} - 3a_{n-2} - a_{n-3}$ with initial conditions $a_0 = 1$, $a_1 = -2$, and $a_2 = -1$.

Solution: The characteristic equation of this recurrence relation is

$r^3 + 3r^2 + 3r + 1 = (r + 1)^3 = 0$, which implies there is a single root $r = -1$ of multiplicity 3 of the characteristic equation. Then the solutions of this recurrence relation are of the form

$$a_n = \alpha_1 (-1)^n + \alpha_2 n (-1)^n + \alpha_3 n^2 (-1)^n.$$

To find the constants α_1 , α_2 , and α_3 , use the initial conditions. This gives

$$a_0 = 1 = \alpha_1$$

$$a_1 = -2 = -\alpha_1 - \alpha_2 - \alpha_3$$

$$a_2 = -1 = \alpha_1 + 2\alpha_2 + 4\alpha_3$$

The simultaneous solution of these three equations is $\alpha_1 = 1$, $\alpha_2 = 3$, and $\alpha_3 = -2$.

Therefore, the solution to this recurrence relation with the given initial conditions is the sequence

$$a_n = (1 + 3n - 2n^2)(-1)^n.$$

Linear Nonhomogeneous Recurrence Relations with Constant Coefficients

Definition: A linear nonhomogeneous recurrence relation with constant coefficients, that is, a recurrence relation of the form

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k} + f(n)$$

where $c_1, c_2, \dots, c_k$ are real numbers and $f(n)$ is a function not identically zero depending only on n . The recurrence relation $a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k}$ is called the associated homogeneous recurrence relation. It plays an important role in the solution of the nonhomogeneous recurrence relation.

Example: Each of the recurrence relations $a_n = a_{n-1} + 2n$, $a_n = a_{n-1} + a_{n-2} + n^2 + n + 1$,

$a_n = 3a_{n-1} + n3^n$, and $a_n = a_{n-1} + a_{n-2} + a_{n-3} + n!$ is a linear nonhomogeneous recurrence relation with constant coefficients. The associated linear homogeneous recurrence relations are $a_n = a_{n-1}$, $a_n = a_{n-1} + a_{n-2}$, $a_n = 3a_{n-1}$, and $a_n = a_{n-1} + a_{n-2} + a_{n-3}$, respectively.

The key fact about linear nonhomogeneous recurrence relations with constant coefficients is that every solution is the sum of a particular solution and a solution of the associated linear homogeneous recurrence relation.

Theorem 4: If a_n^p is a particular solution of the nonhomogeneous linear recurrence relation with constant coefficients $a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k} + f(n)$, then all solution is of the form $a_n = a_n^h + a_n^p$, where a_n^h is a solution of the associated homogeneous recurrence relation $a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k}$.

Example 1: Find all solutions of the recurrence relation $a_n = 3a_{n-1} + 2n$. What is the solution with $a_1 = 3$?

Solution: To solve this linear nonhomogeneous recurrence relation with constant coefficients, we need to solve its associated linear homogeneous equation and to find a particular solution for the given nonhomogeneous equation. The associated linear homogeneous equation is $a_n = 3a_{n-1}$. Its solutions are $a_n^h = \alpha 3^n$ where α is a constant.

We now find a particular solution. Because $f(n) = 2n$ is a polynomial in n of degree one, a reasonable trial solution is a linear function in n , say $An + B$, where A and B are constants. To determine whether there are any solutions of this form, suppose that $a_n^p = An + B$ is such a solution. By substitution on the recurrence relation $a_n = 3a_{n-1} + 2n$, implies

$$An + B = 3(A(n-1) + B) + 2n.$$

Simplifying and combining like terms gives

$$(2 + 2A)n + (2B - 3A) = 0. \text{ It follows}$$

$$2 + 2A = 0 \text{ and}$$

$$2B - 3A = 0.$$

This shows that $An + B$ is a solution if and only if $A = -1$ and $B = -3/2$.

Which implies, $a_n^p = -n - 3/2$ is a particular solution.

Then all solutions are of the form $a_n = a_n^h + a_n^p = \alpha 3^n + -n - 3/2$, where α is a constant.

To find the solution with $a_1 = 3$, let $n = 1$ on general solution. We find that $3 = -1 - 3/2 + 3\alpha$, which implies that $\alpha = 11/6$. Therefore the required solution is

$$a_n = \frac{11}{6} 3^n + -n - 3/2.$$

Example 2: Find all solutions of the recurrence relation $a_n = 5a_{n-1} - 6a_{n-2} + 7^n$.

Solution: This is a linear nonhomogeneous recurrence relation. The solutions of its associated homogeneous recurrence relation $a_n = 5a_{n-1} - 6a_{n-2}$ are $a_n^h = \alpha_1 \cdot 3^n + \alpha_2 \cdot 2^n$, where α_1 and α_2 are constants.

Here $f(n) = 7^n$, a reasonable trial solution is $a_n^p = C \cdot 7^n$, where C is a constant.

Substituting the terms of this sequence into the recurrence relation implies that

$$C \cdot 7^n = 5C \cdot 7^{n-1} - 6C \cdot 7^{n-2} + 7^n.$$

Dividing both sides of this equation by 7^{n-2} becomes $49C = 35C - 6C + 49$,

$$\Rightarrow 20C = 49, \quad \Rightarrow C = 49/20.$$

Hence, $a_n^p = (49/20)7^n$ is a particular solution.

Therefore, all solutions are

$$a_n = \alpha_1 \cdot 3^n + \alpha_2 \cdot 2^n + (49/20)7^n$$

Theorem 5: Suppose that $\{a_n\}$ satisfies the linear nonhomogeneous recurrence relation

$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k} + f(n)$, where $c_1, c_2, \dots, c_k$ are real numbers, and

$f(n) = (b_t n^t + b_{t-1} n^{t-1} + \dots + b_1 n + b_0) s^n$, where $b_0, b_1, \dots, b_t$ and s are real numbers. When s is not a root of the characteristic equation of the associated linear homogeneous recurrence relation, there is a particular solution of the form

$$a_n^p = (p_t n^t + p_{t-1} n^{t-1} + \dots + p_1 n + p_0) s^n$$

When s is a root of this characteristic equation and its multiplicity is m , there is a particular solution of the form

$$a_n^p = n^m (p_t n^t + p_{t-1} n^{t-1} + \dots + p_1 n + p_0) s^n$$

Note that in the case when s is a root of multiplicity m of the characteristic equation of the associated linear homogeneous recurrence relation, the factor n^m ensures that the proposed particular solution will not already be a solution of the associated linear homogeneous recurrence relation.

Example 1: What form does a particular solution of the linear nonhomogeneous recurrence relation $a_n = 6a_{n-1} - 9a_{n-2} + f(n)$ have when

a) $f(n) = 3n$

b) $f(n) = n3^n$,

c) $f(n) = n22^n$

d) $F(n) = (n^2 + 1)3^n$?

Solution: The associated linear homogeneous recurrence relation is $a_n = 6a_{n-1} - 9a_{n-2}$. Its characteristic equation, $r^2 - 6r + 9 = (r - 3)^2 = 0$, has a single root, $r = 3$, of multiplicity two.

- a) If $f(n) = 3n$ the above theorem tells us that a particular solution has the form Cn^23^n .
- b) If $f(n) = n3^n$, a particular solution has the form $n^2(p_1n + p_0)3^n$.
- c) If $f(n) = n^22^n$, a particular solution has the form $(p_2n^2 + p_1n + p_0)2^n$.
- d) if $f(n) = (n^2 + 1)3^n$, a particular solution has the form $n^2(p_2n^2 + p_1n + p_0)3^n$

Example 2: Find all solution of the recurrence relation.

$$a_n = 3a_{n-1} - 2a_{n-2} + n2^n \quad (n \geq 2)$$

with initial values $a_0 = a_1 = 1$. The characteristic equation of the associated homogeneous recurrence relation

$$a_n = 3a_{n-1} - 2a_{n-2}$$

is $r^2 - 3r + 2 = 0$. Its roots are $r = 1$ and $r = 2$. Hence the general solution to the homogeneous equation is $a_n^h = \alpha_1 1^n + \alpha_2 2^n = \alpha_1 + \alpha_2 2^n$. In order to find a particular solution to the non-homogeneous solution, let $a_n^p = (An+B)n2^n$ be a particular solution of the given nonhomogeneous recurrence relation. Then by substitution we have,

$$(An + B)n2^n = 3(A(n-1) + B)(n-1)2^{n-1} - 2(A(n-2) + B)(n-2)2^{n-2} + n2^n.$$

Dividing both sides by 2^{n-1}

$$2An^2 + 2Bn = 2An^2 + 2(B - A + 1)n - (A + B).$$

Comparing coefficients, we obtain

$$B = B - A + 1 \text{ and } A + B = 0.$$

which yields $A = 1$, $B = -1$.

By substituting on

$$a_n = a_n^h + a_n^p$$

The general solution to the recurrence relation is given by

$$a_n = \alpha_1 + \alpha_2 2^n + (n-1)n2^n$$

Using initial conditions, $\alpha_1 = 1$, $\alpha_2 = 0$.

Therefore, the required solution is

$$a_n = 1 + (n-1)n2^n.$$

Example: Solve the recurrence equation $a_n - 5a_{n-1} + 6a_{n-2} = 2^n + n$ with initial condition $a_1 = 0$ and $a_2 = 10$.

Solution: The general to the problem is given by the homogeneous part of the given recurrence equation.

The homogeneous part of the equation is

$$a_n - 5a_{n-1} + 6a_{n-2} = 0$$

The characteristic equation of is given as

$$r^2 - 5r + 6 = 0, \text{ implies } (r - 2)(r - 3) = 0$$

Implies $r = 2$ and $r = 3$.

The two roots 2 and 3 of characteristic equation are distinct. So the homogeneous solution is given by

$$a_n^h = A2^n + B3^n$$

Let particular solution be

$$a_n^p = Cn 2^n + Dn + E.$$

Substituting this particular solution in the given recurrence relation , we have

$$\begin{aligned} Cn 2^n + Dn + E - 5 [C(n-1) 2^{n-1} + D(n-1) + E] \\ + 6[C(n-2) 2^{n-2} + D(n-2) + E] &= 2^n + n \\ C[n 2^n - 5(n-1) 2^{n-1} + 6(n-2) 2^{n-2}] + D[n - 5(n-1) + 6(n-2)] \\ + E[1 - 5 + 6] &= 2^n + n \\ C[4n - 10(n-1) + 6(n-2)] 2^{n-2} + D[n - 5n + 5 + 6n - 12] \\ + 2E &= 2^n + n \\ C[4n - 10n + 10 + 6n - 12] 2^{n-2} + D[2n - 7] + 2E &= 2^n + n \\ C[-2] 2^{n-2} + D[2n - 7] + 2E &= 2^n + n \\ -2C = 4 \quad 2D = 1 \quad \text{and} \quad -7D + 2E &= 0 \\ C = -2 \quad D = \frac{1}{2} \quad \text{and} \quad E = \frac{7}{4}. \end{aligned}$$

Thus, particular solution is

$$a_n = -2n \cdot 2^n + \frac{n}{2} + \frac{7}{4}$$

Total solution is given by

$$a_n = A2^n + B3^n - 2n2^n + \frac{n}{2} + \frac{7}{4}$$

Using initial conditions for $n = 1$ and $n = 2$, we get

$$a_1 = A2^1 + B3^1 - 2 \cdot 1 \cdot 2^1 + \frac{1}{2} + \frac{7}{4}$$

$$0 = 2A + 3B - 4 + \frac{1}{2} + \frac{7}{4}$$

$$2A + 3B = \frac{7}{4}$$

$$a_2 = A2^2 + B3^2 - 2 \cdot 2 \cdot 2^2 + \frac{2}{2} + \frac{7}{4}$$

$$10 = 4A + 9B - 16 + 1 + \frac{7}{4}$$

$$4A + 9B = \frac{93}{4}$$

$$A = -9 \quad \text{and} \quad B = \frac{79}{12}$$

Replacing the values of A and B, we get the closed form formula for the given recurrence equation

$$a_n = -9 \cdot 2^n + \frac{79}{12} 3^n - 2n \cdot 2^n + \frac{n}{2} + \frac{7}{4}.$$

ii) Substitution method

Example 1: Use iteration to solve the recurrence relation $a_n = a_{n-1} + n$ with $a_0 = 4$.

Solution. Again, start by writing down the recurrence relation when $n = 1$. This time, don't subtract the a_{n-1} terms to the other side:

$$a_1 = a_0 + 1.$$

Now $a_2 = a_1 + 2$, but we know what a_1 is. By substitution, we

get

$$a_2 = (a_0 + 1) + 2.$$

Now go to $a_3 = a_2 + 3$, using our known value of a_2 :

$$a_3 = ((a_0 + 1) + 2) + 3.$$

We notice a pattern. Each time, we take the previous term and add the current index. So

$$a_n = (((a_0 + 1) + 2) + 3) + \cdots + n - 1) + n.$$

Regrouping terms, we notice that a_n is just a_0 plus the sum of the integers from 1 to n . So, since $a_0 = 4$,

$$a_n = 4 + \frac{n(n+1)}{2}.$$

Example 2: Solve the recurrence equation $a_n = a_{n-1} + 3$ with $a_1 = 2$.

Solution: Using substitution Method,

$$\begin{aligned} \text{We have } a_n &= a_{n-1} + 3 \\ a_n &= a_{n-2} + 3 + 3 \\ &= a_{n-2} + 2 \times 3 \\ &= a_{n-3} + 3 + 2 \times 3 \\ &= a_{n-3} + 3 \times 3 \\ &= a_{n-4} + 3 + 3 \times 3 \\ &= a_{n-4} + 4 \times 3 \\ &\quad \text{-----} \\ &\quad \text{-----} \\ &= a_{n-(n-1)} + (n-1) \times 3 \\ &= a_1 + 3(n-1) \\ &= 2 + 3(n-1) \\ \therefore a_n &= 2 + 3(n-1) \end{aligned}$$

2.3 Generating Functions

Generating functions are used to represent sequences efficiently by coding the terms of a sequence as coefficients of powers of a variable x in a formal power series. Generating functions can be used to solve many types of counting problems, such as the number of ways to select or distribute objects of different kinds, subject to a variety of constraints, and the number of ways to make change for a dollar using coins of different denominations. Generating functions can be used to

solve recurrence relations by translating a recurrence relation for the terms of a sequence into an equation involving a generating function. This equation can then be solved to find a closed form for the generating function. From this closed form, the coefficients of the power series for the generating function can be found, solving the original recurrence relation.

Definition: The generating function for the sequence $a_0, a_1, \dots, a_k, \dots$ of real numbers is the infinite series

$$G(x) = a_0 + a_1x + \dots + a_kx^k + \dots = \sum_{k=0}^{\infty} a_kx^k.$$

Note: The generating function for $\{a_k\}$ given in the above definition is sometimes called the ordinary generating function of $\{a_k\}$ to distinguish it from other types of generating functions for this sequence.

Example 1: The generating functions for the sequences $\{a_k\}$ with

- a) $a_k = 3$,
- b) $a_k = k + 1$
- c) $a_k = 2k$

are

$$\sum_{k=0}^{\infty} 3x^k, \sum_{k=0}^{\infty} (k+1)x^k, \text{ and } \sum_{k=0}^{\infty} 2kx^k, \text{ respectively.}$$

Example 2: What is the generating function for the sequence 1, 1, 1, 1, 1, 1?

Solution. The generating function of 1, 1, 1, 1, 1, 1 is

$$G(x) = 1 + x + x^2 + x^3 + x^4 + x^5.$$

$$(x^6 - 1)/(x - 1) = 1 + x + x^2 + x^3 + x^4 + x^5, \quad x \neq 1.$$

Consequently, $G(x) = (x^6 - 1)/(x - 1)$ is the generating function.

Example 3: Let m be a positive integer. Let $a_k = C(m, k)$, for $k = 0, 1, 2, \dots, m$. What is the generating function for the sequence $a_0, a_1, \dots, a_m$?

Solution: The generating function for this sequence is

$$G(x) = C(m, 0) + C(m, 1)x + C(m, 2)x^2 + \dots + C(m, m)x^m.$$

The binomial theorem shows that $G(x) = (1 + x)^m$.

Example 4:

The function $f(x) = 1/(1 - x)$ is the generating function of the sequence $1, 1, 1, 1, \dots$, because

$$1/(1 - x) = 1 + x + x^2 + \dots$$

for $|x| < 1$. 

Example 5:

The function $f(x) = 1/(1 - ax)$ is the generating function of the sequence $1, a, a^2, a^3, \dots$ because

$$1/(1 - ax) = 1 + ax + a^2x^2 + \dots$$

when $|ax| < 1$, or equivalently, for $|x| < 1/|a|$ for $a \neq 0$. 

Example 6: Find a generating function for $1, 3, 5, 7, 9, \dots$

Solution: Notice that the sequence of differences is constant. We know how to find the generating function for any constant sequence. So denote the generating function for $1, 3, 5, 7, 9, \dots$ by A . We have

$$\begin{array}{r} A = 1 + 3x + 5x^2 + 7x^3 + 9x^4 + \dots \\ - \quad xA = 0 + x + 3x^2 + 5x^3 + 7x^4 + 9x^5 + \dots \\ \hline (1 - x)A = 1 + 2x + 2x^2 + 2x^3 + 2x^4 + \dots \end{array}$$

We know that $2x + 2x^2 + 2x^3 + 2x^4 + \dots = \frac{2x}{1 - x}$. Thus

$$(1 - x)A = 1 + \frac{2x}{1 - x}.$$

Now solve for A :

$$A = \frac{1}{1 - x} + \frac{2x}{(1 - x)^2} = \frac{1 + x}{(1 - x)^2}.$$

Example 7: Find the generating function for $1, 4, 9, 16, \dots$. Note we take $a_0 = 1$.

Solution: Again we call the generating function for the sequence A . Using differencing

$$\begin{array}{r}
A = 1 + 4x + 9x^2 + 16x^3 + \cdots \\
- \quad xA = 0 + x + 4x^2 + 9x^3 + 16x^4 + \cdots \\
\hline
(1-x)A = 1 + 3x + 5x^2 + 7x^3 + \cdots
\end{array}$$

Since $1 + 3x + 5x^2 + 7x^3 + \cdots = \frac{1+x}{(1-x)^2}$ we have $A = \frac{1+x}{(1-x)^3}$.

Example 8: The sequence 1, 3, 7, 15, 31, 63, ... satisfies the recurrence relation $a_n = 3a_{n-1} - 2a_{n-2}$. Find the generating function for the sequence.

Solution: Call the generating function for the sequence A . We have

$$\begin{array}{r}
A = 1 + 3x + 7x^2 + 15x^3 + 31x^4 + \cdots + a_n x^n + \cdots \\
-3xA = 0 - 3x - 9x^2 - 21x^3 - 45x^4 - \cdots - 3a_{n-1}x^n - \cdots \\
+ \quad 2x^2A = 0 + 0x + 2x^2 + 6x^3 + 14x^4 + \cdots + 2a_{n-2}x^n + \cdots \\
\hline
(1 - 3x + 2x^2)A = 1
\end{array}$$

Thus, the generating function is

$$A = \frac{1}{1 - 3x + 2x^2}.$$

Definition:

Let u be a real number and k a nonnegative integer. Then the *extended binomial coefficient* $\binom{u}{k}$ is defined by

$$\binom{u}{k} = \begin{cases} u(u-1)\cdots(u-k+1)/k! & \text{if } k > 0, \\ 1 & \text{if } k = 0. \end{cases}$$

Example 1: Find the values of the extended binomial coefficients $\binom{-2}{3}$ and $\binom{1/2}{3}$.

Solution: Taking $u = -2$ and $k = 3$ in definition above gives us

$$\binom{-2}{3} = \frac{(-2)(-3)(-4)}{3!} = -4.$$

Similarly, taking $u = 1/2$ and $k = 3$ gives us

$$\binom{1/2}{3} = \frac{(1/2)(1/2-1)(1/2-2)}{3!}$$

$$\begin{aligned}
 &= (1/2)(-1/2)(-3/2)/6 \\
 &= 1/16.
 \end{aligned}$$

EXAMPLE 2: When the top parameter is a negative integer, the extended binomial coefficient can be expressed in terms of an ordinary binomial coefficient. To see that this is the case, note that

$$\begin{aligned}
 \binom{-n}{r} &= \frac{(-n)(-n-1)\cdots(-n-r+1)}{r!} && \text{by definition of extended binomial coefficient} \\
 &= \frac{(-1)^r n(n+1)\cdots(n+r-1)}{r!} && \text{factoring out } -1 \text{ from each term in the numerator} \\
 &= \frac{(-1)^r (n+r-1)(n+r-2)\cdots n}{r!} && \text{by the commutative law for multiplication} \\
 &= \frac{(-1)^r (n+r-1)!}{r!(n-1)!} && \text{multiplying both the numerator and denominator} \\
 &&& \text{by } (n-1)! \\
 &= (-1)^r \binom{n+r-1}{r} && \text{by the definition of binomial coefficients} \\
 &= (-1)^r C(n+r-1, r). && \text{using alternative notation for binomial} \\
 &&& \text{coefficients}
 \end{aligned}$$

Theorem 6: The extended binomial theorem

Let x be a real number with $|x| < 1$ and let u be a real number. Then

$$(1+x)^u = \sum_{k=0}^{\infty} \binom{u}{k} x^k.$$

Example 1: Find the generating functions for $(1+x)^{-n}$ and $(1-x)^{-n}$, where n is a positive integer, using the extended binomial theorem.

Solution: By the extended binomial theorem, it follows that

$$(1+x)^{-n} = \sum_{k=0}^{\infty} \binom{-n}{k} x^k.$$

Using the above example

$$(1+x)^{-n} = \sum_{k=0}^{\infty} (-1)^k C(n+k-1, k) x^k.$$

Replacing x by $-x$, we find that

$$(1-x)^{-n} = \sum_{k=0}^{\infty} C(n+k-1, k) x^k.$$

Useful Generating Functions

$G(x)$	a_k
$(1+x)^n = \sum_{k=0}^n C(n, k) x^k$ $= 1 + C(n, 1)x + C(n, 2)x^2 + \cdots + x^n$	$C(n, k)$
$(1+ax)^n = \sum_{k=0}^n C(n, k) a^k x^k$ $= 1 + C(n, 1)ax + C(n, 2)a^2x^2 + \cdots + a^n x^n$	$C(n, k) a^k$
$(1+x^r)^n = \sum_{k=0}^n C(n, k) x^{rk}$ $= 1 + C(n, 1)x^r + C(n, 2)x^{2r} + \cdots + x^{rn}$	$C(n, k/r)$ if $r \mid k$; 0 otherwise
$\frac{1-x^{n+1}}{1-x} = \sum_{k=0}^n x^k = 1 + x + x^2 + \cdots + x^n$	1 if $k \leq n$; 0 otherwise
$\frac{1}{1-x} = \sum_{k=0}^{\infty} x^k = 1 + x + x^2 + \cdots$	1
$\frac{1}{1-ax} = \sum_{k=0}^{\infty} a^k x^k = 1 + ax + a^2x^2 + \cdots$	a^k
$\frac{1}{1-x^r} = \sum_{k=0}^{\infty} x^{rk} = 1 + x^r + x^{2r} + \cdots$	1 if $r \mid k$; 0 otherwise
$\frac{1}{(1-x)^2} = \sum_{k=0}^{\infty} (k+1)x^k = 1 + 2x + 3x^2 + \cdots$	$k+1$
$\frac{1}{(1-x)^n} = \sum_{k=0}^{\infty} C(n+k-1, k) x^k$ $= 1 + C(n, 1)x + C(n+1, 2)x^2 + \cdots$	$C(n+k-1, k) = C(n+k-1, n-1)$

$\frac{1}{(1+x)^n} = \sum_{k=0}^{\infty} C(n+k-1, k)(-1)^k x^k$ $= 1 - C(n, 1)x + C(n+1, 2)x^2 - \dots$	$(-1)^k C(n+k-1, k) = (-1)^k C(n+k-1, n-1)$
$\frac{1}{(1-ax)^n} = \sum_{k=0}^{\infty} C(n+k-1, k)a^k x^k$ $= 1 + C(n, 1)ax + C(n+1, 2)a^2 x^2 + \dots$	$C(n+k-1, k)a^k = C(n+k-1, n-1)a^k$

Example 2: Since Heather's neighbors just moved in, she thought it would be nice to put together a fruit basket for them in the basket she recently made at summer camp. When Heather's mother returned from the grocery store, she had a bag containing apples, bananas, and oranges. How many different possibilities are there for a fruit basket consisting of 36 pieces of fruit if

- there is an ample supply of apples and bananas, but there are only 4 oranges?
- there are 16 of each kind of fruit?
- there is an ample supply of bananas and oranges, and there are 4 distinguishable apples (one Cortland, one Delicious, one Empire, and one Macintosh)?

In each case, fruit baskets are distinguished solely by how many of each type of fruit they contain.

Solution:

a) The generating function that accounts for fruit baskets of all possible sizes is

$$g(x) = (1 + x + x^2 + \dots)(1 + x + x^2 + \dots)(1 + x + x^2 + x^3 + x^4).$$

$$g(x) = \left(\frac{1}{1-x} \right)^2 \frac{1-x^5}{1-x} = \frac{1}{(1-x)^3} (1-x^5).$$

If we let a_i and b_i denote the coefficients of x^i in $\frac{1}{(1-x)^3}$ and $1-x^5$,

with $n = 3$ tell us that the coefficient of x^{36} in $g(x)$ is

$$a_{31}b_5 + a_{36}b_0 = \binom{31+3-1}{31}(-1) + \binom{36+3-1}{36}(1) = -\binom{33}{31} + \binom{38}{36} = 175.$$

b) When there are 16 of each type of fruit, the relevant generating function is

$$\begin{aligned} h(x) &= (1 + x + x^2 + \cdots + x^{16})^3 \\ &= \left(\frac{1 - x^{17}}{1 - x} \right)^3 \\ &= \frac{1}{(1 - x)^3} (1 - x^{17})^3 \\ &= \frac{1}{(1 - x)^3} (1 - 3x^{17} + 3x^{34} - x^{51}). \end{aligned}$$

Here, the coefficient of x^{36} in $h(x)$ is

$$a_2(3) + a_{19}(-3) + a_{36}(1) = \binom{4}{2}(3) + \binom{21}{19}(-3) + \binom{38}{36}(1) = 91.$$

c) Effectively, there are now 6 types of fruit; each of the 4 apples is its own type. The relevant generating function is

$$\begin{aligned} f(x) &= (1 + x + x^2 + \cdots)^2 (1 + x)^4 \\ &= \frac{1}{(1 - x)^2} (1 + x)^4. \end{aligned}$$

From (7.5) and Theorem 7.6, it follows that the coefficient of x^{36} in $f(x)$ is

$$\binom{33}{32} \binom{4}{4} + \binom{34}{33} \binom{4}{3} + \binom{35}{34} \binom{4}{2} + \binom{36}{35} \binom{4}{1} + \binom{37}{36} \binom{4}{0} = 560.$$

iii) Using Generating Functions to Solve Recurrence Relations

Example 1: Solve the recurrence relation $a_k = 3a_{k-1}$ for $k = 1, 2, 3, \dots$ and initial condition $a_0 = 2$.

Let $G(x)$ be the generating function for the sequence $\{a_k\}$, that is,

$$\begin{aligned} G(x) &= \sum_{k=0}^{\infty} a_k x^k. \\ xG(x) &= \sum_{k=0}^{\infty} a_k x^{k+1} = \sum_{k=1}^{\infty} a_{k-1} x^k. \end{aligned}$$

Using the recurrence relation, we see that

$$G(x) - 3xG(x) = \sum_{k=0}^{\infty} a_k x^k - 3 \sum_{k=1}^{\infty} a_{k-1} x^k$$

$$\begin{aligned}
&= a_0 + \sum_{k=1}^{\infty} (a_k - 3a_{k-1})x^k \\
&= 2,
\end{aligned}$$

because $a_0 = 2$ and $a_k = 3a_{k-1}$. Thus,

$$G(x) - 3xG(x) = (1 - 3x)G(x) = 2.$$

Solving for $G(x)$ shows that $G(x) = 2/(1 - 3x)$. Using the identity $1/(1 - ax) = \sum_{k=0}^{\infty} a^k x^k$, from Table 1, we have

$$G(x) = 2 \sum_{k=0}^{\infty} 3^k x^k = \sum_{k=0}^{\infty} 2 \cdot 3^k x^k.$$

Consequently, $a_k = 2 \cdot 3^k$. 

Example 2: Use generating functions to find an explicit formula for a_n for the recurrence relation

$$a_n = 8a_{n-1} + 10^{n-1} \quad \text{and the initial condition } a_1 = 9.$$

Solution:

We multiply both sides of the recurrence relation by x^n to obtain

$$a_n x^n = 8a_{n-1} x^n + 10^{n-1} x^n.$$

Let $G(x) = \sum_{n=0}^{\infty} a_n x^n$ be the generating function of the sequence $a_0, a_1, a_2, \dots$. We sum both sides of the last equation starting with $n = 1$, to find that

$$\begin{aligned}
G(x) - 1 &= \sum_{n=1}^{\infty} a_n x^n = \sum_{n=1}^{\infty} (8a_{n-1} x^n + 10^{n-1} x^n) \\
&= 8 \sum_{n=1}^{\infty} a_{n-1} x^n + \sum_{n=1}^{\infty} 10^{n-1} x^n \\
&= 8x \sum_{n=1}^{\infty} a_{n-1} x^{n-1} + x \sum_{n=1}^{\infty} 10^{n-1} x^{n-1}
\end{aligned}$$

$$\begin{aligned}
&= 8x \sum_{n=0}^{\infty} a_n x^n + x \sum_{n=0}^{\infty} 10^n x^n \\
&= 8xG(x) + x/(1 - 10x),
\end{aligned}$$

$$G(x) - 1 = 8xG(x) + x/(1 - 10x).$$

Solving for $G(x)$ shows that

$$G(x) = \frac{1 - 9x}{(1 - 8x)(1 - 10x)}.$$

Expanding the right-hand side of this equation into partial fractions (as is done in the integration of rational functions studied in calculus) gives

$$G(x) = \frac{1}{2} \left(\frac{1}{1 - 8x} + \frac{1}{1 - 10x} \right).$$

$$\begin{aligned}
G(x) &= \frac{1}{2} \left(\sum_{n=0}^{\infty} 8^n x^n + \sum_{n=0}^{\infty} 10^n x^n \right) \\
&= \sum_{n=0}^{\infty} \frac{1}{2} (8^n + 10^n) x^n.
\end{aligned}$$

Consequently, we have shown that

$$a_n = \frac{1}{2} (8^n + 10^n).$$

Example: Use generating functions to solve the recurrence relation

$$a_n = 3a_{n-1} + 2, a_0 = 1$$

Solution. Let $G(x) = \sum_{n=0}^{\infty} a_n x^n$, where $G(x)$ is the generating function for the sequence $\{a_n\}$.

Multiplying each term in the given recurrence relation by x^n and summing from 1 to ∞ , we get

$$\sum_{n=1}^{\infty} a_n x^n = 3 \sum_{n=1}^{\infty} a_{n-1} x^n + 2 \sum_{n=1}^{\infty} x^n$$

$$G(x) - a_0 = 3x G(x) + 2 \left[\frac{1}{1-x} - 1 \right]$$

$$\text{Since } xG(x) = \sum_{n=0}^{\infty} a_n x^{n+1} = \sum_{n=1}^{\infty} a_{n-1} x^n$$

$$G(x) - 3xG(x) = 1 + \frac{2x}{1-x}$$

$$G(x) = \frac{1+x}{(1-x)(1-3x)}$$

$$= \frac{2}{1-3x} - \frac{1}{1-x}$$

$$\sum_{n=0}^{\infty} a_n x^n = 2 \sum_{n=0}^{\infty} 3^n x^n - \sum_{n=0}^{\infty} x^n.$$

Hence $a_n = 2 \cdot 3^n - 1$, which is the required solution.

Check list

To check your mastery the topic put (✓) if you can perform it.; otherwise put (X) in front of each statement.

Important points	I Can	I Can't
1. Solve linear recurrence relations with constant coefficients using a) Characteristic roots method b) Substitution method. c) Generating function. 2. Define generating function. 3. Find a generating function for a given sequence.		

Is your response positive for the above questions? If your answer is “I Can”, go to the next section. If your answer is “I Can’t”, go back and read once again.

Exercise 2.2

- For each recurrence relation and initial conditions, find: (i) general solution; (ii) unique solution with the given initial conditions:

$$\begin{array}{ll}
 \text{(a)} & a_n = 3a_{n-1} + 10a_{n-2}; a_0 = 5, a_1 = 11 \\
 \text{(b)} & a_n = 4a_{n-1} + 21a_{n-2}; a_0 = 9, a_1 = 13 \\
 \text{(c)} & a_n = 3a_{n-1} - 2a_{n-2}; a_0 = 5, a_1 = 8 \\
 \text{(d)} & a_n = 5a_{n-1} - 6a_{n-2}; a_0 = 2, a_1 = 8 \\
 \text{(e)} & a_n = 3a_{n-1} - a_{n-2}; a_0 = 0, a_1 = 1 \\
 \text{(f)} & a_n = 5a_{n-1} - 3a_{n-2}; a_0 = 0, a_1 = 1
 \end{array}$$

- Solve the following recurrence relations with the initial conditions.

- (a) $a_n = 10a_{n-1} - 32a_{n-2} + 32a_{n-3}$ with $a_0 = 5, a_1 = 18, a_2 = 76$
 (b) $a_n = 9a_{n-1} - 27a_{n-2} + 27a_{n-3}$ with $a_0 = 5, a_1 = 24, a_2 = 117$

3. Find the unique solution to each recurrence relation with the given initial conditions:

a)

$$a_n = 10a_{n-1} - 32a_{n-2} + 32a_{n-3} \text{ with } a_0 = 5, a_1 = 18, a_2 = 76$$

b)

$$a_n = 9a_{n-1} - 27a_{n-2} + 27a_{n-3} \text{ with } a_0 = 5, a_1 = 24, a_2 = 117$$

4.

a) Find all solutions of the recurrence relation

$$a_n = 2a_{n-1} + 2n^2.$$

b) Find the solution of the recurrence relation in part (a) with initial condition $a_1 = 4$.

5.

a) Find all solutions of the recurrence relation

$$a_n = 2a_{n-1} + 3^n.$$

b) Find the solution of the recurrence relation in part (a) with initial condition $a_1 = 5$.

6. What is the general form of the particular solution of the linear non homogeneous recurrence relation

$$a_n = 6a_{n-1} - 12a_{n-2} + 8a_{n-3} + F(n) \text{ if}$$

a) $F(n) = n^2?$

b) $F(n) = 2^n?$

c) $F(n) = n2^n?$

d) $F(n) = (-2)^n?$

e) $F(n) = n^2 2^n?$

f) $F(n) = n^3(-2)^n?$

g) $F(n) = 3?$

7. Find the generating function for the finite sequence 2, 2, 2, 2, 2, 2.

8. Find the generating function for the finite sequence 1, 4, 16, 64, 256.

9. Find a closed form for the generating function for each of these sequences. (For each sequence, use the most obvious choice of a sequence that follows the pattern of the initial terms listed.)

a) 0, 2, 2, 2, 2, 2, 2, 0, 0, 0, 0, 0, ...

b) 0, 0, 0, 1, 1, 1, 1, 1, 1, ...

c) 0, 1, 0, 0, 1, 0, 0, 1, 0, 0, 1, ...

d) 2, 4, 8, 16, 32, 64, 128, 256, ...

e) 2, -2, 2, -2, 2, -2, 2, -2, ...

- f) 1, 1, 0, 1, 1, 1, 1, 1, 1, 1, ...
- g) 0, 0, 0, 1, 2, 3, 4, ...
10. Find a closed form for the generating function for the sequence $\{a_n\}$, where
- $a_n = 5$ for all $n = 0, 1, 2, \dots$
 - $a_n = 3n$ for all $n = 0, 1, 2, \dots$
 - $a_n = 2$ for $n = 3, 4, 5, \dots$ and $a_0 = a_1 = a_2 = 0$.
 - $a_n = 2n + 3$ for all $n = 0, 1, 2, \dots$
11. Find a generating function for the sequence with recurrence relation $a_n = 3a_{n-1} - a_{n-2}$ with initial terms $a_0 = 1$ and $a_1 = 5$.
12. Find the generating function for the sequence with closed formula $a_n = 2(5^n) + 7(-3)^n$.
13. Find a closed formula for the n th term of the sequence with generating function
- $$G(x) = \frac{3x}{1-4x} + \frac{1}{1-x}.$$
14. For each of these generating functions, provide a closed formula for the sequence it determines.
- $(3x - 4)^3$
 - $(x^3 + 1)^3$
 - $1/(1 - 5x)$
 - $x^3/(1 + 3x)$
15. Use generating functions to solve the recurrence relation $a_k = 5a_{k-1} - 6a_{k-2}$ with initial conditions $a_0 = 6$ and $a_1 = 30$.
16. Use generating functions to solve the recurrence relation $a_k = a_{k-1} + 2a_{k-2} + 2k$ with initial conditions $a_0 = 4$ and $a_1 = 12$.

Review Exercise

- Find the first four terms of the sequence defined by each of the recurrence relations and initial conditions
 - $a_n = 2a_{n-1} + n, \quad n \geq 2, a_1 = 1$
 - $a_n = a_{n-1}^2, \quad n \geq 2, a_1 = 12$
 - $a_n = n(a_{n-1})^2, \quad n \geq 1, a_0 = 1$
 - $b_k = b_{k-1} + 2b_{k-2}, \quad k \geq 2, b_0 = 1, b_1 = 1$
 - $b_k = kb_{k-1} + k^2b_{k-2}, \quad k \geq 2, b_0 = 1, b_1 = 1.$

2. Show that the sequence $2, 3, 4, 5, \dots, 2n+1, \dots$ for $n \geq 0$, satisfies the recurrence relation $a_k = 2a_{k-1} - a_{k-2}$, $k \geq 2$.
3. Show that the sequence $0, 1, 3, 7, \dots, 2^m - 1, \dots$ for $n \geq 0$, satisfies the recurrence relation $a_k = 3a_{k-1} - 2a_{k-2}$, $k \geq 2$.
4. Show that $a_n = C_1 + C_2 2^n - n$ is a solution of the recurrence relation $a_n - 3a_{n-1} + 2a_{n-2} = 1$.
5. Show that $a_n = 0$, $a_n = 4^n$, $a_n = n4^n$ and $a_n = 2 \cdot 4^n + 3n4^n$ are all solutions of the same recurrence relation

$$a_n = 8a_{n-1} - 16a_{n-2}.$$

6. Show that $a_n = C_1 2^n + C_2 4^n$ is a solution of the recurrence relation $a_n - 6a_{n-1} + 8a_{n-2} = 0$.
7. Show that $a_n = (C_1 + C_2 n) \cdot 2^n$ is a solution of the recurrence relation $a_n - 4a_{n-1} + 4a_{n-2} = 0$.
8. Prove that if $S_n = aS_{n-1}$ for $n \geq 1$ and $a \neq 0$ then $S_n = a^n S_0$.
9. Use the iterative method to find the solution to each of the following recurrence relations with initial conditions.
 - (a) $a_n = a_{n-1} + 2$, $a_0 = 1$
 - (b) $a_n = 2a_{n-1} + 1$, $a_1 = 7$
 - (c) $a_n = a_{n-1} + n$, $a_1 = 4$
 - (d) $a_n = na_{n-1}$, $a_0 = 5$.
10. Solve the recurrence relations together with initial conditions given
 - (a) $a_n = a_{n-1} + 6a_{n-2}$, $n \geq 2$, $a_0 = 1$, $a_1 = 1$
 - (b) $a_n = 3a_{n-1} - 3a_{n-2} + a_{n-3}$, $a_0 = 0$, $a_3 = 3$, $a_5 = 10$
 - (c) $a_n = 7a_{n-1} - 10a_{n-2}$, $a_0 = 4$, $a_1 = 17$
 - (d) $a_n - 8a_{n-1} + 16a_{n-2} = 0$, $a_2 = 16$, $a_3 = 80$
 - (e) $a_n = a_{n-1} + 2a_{n-2}$, $a_0 = 2$, $a_1 = 7$.
11. Solve each of the following recurrence relations that satisfies the initial conditions.
 - (a) $a_n - a_{n-1} = n$, $a_0 = 2$
 - (b) $a_n - 2a_{n-1} = 6n$, $a_1 = 2$
 - (c) $a_n - a_{n-1} = 6_{n+1}$, $a_0 = 1$
 - (d) $a_n - a_{n-1} = 3^n$, $a_0 = 1$
 - (e) $a_{n+1} - 2a_n = 3 + 4^n$, $a_0 = 2$

Chapter Three

Elements of Graph Theory

Chapter Objectives

On completion of this chapter, students will be able to

- Understand a graph.
- Understand simple graph and multi graph.
- Know different types of simple graphs.
- Understand isomorphism of two graphs.
- Understand walk and path on a graph.
- Understand connectivity of graphs.
- Understand Eulerian graphs and Hamiltonian graphs.
- Understand planar graph.
- Understand tree.
- Understand directed graph.

Introduction

Graphs are discrete structures consisting of vertices and edges that connect these vertices. There are different kinds of graphs, depending on whether edges have directions, whether multiple edges can connect the same pair of vertices, and whether loops are allowed. Problems in almost every conceivable discipline can be solved using graph models.

Objectives: On completion of subtopic, students will be able to.

- ✓ Define a graph.
- ✓ Identify simple graph and multi graph.
- ✓ Identify different types of simple graphs.
- ✓ Find degree of a vertex in an undirected graph.
- ✓ State hand shaking theorem and verify it for any finite undirected graph.

3.1 Definitions of terms of graphs and types of graphs

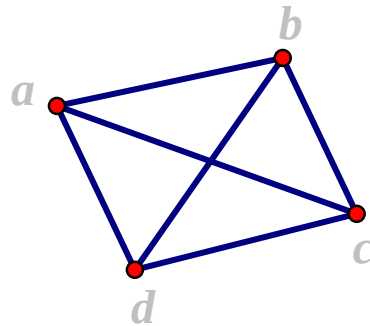
Definition: A graph $G = (V, E)$ consists of V , a nonempty set of vertices (or nodes) and E , a set of edges. Each edge has either one or two vertices associated with it, called its endpoints. An edge is said to connect its endpoints.

Example 1: Let $G = (V, E)$ is a graph with $V = \{a, b, c, d\}$ and

$$E = \{(a, b), (a, c), (a, d), (b, c), (b, d), (c, d)\}$$

Then draw the graph G .

Solution: G is the graph below



Note: The set of vertices V of a graph G may be infinite. A graph with an infinite vertex set or an infinite number of edges is called an infinite graph, and in comparison, a graph with a finite vertex set and a finite edge set is called a finite graph.

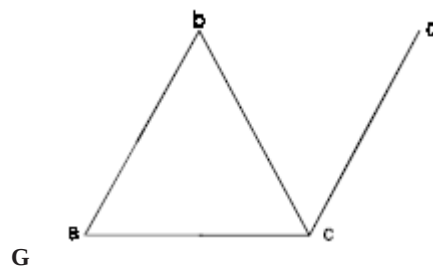
Undirected graph

An undirected graph G consists of set V of vertices and a set E of edges such that each edge $e \in E$ is associated with an unordered pair of vertices.

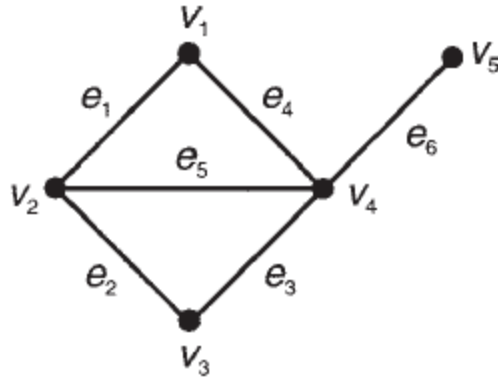
In other words, if each edge of the graph G has no direction then the graph is called undirected graph.

We can refer to an edge joining the vertex pair i and j as either (i, j) or (j, i) .

Example: The graph G below is an undirected graph.



Example 2: The following is the graph with 5 vertices and 6 edges.



Graph Terminology and Special Types of Graphs

Definition: Two vertices u and v in an undirected graph G are called adjacent (or neighbors) in G if u and v are endpoints of an edge e of G . Such an edge e is called incident with the vertices u and v and e is said to connect u and v .

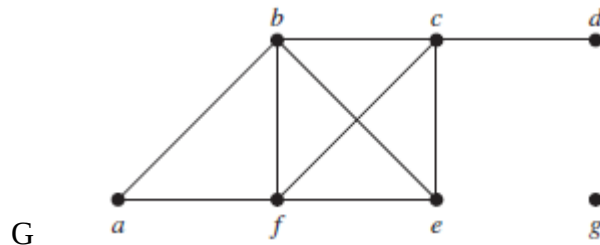
Example 1: The vertices a and b , b and c are adjacent vertices and the respective edges

Definition: The set of all neighbors of a vertex v of $G = (V, E)$, denoted by $N(v)$, is called the neighborhood of v . If A is a subset of V , we denote by $N(A)$ the set of all vertices in G that are adjacent to at least one vertex in A . So,

$$N(A) = \bigcup_{v \in A} N(v).$$

Definition: The degree of a vertex in an undirected graph is the number of edges incident with it, except that a loop at a vertex contributes twice to the degree of that vertex. The degree of the vertex v is denoted by $\deg(v)$.

Example 1: what are the degrees and what are the neighborhoods of the vertices in the graph G displayed in Figure below?

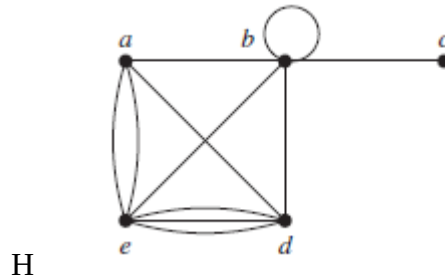


Solution: $\deg(a) = 2$, $\deg(b) = \deg(c) = \deg(f) = 4$, $\deg(d) = 1$, $\deg(e) = 3$, and $\deg(g) = 0$.

The neighborhoods of these vertices are $N(a) = \{b, f\}$, $N(b) = \{a, c, e, f\}$,

$N(c) = \{b, d, e, f\}$, $N(d) = \{c\}$, $N(e) = \{b, c, f\}$, $N(f) = \{a, b, c, e\}$, and $N(g) = \emptyset$.

Example 2: what are the degrees and what are the neighborhoods of the vertices in the graph G displayed in figure below?



Solution: $\deg(a) = 4$, $\deg(b) = \deg(e) = 6$, $\deg(c) = 1$, and $\deg(d) = 5$. The neighborhoods of these vertices are $N(a) = \{b, d, e\}$, $N(b) = \{a, b, c, d, e\}$, $N(c) = \{b\}$, $N(d) = \{a, b, e\}$, and $N(e) = \{a, b, d\}$.

Isolated and pendent vertices

Isolated vertex

A vertex having no incident edge is called an isolated vertex. In other words, isolated vertices are those with zero degree.

Pendent or end vertex

A vertex of degree one, is called a pendent vertex or an end vertex. In the above Figure, v_5 is a pendent vertex.

In degree and out degree

In a graph G, the out degree of a vertex v_i of G, denoted by $\text{out deg}_G(v_i)$ or $\deg_G^+(v_i)$, is the number of edges beginning at v_i and the in degree of v_i , denoted by $\text{in deg}_G(v_i)$ or $\deg_G^-(v_i)$, is the number of edges ending at v_i .

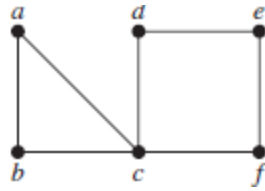
The sum of the in degree and out degree of a vertex is called the total degree of the vertex.

Theorem 1: Handshaking theorem

If $G = (V, E)$ be an undirected graph with e edges .then the sum of degrees of the vertices of G is equal to twice the number of edges. i.e

$$\sum_{v \in V} \deg_G(v) = 2e$$

Example 1: Verify handshaking theorem for the graph G below.



Solution: $\deg(a) = 2$, $\deg(b) = 2$, $\deg(c) = 4$, $\deg(d) = 2$, $\deg(e) = 2$ and $\deg(f) = 2$.

$$e = 7$$

$$\deg(a) + \deg(b) + \deg(c) + \deg(d) + \deg(e) + \deg(f) = 2 \cdot e$$

$$2 + 2 + 4 + 2 + 2 + 2 = 14 = 2 \cdot 7.$$

Example 2: How many vertices and edges must a graph have if its degree sequence is (4, 4, 3, 3, 3, 2, 1)?

Solution: The number of vertices is easy to find, it is the number of degrees in the sequence: that is 7. To find the number of edges, we compute the degree sum:

$$4 + 4 + 3 + 3 + 3 + 2 + 1 = 20,$$

so the number of edges is half this. Which is 10.

Example 3: Determine the number of edges in a graph with 6 vertices, 2 of degree 4 and 4 of degree 2. Draw two such graphs.

Solution: Suppose the graph with 6 vertices has e number of edges. Therefore by Handshaking theorem, we have

$$\sum_{i=1}^6 \deg(v_i) = 2e$$

$$\Rightarrow d(v_1) + d(v_2) + d(v_3) + d(v_4) + d(v_5) + d(v_6) = 2e$$

Now, given 2 vertices are of degree 4 and 4 vertices are of degree 2.

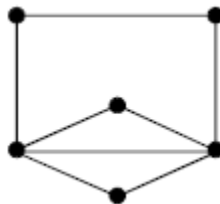
Hence the above equation,

$$(4 + 4) + (2 + 2 + 2 + 2) = 2e$$

$$\Rightarrow 16 = 2e \quad \Rightarrow e = 8.$$

Hence the number of edges in a graph with 6 vertices with given condition is 8.

The graph is as follows



Corollary 1: In a non-directed graph, the total number of odd degree vertices is even.

Example 4: How many edges are there in a graph with 10 vertices each of degree six?

Solution: Because the sum of the degrees of the vertices is $6 \cdot 10 = 60$, it follows that $2e = 60$

where m is the number of edges.

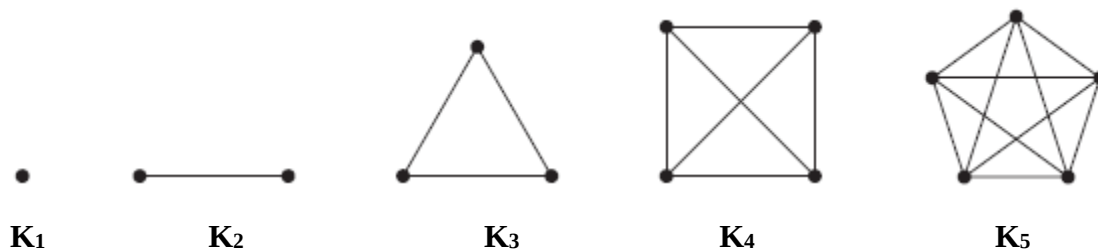
Therefore, $e = 30$.

Some Special Simple Graphs

i) Complete Graphs

Definition: A complete graph on n vertices is a simple graph that contains exactly one edge between each pair of distinct vertices. Denoted by K_n ,

Example 1: The graphs K_n , for $n = 1, 2, 3, 4, 5$ are given below.

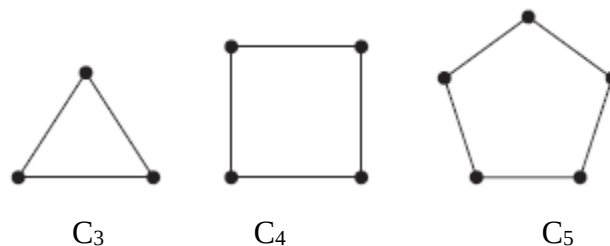


Note: A complete graph with n vertices has $C(n, 2) = \frac{n(n-1)}{2}$ edges.

ii) Cycles

Definition: A cycle graph C_n , $n \geq 3$, consists of n vertices $v_1, v_2, \dots, v_n$ and edges $(v_1, v_2), (v_2, v_3), \dots, (v_{n-1}, v_n)$, and $\{v_n, v_1\}$.

Example 2: The cycles C_3 , C_4 and C_5 are given below



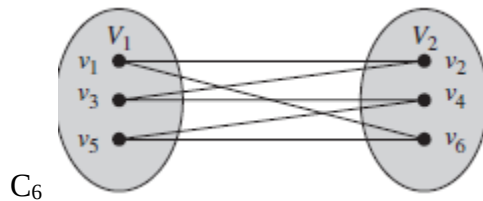
iii) Bipartite Graphs

Definition: A simple graph G is called bipartite if its vertex set V can be partitioned into two disjoint sets V_1 and V_2 such that every edge in the graph connects a vertex in V_1 and a vertex in V_2 (so that no edge in G connects either two vertices in V_1 or two vertices in V_2).

When the condition holds in the above definition we call the pair (V_1, V_2) a bipartition of the vertex set V of G .

Example 3: C_6 is bipartite, as shown in diagram below.

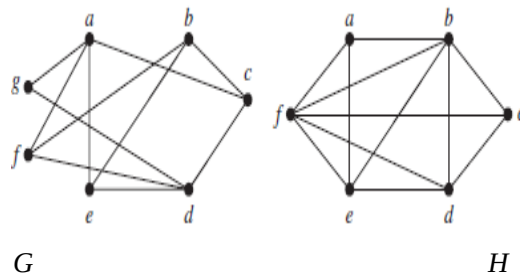
Because its vertex set can be partitioned into the two sets $V_1 = \{v_1, v_3, v_5\}$ and $V_2 = \{v_2, v_4, v_6\}$, and every edge of C_6 connects a vertex in V_1 and a vertex in V_2 .



Example 4: Show that K_3 is not bipartite.

Solution: To verify this, note that if we divide the vertex set of K_3 into two disjoint sets, one of the two sets must contain two vertices. If the graph were bipartite, these two vertices could not be connected by an edge, but in K_3 each vertex is connected to every other vertex by an edge.

Example 5: Are the graphs G and H displayed in Figure below bipartite?



Solution: Graph G is bipartite because its vertex set is the union of two disjoint sets, $\{a, b, d\}$ and $\{c, e, f, g\}$, and each edge connects a vertex in one of these subsets to a vertex in the other subset. Graph H is not bipartite because its vertex set cannot be partitioned into two subsets so that edges do not connect two vertices from the same subset.

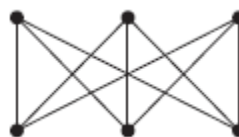
Definition: A complete bipartite graph $K_{m,n}$ is a graph that has its vertex set partitioned into two subsets of m and n vertices, respectively with an edge between two vertices if and only if one vertex is in the first subset and the other vertex is in the second subset.

Example 6: Draw the complete bipartite graphs $K_{2,3}$ and $K_{3,3}$.

Solution:



$K_{2,3}$

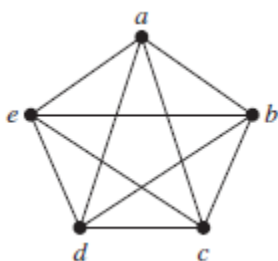


$K_{3,3}$

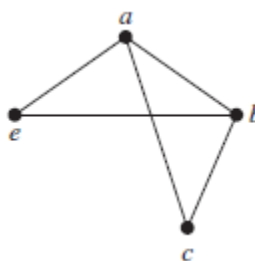
Definition: A subgraph of a graph $G = (V, E)$ is a graph $H = (W, F)$, where $W \subseteq V$ and $F \subseteq E$.

Example 7: The graph G is a subgraph of K_5 .

Solution: K_5 and its subgraph G are given below.



K_5



G

Check list

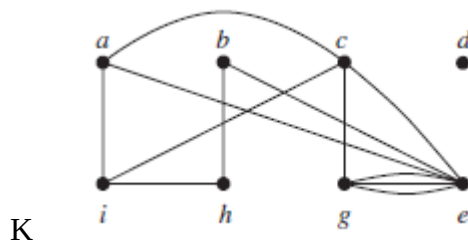
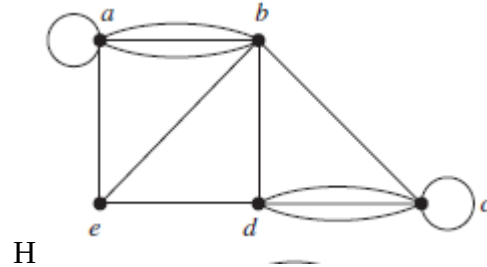
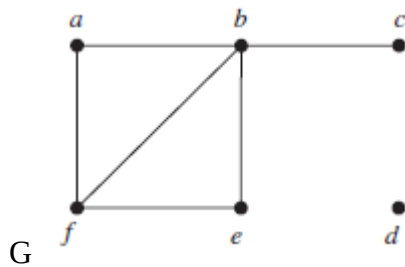
To check your mastery the topic put ($\checkmark$) if you can perform it.; otherwise put (X) in front of each statement.

Important points	I Can	I Can't
1. Define a graph (undirected graph). 2. Define simple graph and multi graph. 3. Define simple graphs (regular graph, complete graph, bipartite graph). 4. Find degree of a vertex in an undirected graph. 5. State handshaking theorem and verify it for any finite undirected graph.		

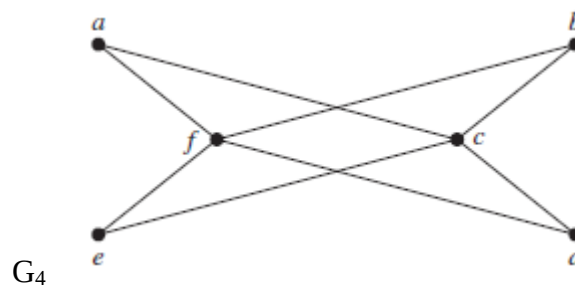
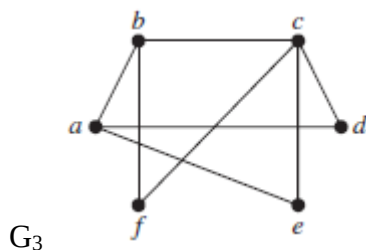
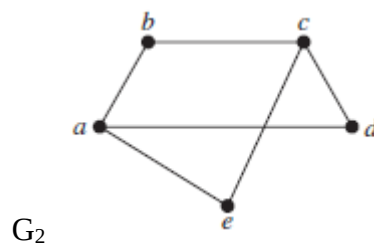
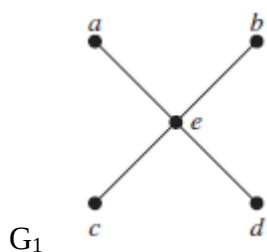
Is your response positive for the above questions? If your answer is “I Can”, go to the next section. If your answer is “I Can’t”, go back and read once again.

Exercise 3.1

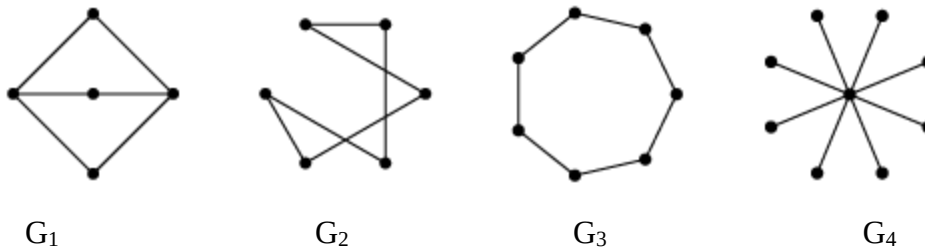
- Find the number of vertices, the number of edges, and the degree of each vertex in the given undirected graph. Identify all isolated and pendant vertices.



- Find the sum of the degrees of the vertices of each graph in question no.1 and verify that it equals twice the number of edges in the graph.
- Draw the following graphs.
 - K_7
 - $K_{1,5}$
 - $K_{4,4}$
 - C_7
- Determine whether the following graphs are bipartite or not.



- Which of the graphs below are bipartite? Justify your answers.



6. For which values of n are these graphs bipartite.
 - a) K_n
 - b) C_n
7. How many vertices and how many edges do these graphs have?
 - a) K_n
 - b) C_n
8. How many edges does a graph have if its degree sequence is 4, 3, 3, 2, 2? Draw such a graph.
9. How many edges does a graph have if its degree sequence is 5, 2, 2, 2, 2, 1? Draw such a graph.
10. For which values of n are these graphs regular?
 - a) K_n
 - b) C_n
11. For which values of m and n is $K_{m,n}$ regular?
12. If 10 people each shake hands with each other, how many handshakes took place? What does this question have to do with graph theory?

3.2 Isomorphism of Graphs and Matrix Representations of Graphs

Objectives: On completion of subtopic, students will be able to

- ✓ Define isomorphism of two graphs.
- ✓ Determine whether the given two graphs are isomorphic or not.
- ✓ List the invariant properties of isomorphic graphs.
- ✓ Define incidence matrix and adjacency matrix of a graph.
- ✓ Find the incidence matrix and adjacency matrix of a graph.

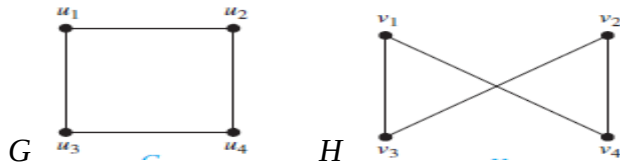
Isomorphism of Graph

Definition: The simple graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ are isomorphic if there exists a one-to-one and onto function f from V_1 to V_2 with the property that a and b are adjacent in G_1 if and only if $f(a)$ and $f(b)$ are adjacent in G_2 , for all a and b in V_1 . Denoted by $G_1 \cong G_2$.

Two simple graphs that are not isomorphic are called non isomorphic. In other words, when two simple graphs are isomorphic, there is a one-to-one correspondence

between vertices of the two graphs that preserves the adjacency relationship.

Example 1: Show that the graphs $G = (V, E)$ and $H = (W, F)$ are isomorphic.



Solution: Take the function f with $f(u_1) = v_1$, $f(u_2) = v_4$,

$$f(u_3) = v_3, \text{ and } f(u_4) = v_2$$

is a one to-one and onto function between V and W . To see that this correspondence preserves adjacency.

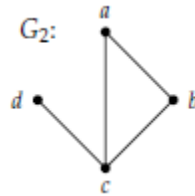
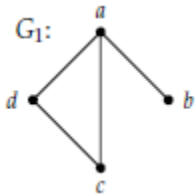
In graph G u_1 and u_2 , u_1 and u_3 , u_2 and u_4 and u_3 and u_4 , and each of the pairs $f(u_1) = v_1$ and $f(u_2) = v_4$, $f(u_1) = v_1$ and $f(u_3) = v_3$, $f(u_2) = v_4$ and $f(u_4) = v_2$ and $f(u_3) = v_3$ and $f(u_4) = v_2$ consists of two adjacent vertices in H .

Therefore, the graphs G and H are isomorphic.

Example 2: Decide whether the graphs $G_1 = \{V_1, E_1\}$ and $G_2 = \{V_2, E_2\}$ are isomorphic.

$$V_1 = \{a, b, c, d\}, E_1 = \{\{a, b\}, \{a, c\}, \{a, d\}, \{c, d\}\}$$

$$V_2 = \{a, b, c, d\}, E_2 = \{\{a, b\}, \{a, c\}, \{b, c\}, \{c, d\}\}$$



Solution: We can try to build an isomorphism. How about we say $f(a) = b$, $f(b) = c$, $f(c) = d$ and $f(d) = a$. This is definitely a bijection, but to make sure that the function is an isomorphism.

Alternatively, notice that in G_1 , the vertex a is adjacent to every other vertex. In G_2 , there is also a vertex with this property: c . So build the bijection $g : V_1 \rightarrow V_2$ by defining $g(a) = c$ to start with. Next, where should we send b ? In G_1 , the vertex b is only adjacent to vertex a . There is exactly one vertex like this in G_2 , namely d . So let $g(b) = d$. As for the last two, in this example, we have a free choice: let $g(c) = b$ and $g(d) = a$.

We should check that this really is an isomorphism. It is definitely a bijection. We must make sure that the edges are respected.

The four edges in G_1 are $\{a, b\}$, $\{a, c\}$, $\{a, d\}$, $\{c, d\}$.

Under the proposed isomorphism these become

$\{g(a), g(b)\}, \{g(a), g(c)\}, \{g(a), g(d)\}, \{g(c), g(d)\}$ i.e $\{c, d\}, \{c, b\}, \{c, a\}, \{b, a\}$,

which are precisely the edges in G_2 . Thus g is an isomorphism, so G_1 is isomorphic to G_2

Note: A property preserved by isomorphism of graphs is called a graph invariant.

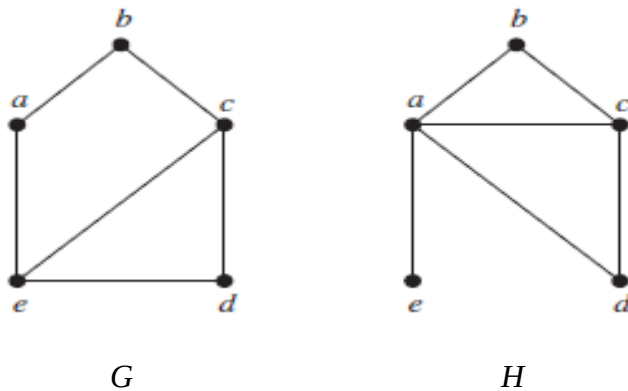
Isomorphic simple graphs must have the same number of vertices, because there is a one-to-one correspondence between the sets of vertices of the graphs.

Isomorphic simple graphs also must have the same number of edges, because the one-to-one correspondence between vertices establishes a one-to-one correspondence between edges. In addition, the degrees of the vertices in isomorphic simple graphs must be the same. That is, a vertex v of degree d in G must correspond to a vertex $f(v)$ of degree d in H , because a vertex w in G is adjacent to v if and only if $f(v)$ and $f(w)$ are adjacent in H . i.e.

The necessary conditions for two graphs to be isomorphic are

- i. both must have the same number of vertices.
- ii. both must have the same number of edges.
- iii. both must have equal number of vertices with the same degree.
- iv. they must have the same degree sequence and same cycle vector $(c_1, \dots, c_n)$, where c_i is the number of cycles of length i .

Example 3: Show that the graphs G and H are not isomorphic.



Solution: Both G and H have five vertices and six edges. However, H has a vertex of degree one, namely, e , whereas G has no vertices of degree one. It follows that G and H are not isomorphic.

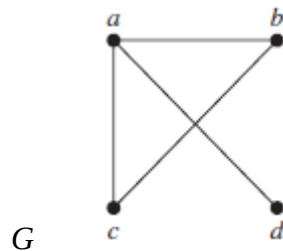
Matrix Representations of Graphs

- i) Adjacency Matrix
- ii) Incidence Matrix

Definition: Suppose that $G = (V, E)$ is a simple graph where $|V| = n$. Suppose that the vertices of G are listed arbitrarily as $v_1, v_2, \dots, v_n$. The adjacency matrix A (or AG) of G , with respect to this listing of the vertices, is the $n \times n$ zero–one matrix with 1 as its (i, j) th entry when v_i and v_j are adjacent, and 0 as its (i, j) th entry when they are not adjacent. In other words, if its adjacency matrix is $A = [a_{ij}]$, then

$$a_{ij} = \begin{cases} 1 & \text{if } \{v_i, v_j\} \text{ is an edge of } G, \\ 0 & \text{otherwise.} \end{cases}$$

Example 4: Find the adjacency matrix of the graph G .



Solution: Using the definition the adjacency matrix of G is

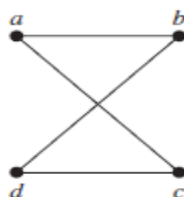
$A(G) =$

$$\begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

Example 5: Draw a graph with the adjacency matrix with respect to the ordering of vertices a, b, c, d .

$$\begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

Solution: The graph with given adjacency matrix is

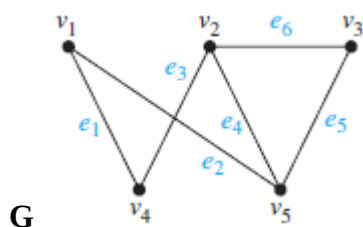


Definition: Let $G = (V, E)$ be an undirected graph. Suppose that $v_1, v_2, \dots, v_n$ are the vertices and $e_1, e_2, \dots, e_m$ are the edges of G . Then the incidence matrix with respect to this ordering of V and E is the $n \times m$

matrix $M = [m_{ij}]$, where

$$m_{ij} = \begin{cases} 1 & \text{when edge } e_j \text{ is incident with } v_i, \\ 0 & \text{otherwise.} \end{cases}$$

Example 6: Represent the graph G shown below with an incidence matrix.

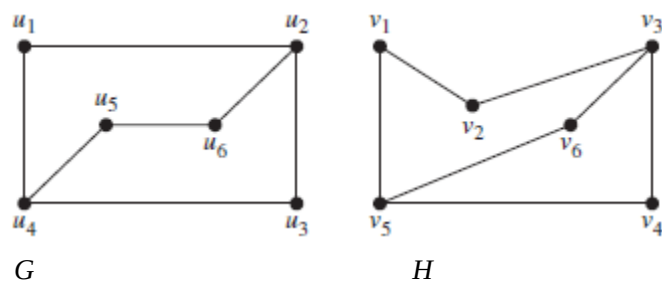


Solution: The incidence matrix of G is

$$\begin{matrix} & e_1 & e_2 & e_3 & e_4 & e_5 & e_6 \\ \begin{matrix} v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \end{matrix} & \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 & 0 \end{bmatrix} \end{matrix}.$$

Note: If the adjacency matrix of G is the same as the adjacency matrix of H , then the two graphs are isomorphic.

Example 7: Determine whether the graphs G and H are isomorphic.



Solution: Both G and H have six vertices and seven edges. Both have four vertices of degree two and two vertices of degree three. It is also easy to see that the subgraphs of G and H consisting of all vertices of degree two and the edges connecting them are isomorphic. Because G and H agree with respect to these invariants,

The adjacency matrix of G is

$$\mathbf{A}_G = \begin{matrix} & \begin{matrix} u_1 & u_2 & u_3 & u_4 & u_5 & u_6 \end{matrix} \\ \begin{matrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \\ u_6 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 \end{bmatrix} \end{matrix},$$

and the adjacency matrix of H is

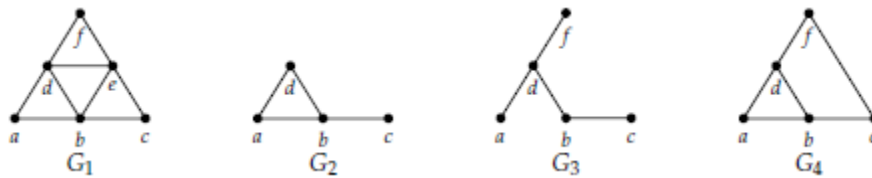
$$\mathbf{A}_H = \begin{matrix} & \begin{matrix} v_6 & v_3 & v_4 & v_5 & v_1 & v_2 \end{matrix} \\ \begin{matrix} v_6 \\ v_3 \\ v_4 \\ v_5 \\ v_1 \\ v_2 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 \end{bmatrix} \end{matrix}.$$

This implies $\mathbf{A}_G = \mathbf{A}_H$, it follows that G and H are isomorphic.

Subgraphs

We say that $G' = (V', E')$ is a subgraph of $G = (V, E)$, and write $G' \subseteq G$, provided $V' \subseteq V$ and $E' \subseteq E$. We say that $G' = (V', E')$ is an induced subgraph of $G = (V, E)$ provided $V' \subseteq V$ and every edge in E whose vertices are still in V' is also an edge in E' .

Example 1: Consider the graphs



Here both G_2 and G_3 are subgraphs of G_1 . But only G_2 is an induced subgraph. Every edge in G_1 that connects vertices in G_2 is also an edge in G_2 . In G_3 , the edge $\{a, b\}$ is in E_1 but not E_3 , even though vertices a and b are in V_3 .

The graph G_4 is not a subgraph of G_1 , even though it looks like all we did is remove vertex e . The reason is that in E_4 we have the edge $\{c, f\}$ but this is not an element of E_1 , so we don't have the required $E_4 \subseteq E_1$.

3.3 Path and Connectivity

Objectives: On completion of subtopic, students will be able to

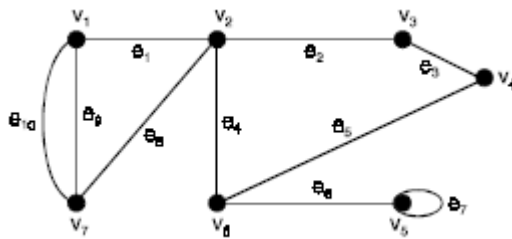
- ✓ Define walk and path on undirected graph.
- ✓ Find walk and path on a graph.
- ✓ Define connectivity of graphs.

Determine whether the given graph is connected or disconnected.

Many problems can be modeled with paths formed by traveling along the edges of graphs. For instance, the problem of determining whether a message can be sent between two computers using intermediate links can be studied with a graph model. Problems of efficiently planning routes for mail delivery, garbage pickup, diagnostics in computer networks, and so on can be solved using models that involve paths in graphs.

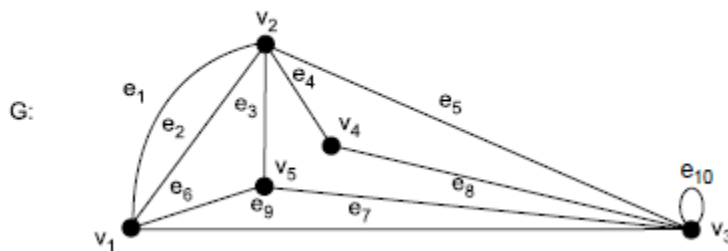
Definition: A walk is defined as a finite alternative sequence of vertices and edges which begins and ends with vertices, such that no edge appears more than once in the sequence.

Example 1: In the graph shown below the sequences $v_2 e_4 v_6 e_5 v_4 e_3 v_3$ is a walk.



Example 2:

Consider the graph G as



Consider the following walks

$$W_1 = v_1 e_1 v_2 e_2 v_1 e_6 v_5 e_7 v_3 e_{10} v_3 e_8 v_4$$

$$W_2 = v_1 e_1 v_2 e_1 v_1 e_1 v_2 e_2 v_1 e_1 v_2$$

$$W_3 = v_3 e_{10} v_3 e_9 v_1 e_1 v_2 e_2 v_1$$

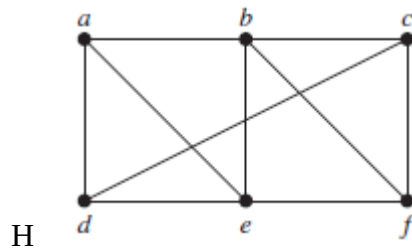
$$W_4 = v_1 e_2 v_2 e_5 v_3 e_7 v_5$$

Definition: Let n be a nonnegative integer and G an undirected graph. A path of length n from u to v in G is a sequence of n edges $e_1, e_2, \dots, e_n$ of G for which there exists a sequence $x_0 = u, x_1, \dots, x_{n-1}, x_n = v$ of vertices such that e_i has, for $i = 1, \dots, n$, the endpoints x_{i-1} and x_i . When the graph is simple, we denote this path by its vertex sequence $x_0, x_1, \dots, x_n$ (because listing these vertices uniquely determines the path).

The path is a circuit if it begins and ends at the same vertex, that is, if $u = v$, and has length greater than zero. The path or circuit is said to pass through the vertices $x_1, x_2, \dots, x_{n-1}$ or traverse the edges $e_1, e_2, \dots, e_n$.

A path or circuit is simple if it does not contain the same edge more than once.

Example 1: In the simple graph H shown below a, d, c, f, e is a simple path of length 4, because $\{a, d\}, \{d, c\}, \{c, f\}$, and $\{f, e\}$ are all edges. However, d, e, c, a is not a path, because $\{e, c\}$ is not an edge. Note that b, c, f, e, b is a circuit of length 4 because $\{b, c\}, \{c, f\}, \{f, e\}$, and $\{e, b\}$ are edges, and this path begins and ends at b . The path a, b, e, d, a, b , which is of length 5, is not simple because it contains the edge $\{a, b\}$ twice.



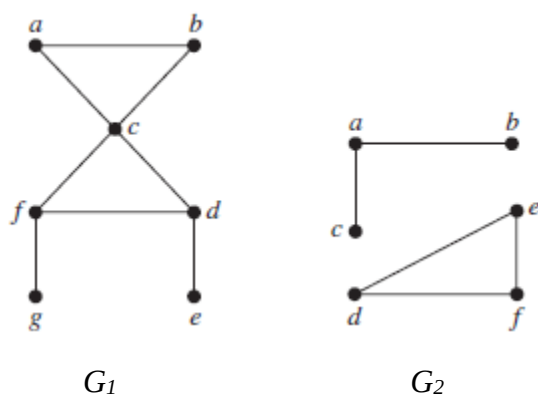
A path or circuit is called simple if it does not contain the same edge more than once.

Connectedness in Undirected Graphs

Definition: An undirected graph is called connected if there is a path between every pair of distinct vertices of the graph. An undirected graph that is not connected is called disconnected. We say that we disconnect a graph when we remove vertices or edges, or both, to produce a disconnected subgraph.

Thus, any two computers in the network can communicate if and only if the graph of this network is connected.

Example 1: The graph G_1 given below is connected, because for every pair of distinct vertices there is a path between them. However, the graph G_2 is not connected. For instance, there is no path in G_2 between vertices a and d .

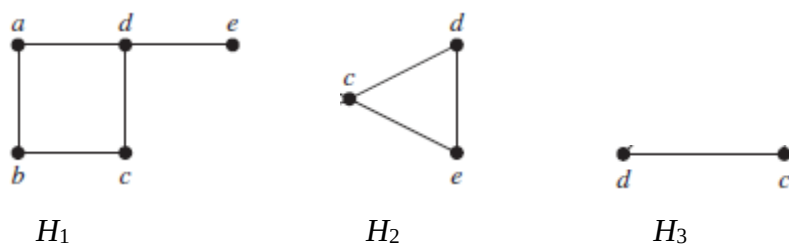


Theorem 2: There is a simple path between every pair of distinct vertices of a connected undirected graph.

A connected component of a graph G is a connected subgraph of G that is not a proper subgraph of another connected subgraph of G . That is, a connected component of a graph G is a maximal connected subgraph of G . A graph G that is not connected has two or more connected components that are disjoint and have G as their union.

Example 2: What are the connected components of the graph H ?

Solution: The graph H is the union of three disjoint connected subgraphs H_1, H_2 , and H_3 . These three subgraphs are the connected components of H given below.



How Connected is a Graph?

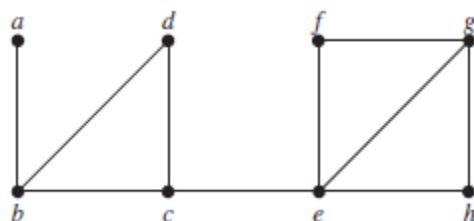
Suppose that a graph represents a computer network. Knowing that this graph is connected tells us that any two computers on the network can communicate. However, we would also like to understand how reliable this network is. For instance, will it still be possible for all computers to communicate after a router or a communications link fails? To answer this and similar questions, we now develop some new concepts.

Sometimes the removal from a graph of a vertex and all incident edges produces a subgraph with more connected components. Such vertices are called cut vertices (or articulation points).

The removal of a cut vertex from a connected graph produces a subgraph that is not connected.

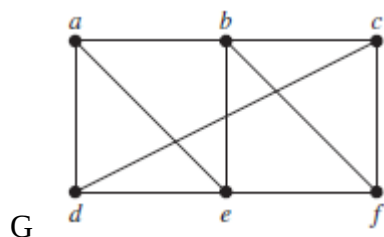
Analogously, an edge whose removal produces a graph with more connected components than in the original graph is called a cut edge or bridge. Note that in a graph representing a computer network, a cut vertex and a cut edge represent an essential router and an essential link that cannot fail for all computers to be able to communicate.

Example 1: Find the cut vertices and cut edges in the graph G .



Solution: The cut vertices of G are b , c , and e . The removal of one of these vertices (and its adjacent edges) disconnects the graph. The cut edges are $\{a, b\}$ and $\{c, e\}$. Removing either one of these edges disconnects G .

A subset U of the vertex set V of $G = (V, E)$ is a vertex cut, or separating set, if $G - U$ is disconnected. For instance, in the following graph G , the set $\{b, c, e\}$ is a vertex cut with three vertices.



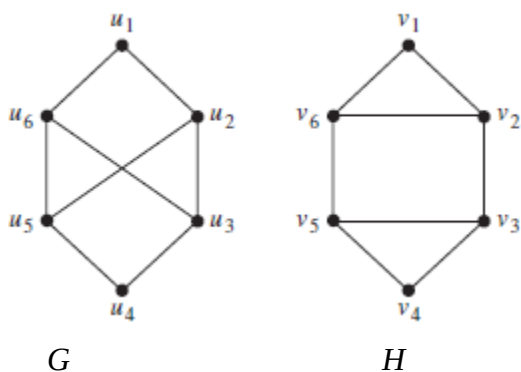
Paths and Isomorphism

There are several ways that paths and circuits can help determine whether two graphs are isomorphic.

For example, the existence of a simple circuit of a particular length is a useful invariant that can be used to show that two graphs are not isomorphic. In addition, paths can be used to construct mappings that may be isomorphism.

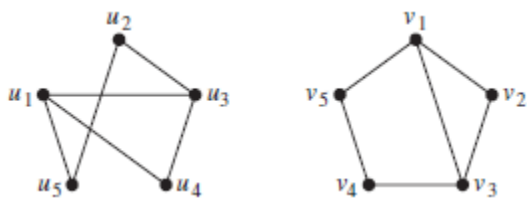
As we mentioned, a useful isomorphic invariant for simple graphs is the existence of a simple circuit of length k , where k is a positive integer greater than 2.

Example 2: Determine whether the graphs G and H shown below are isomorphic.



Solution: Both G and H have six vertices and eight edges. Each has four vertices of degree three, and two vertices of degree two. So, the three invariants number of vertices, number of edges, and degrees of vertices all agree for the two graphs. However, H has a simple circuit of length three, namely, v_1, v_2, v_6, v_1 , whereas G has no simple circuit of length three, as can be determined by inspection (all simple circuits in G have length at least four). Because the existence of a simple circuit of length three is an isomorphic invariant, G and H are not isomorphic.

Example 3: Determine whether the graphs G_1 and G_2 are isomorphic.



Solution: Both G and H have five vertices and six edges, both have two vertices of degree three and three vertices of degree two, and both have a simple circuit of length three, a simple circuit of length four, and a simple circuit of length five. Because all these isomorphic invariants agree, G and H may be isomorphic.

To find a possible isomorphism, we can follow paths that go through all vertices so that the corresponding vertices in the two graphs have the same degree. For example, the paths u_1, u_4, u_3, u_2, u_5 in G and v_3, v_2, v_1, v_5, v_4 in H both go through every vertex in the graph; start at a vertex of degree three; go through vertices of degrees two, three, and two, respectively; and end at a vertex of degree two. By following these paths through the graphs, we define the mapping f with $f(u_1) =$

$v_3, f(u_4) = v_2, f(u_3) = v_1, f(u_2) = v_5, \text{ and } f(u_5) = v_4$. This show that f is an isomorphism, so G and H are isomorphic.

Check list

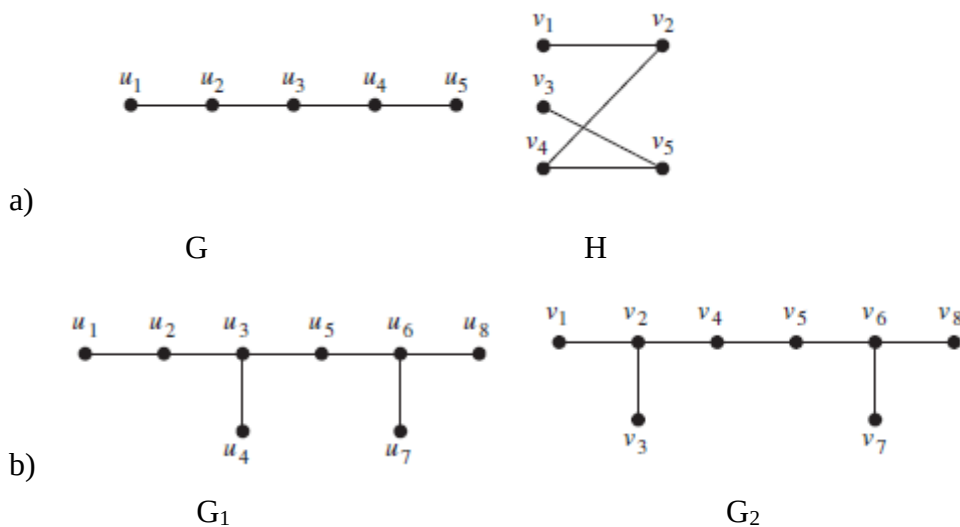
To check your mastery the topic put (✓) if you can perform it.; otherwise put (X) in front of each statement.

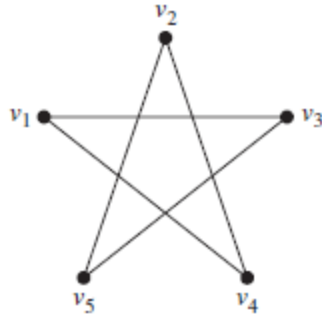
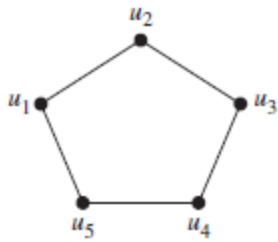
Important points	I Can	I Can't
1. Define isomorphism of two graphs. 2. Determine whether the given two graphs are isomorphic or not. 3. List the invariant properties of isomorphic graphs. 4. Define incidence matrix and adjacency matrix of a graph. 5. Find the incidence matrix and adjacency matrix of a graph. 6. Define walk and path on undirected graph. 7. Find walk and path on a graph. 8. Define connectivity of graphs. 9. Determine whether the given graph is connected or disconnected.		

Is your response positive for the above questions? If your answer is “I Can”, go to the next section. If your answer is “I Can’t”, go back and read once again.

Exercise 3.2

1. Define isomorphism of two graphs
2. Determine whether the given pair of graphs isomorphic.

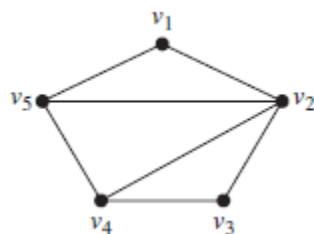
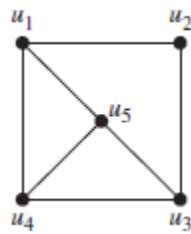




c)

G

H



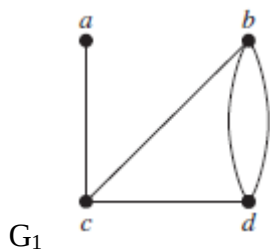
d)

G₁

G₂

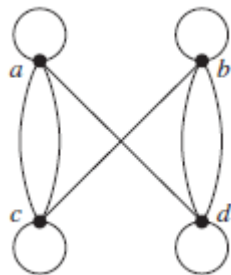
3. Represent the given graphs using

- Adjacency matrix.
- Incidence matrix



G₁

G₂



G₃

c) Draw an undirected graph represented by the given adjacency matrix.

$$A_1 = \begin{bmatrix} 1 & 3 & 2 \\ 3 & 0 & 4 \\ 2 & 4 & 0 \end{bmatrix} \quad A_2 = \begin{bmatrix} 1 & 2 & 0 & 1 \\ 2 & 0 & 3 & 0 \\ 0 & 3 & 1 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}$$

$$A_3 = \begin{bmatrix} 0 & 1 & 3 & 0 & 4 \\ 1 & 2 & 1 & 3 & 0 \\ 3 & 1 & 1 & 0 & 1 \\ 0 & 3 & 0 & 0 & 2 \\ 4 & 0 & 1 & 2 & 3 \end{bmatrix}$$

4. Are the simple graphs with the following adjacency matrices isomorphic?

a) $\begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}$

b) $\begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}$

5. Describe the row and column of an adjacency matrix of a graph corresponding to an isolated vertex.

6. Describe the row of an incidence matrix of a graph corresponding to an isolated vertex.

7. Does each of these lists of vertices form a path in the following graph? Which are circuits?

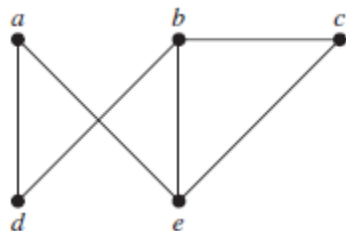
What are the lengths of those that are paths?

a) a, e, b, c, b

b) a, e, a, d, b, c, a

c) e, b, a, d, b, e

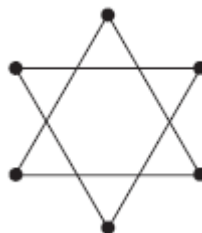
d) c, b, d, a, e, c

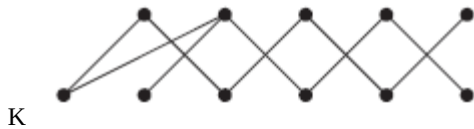


8. Determine whether the given graph is connected or disconnected.

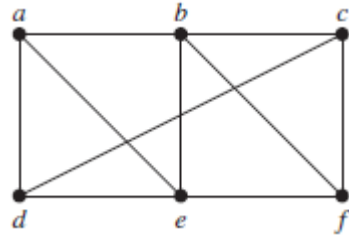


H

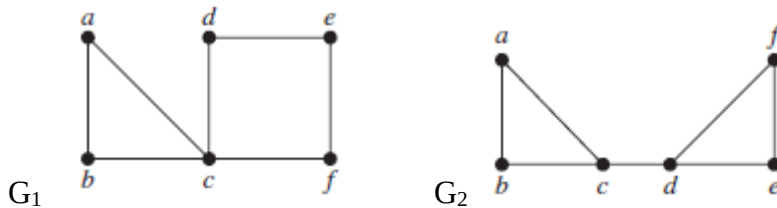




9. Find the number of paths between c and d in the graph G of length
 a) 2 b) 3 c) 4 d) 5 e) 6 f) 7



10. Find all the cut vertices of the following graphs.



11. Find all the cut edges in the graphs G_1 and G_2 on Q no. 10 above.
 12. Find $\kappa(Km,n)$ and $\lambda(Km,n)$, where m and n are positive integers.
 13. Draw a graph G with $\kappa(G) = 1$, $\lambda(G) = 2$, and $\min_{v \in V} \deg(v) = 3$.

3.4 Eulerian and Hamiltonian Graph

Objectives: On completion of subtopic, students will be able to

- ✓ Define Eulerian graphs and Hamiltonian graphs.
- ✓ Determine whether a graph is Eulerian or not.
- ✓ Determine whether a graph is Hamiltonian or not.

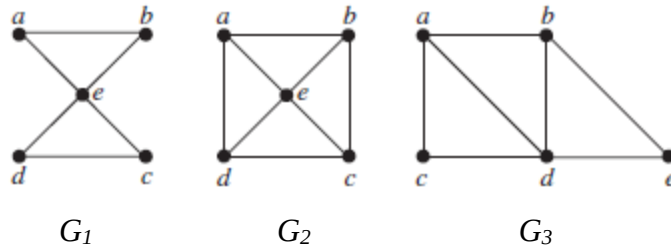
Euler circuit and Euler path

Definition: An Euler circuit in a graph G is a simple circuit containing every edge of G . An Euler path in G is a simple path containing every edge of G .

Definition: A graph G is said to be Eulerian if it has an Eulerian circuit.

The following examples illustrate the concept of Euler circuits and paths.

Example: Which of the undirected graphs given below are Eulerian graph?



Solution: The graph G_1 has an Euler circuit, for example, a, e, c, d, e, b, a . Neither of the graphs G_2 or G_3 has an Euler circuit.

However, G_3 has an Euler path a, c, d, e, b, d, a, b . G_2 does not have an Euler path.

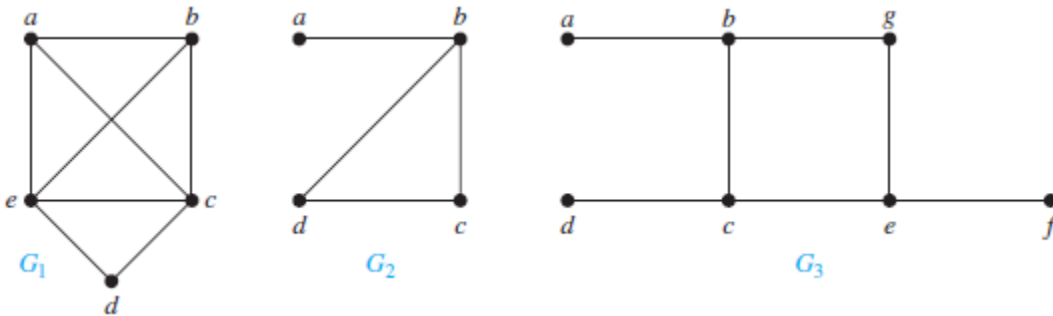
Therefore, G_1 is Eulerian graph.

Hamilton Paths and Circuits

Definition: A simple path in a graph G that passes through every vertex exactly once is called a Hamilton path, and a simple circuit in a graph G that passes through every vertex exactly once is called a Hamilton circuit.

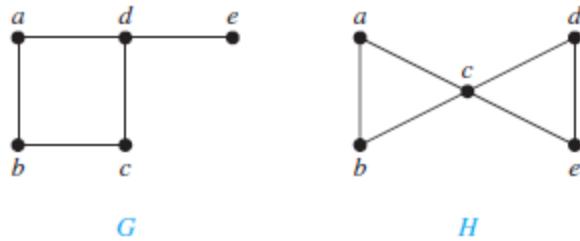
Definition: A graph is said to be a Hamiltonian graph if it contains Hamiltonian cycle i.e. A graph G is Hamiltonian if there exist a cycle containing every vertex of G .

Example 1: Which of the simple graphs are Hamilton graph?



Solution: G_1 has a Hamilton circuit: a, b, c, d, e, a . This implies G_1 is Hamiltonian graph. There is no Hamilton circuit in G_2 (this can be seen by noting that any circuit containing every vertex must contain the edge $\{a, b\}$ twice), but G_2 does have a Hamilton path, namely, a, b, c, d . G_3 has neither a Hamilton circuit nor a Hamilton path, because any path containing all vertices must contain one of the edges $\{a, b\}, \{e, f\}$, and $\{c, d\}$ more than once.

Example 2: Show that neither graph displayed below has a Hamilton circuit.



Solution: There is no Hamilton circuit in G because G has a vertex of degree one, namely, e .

Now consider H . Because the degrees of the vertices a , b , d , and e are all two, every edge incident with these vertices must be part of any Hamilton circuit. It is now easy to see that no Hamilton circuit can exist in H , for any Hamilton circuit would have to contain four edges incident with c , which is impossible.

Example 3: Show that K_5 has a Hamilton circuit whenever $n \geq 3$.

Solution: We can form a Hamilton circuit in K_5 beginning at any vertex. Such a circuit can be built by visiting vertices in any order we choose, as long as the path begins and ends at the same vertex and visits each other vertex exactly once. This is possible because there are edges in K_5 between any two vertices.

3.5 Planar Graph

Objectives: At the end of this lesson, students will be able to

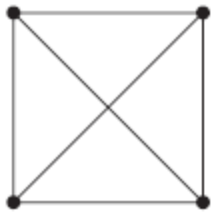
- ✓ Define planar graph.
- ✓ Determine whether a graph is planar or nonplanar.
- ✓ State Euler's formula.
- ✓ Verify Euler's formula for a connected planar graph

Definition: A graph is called planar if it can be drawn in the plane without any edges crossing, where a crossing of edges is the intersection of the lines or arcs representing them at a point other than their common endpoint. Such a drawing is called a planar representation of the graph.

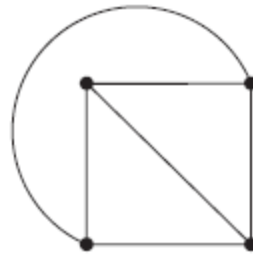
A graph may be planar even if it is usually drawn with crossings, because it may be possible to draw it in a different way without crossings.

Example 1: Is K_4 planar?

Solution: K_4 is planar because it can be drawn without crossings as shown below.



K_4 with edge crossing



K_4 without edge crossing (planar representation)

Applications of planar graphs

Planarity of graphs plays an important role in the design of electronic circuits. We can model a circuit with a graph by representing components of the circuit by vertices and connections between them by edges. We can print a circuit on a single board with no connections crossing if the graph representing the circuit is planar. When this graph is not planar, we must turn to more expensive options. For example, we can partition the vertices in the graph representing the circuit into planar subgraphs. We then construct the circuit using multiple layers. We can construct the circuit using insulated wires whenever connections cross. In this case, drawing the graph with the fewest possible crossings is important.

The planarity of graphs is also useful in the design of road networks. Suppose we want to connect a group of cities by roads. We can model a road network connecting these cities using a simple graph with vertices representing the cities and edges representing the highways connecting them. We can build this road network without using underpasses or overpasses if the resulting graph is planar.

Euler's Formula

A planar representation of a graph splits the plane into regions, including an unbounded region. Euler showed that all planar representations of a graph split the plane into the same number of regions. He accomplished this by finding a relationship among the number of regions, the number of vertices, and the number of edges of a planar graph.

Theorem 3: Euler's formula

Let G be a connected planar simple graph with e edges and v vertices. Let r be the number of regions in a planar representation of G . Then $r = e - v + 2$.

Example 1: Suppose that a connected planar simple graph has 20 vertices, each of degree 3. Into how many regions does a representation of this planar graph split the plane?

Solution: This graph has 20 vertices, each of degree 3, so $v = 20$. Because the sum of the degrees of the vertices, $3v = 3 \cdot 20 = 60$, is equal to twice the number of edges($2e$)

we have $2e = 60$, This implies $e = 30$.

Consequently, from Euler's formula, the number of regions is

$$r = e - v + 2 = 30 - 20 + 2 = 12.$$

Corollary 2: If G is a connected planar simple graph with e edges and v vertices, where $v \geq 3$, then $e \leq 3v - 6$.

Example 1: Show that K_5 is nonplanar using the above corollary.

Solution: The graph K_5 has five vertices and 10 edges. However, the inequality $e \leq 3v - 6$ is not satisfied for this graph because $e = 10$ and $3v - 6 = 3(5) - 6 = 9$. Implies $10 > 9$. Therefore, K_5 is not planar.

Theorem 4: Kuratowski's Theorem

A graph is nonplanar if and only if it contains a subgraph which is a subdivision of $K_{3,3}$ or K_5 .

3.6 Trees and forest

Objectives: At the end of this lesson, students will be able to

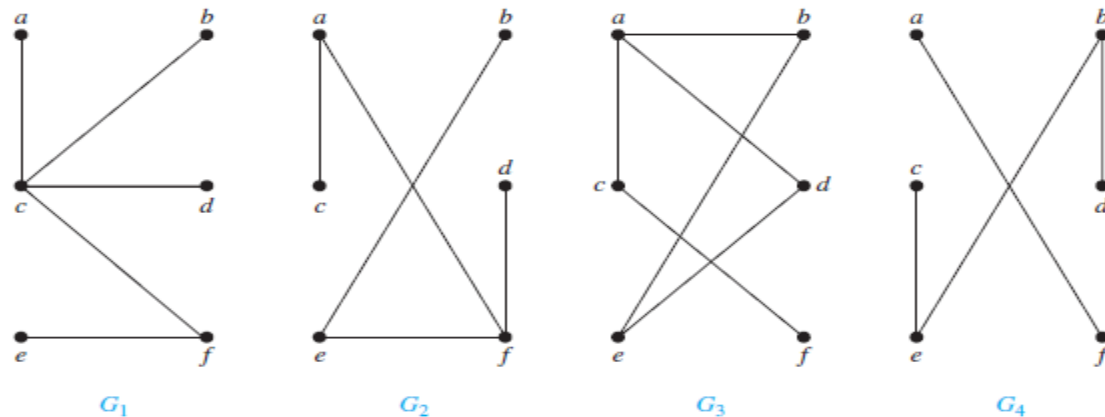
- ✓ Define tree graph.
- ✓ Determine whether a graph is a tree or not.
- ✓ Define rooted tree.

Trees are particularly useful in computer science, where they are employed in a wide range of algorithms. For instance, trees are used to construct efficient algorithms for locating items in a list. They can be used in algorithms, such as Huffman coding, that construct efficient codes saving costs in data transmission and storage. Trees can be used to study games such as checkers and chess and can help determine winning strategies for playing these games. Trees can be used to model procedures carried out using a sequence of decisions. Constructing these models can help determine the computational complexity of algorithms based on a sequence of decisions, such as sorting algorithms.

We can assign weights to the edges of a tree to model many problems. For example, using weighted trees we can develop algorithms to construct networks containing the least expensive set of telephone lines linking different network nodes.

Definition: A tree is a simple connected undirected graph with no simple circuits.

Example 1: Which of the graphs shown below are trees?



Solution: G_1 and G_2 are trees, because both are connected graphs with no simple circuits. G_3 is not a tree because e, b, a, d, e is a simple circuit in this graph. Finally, G_4 is not a tree because it is not connected.

Trees are often defined as undirected graphs with the property that there is a unique simple path between every pair of vertices.

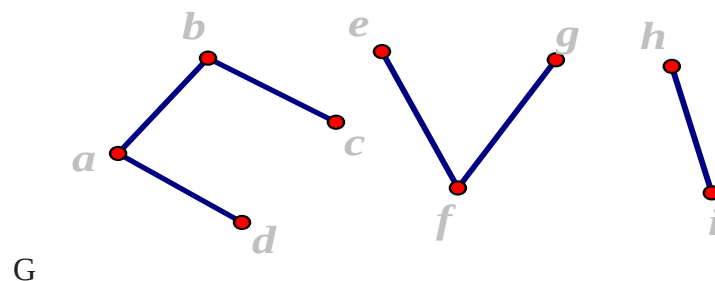
THEOREM 5: An undirected graph is a tree if and only if there is a unique simple path between any two of its vertices.

Theorem 6: A tree T with n vertices has exactly $(n - 1)$ edges.

Example: A graph G_1 on example1 above is tree with $n = 6$ vertices and $6 - 1 = 5$ edges.

Definition: A forest is a graph that contains no cycles.

Example: The graph G given below is forest with 9 vertices and 6 edges.



Note: Each components of a forest is a tree.

Definition: If G is connected graph and u and v are any two vertices of G , the length of the shortest path between u and v is called the distance between u and v and is denoted by $d(u, v)$. The distance function on defined above has the following properties. If u, v and w are any three vertices of a connected graph then.

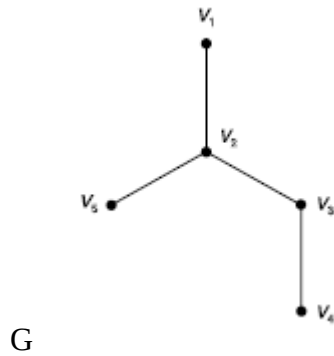
- i. $d(u, v) \geq 0$ and $d(u, v) = 0$ iff $u = v$
- ii. $d(u, v) = d(v, u)$
- iii. $d(u, v) \leq d(u, w) + d(w, v)$

Definition: Let G be a connected graph. For any vertex v on G , the eccentricity of v denoted by $e(v)$ is

$$e(v) = \max \{d(u, v) : u, v \in V\}$$

Note: $e(v)$ is the length of the longest path in G starting from the vertex v .

Example: In the graph G below, $e(v_1) = 3$.



Definition: The diameter of a connected r graph G is defined as the maximum eccentricity among all vertices of the graph G . It is denoted by d .

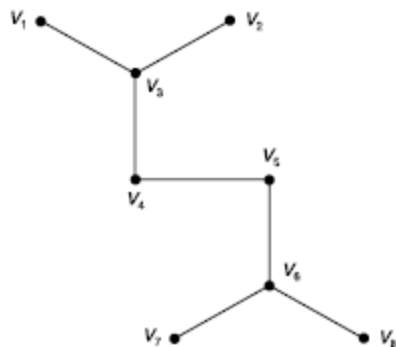
Hence, $d = \text{diameter of } G = \max \{e(v) : v \in V\}$

Definition: The radius of a connected graph G is defined as the minimum eccentricity among all vertices of the graph. It is denoted by r .

Thus $r = \text{radius of } G = \min \{e(v) : v \in V\}$

Note: The radius of connected graph may not be half of its diameter.

Example: Consider the tree T below.



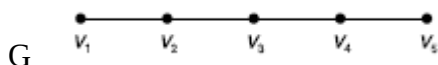
Solution: $e(v_1) = 5, e(v_2) = 5, e(v_3) = 4,$
 $e(v_4) = 3, e(v_5) = 3, e(v_6) = 4,$
 $e(v_7) = e(v_8) = 5$

The radius of $T = r = 3$

Definition: The center of connected graph G is defined as the set of vertices having minimum eccentricity among all vertices of the graph. It is denoted by C or $C(G)$.

$$C = C(G) = \text{center of } G = \{v \in V : e(v) = r\}$$

Example 1: Consider the graph G below.



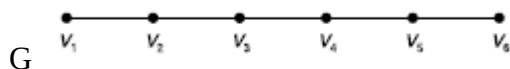
Solution: $e(v_1) = 4, e(v_2) = 3, e(v_3) = 2,$
 $e(v_4) = 3, e(v_5) = 4, \text{ radius of } G = i = 2.$

Hence center of $G = \{v_3\}$.

Note:

- i. The distance between two adjacent vertices of a connected graph G is 1.
- ii. The maximum distance from each vertex of G occurs at a pendant vertices of G .
- iii. If $C(G) = V(G)$ then G is called self-centred graph.
- iv. If P is a path of even length the P has only one vertex at the centre.
- v. If P is a path of odd length this centre of P contains two adjacent vertices.

Example 2: In the graph G below.



Solution: Centre of $G = \{v_3, v_4\}$

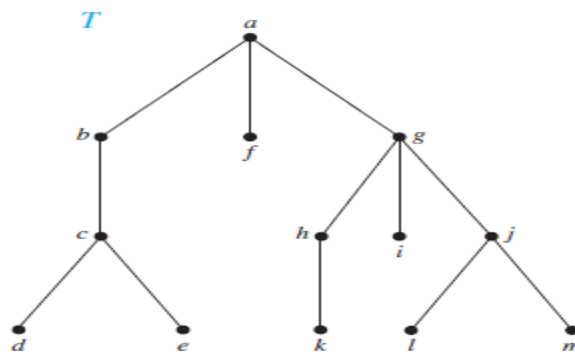
Rooted Trees

Definition: A rooted tree is a tree in which one vertex has been designated as the root and every edge is directed away from the root.

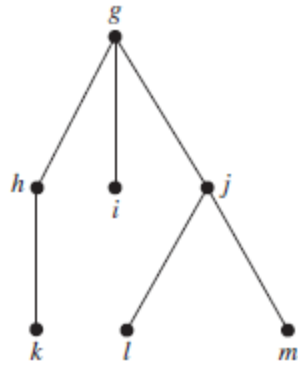
The terminology for trees has botanical and genealogical origins. Suppose that T is a rooted tree. If v is a vertex in T other than the root, the parent of v is the unique vertex u such that there is a directed edge from u to v (the reader should show that such a vertex is unique). When u is the parent of v , v is called a child of u . Vertices with the same parent are called siblings. The ancestors of a vertex other than the root are the vertices in the path from the root to this vertex, excluding the vertex itself and including the root (that is, its parent, its parent's parent, and so on, until the root is reached). The descendants of a vertex v are those vertices that have v as an ancestor. A vertex of a rooted tree is called a leaf if it has no children. Vertices that have children are called internal vertices. The root is an internal vertex unless it is the only vertex in the graph, in which case it is a leaf.

If a is a vertex in a tree, the subtree with a as its root is the subgraph of the tree consisting of a and its descendants and all edges incident to these descendants.

Example 1: In the rooted tree T (with root a) shown below, find the parent of c , the children of g , the siblings of h , all ancestors of e , all descendants of b , all internal vertices, and all leaves. What is the subtree rooted at g ?



Solution: The parent of c is b . The children of g are h , i , and j . The siblings of h are i and j . The ancestors of e are c , b , and a . The descendants of b are c , d , and e . The internal vertices are a , b , c , g , h , and j . The leaves are d , e , f , i , k , l , and m . The subtree rooted at g is shown below

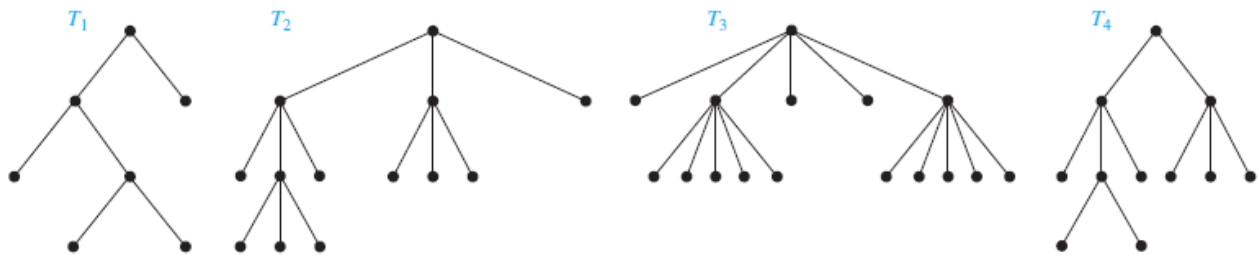


Note: Rooted trees with the property that all of their internal vertices have the same number of children are used in many different applications, like searching, sorting, and coding.

Definition: A rooted tree is called an m -ary tree if every internal vertex has no more than m children.

The tree is called a full m -ary tree if every internal vertex has exactly m children. An m -ary tree with $m = 2$ is called a binary tree.

Example 1: Are the rooted trees given below full m -ary trees for some positive integer m ?



Solution: T_1 is a full binary tree because each of its internal vertices has two children. T_2 is a full 3-ary tree because each of its internal vertices has three children. In T_3 each internal vertex has five children, so T_3 is a full 5-ary tree. T_4 is not a full m -ary tree for any m because some of its internal vertices have two children and others have three children.

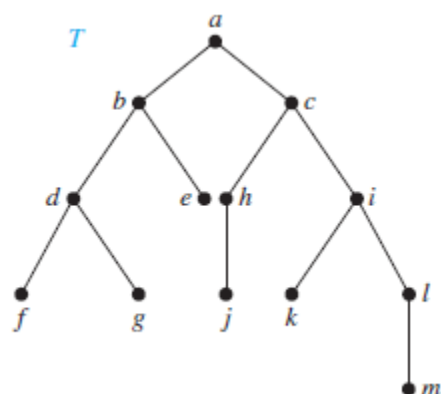
Ordered rooted trees

Definition: An ordered rooted tree is a rooted tree where the children of each internal vertex are ordered. Ordered rooted trees are drawn so that the children of each internal vertex are shown in order from left to right. Note that a representation of a rooted tree in the conventional way determines an ordering for its edges. We will use such orderings of edges in drawings without explicitly mentioning that we are considering a rooted tree to be ordered.

In an ordered binary tree (usually called just a binary tree), if an internal vertex has two children, the first child is called the left child and the second child is called the right child.

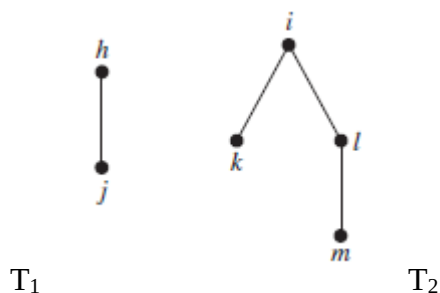
The tree rooted at the left child of a vertex is called the left subtree of this vertex, and the tree rooted at the right child of a vertex is called the right subtree of the vertex. The reader should note that for some applications every vertex of a binary tree, other than the root, is designated as a right or a left child of its parent. This is done even when some vertices have only one child.

Example 1: What are the left and right children of d in the binary tree T shown below (where the order is that implied by the drawing)? What are the left and right subtrees of c ?



Solution: The left child of d is f and the right child is g .

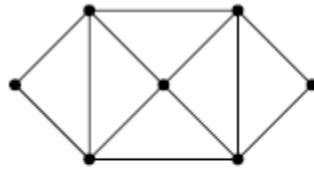
The left and right subtrees T_1 and T_2 respectively are shown as follows



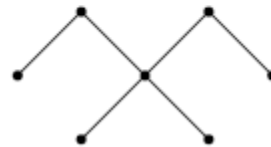
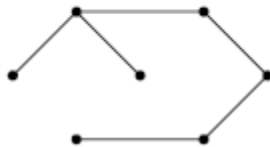
Definition: Let G be a simple graph. A spanning tree of G is a subgraph of G that is a tree containing every vertex of G .

A simple graph with a spanning tree must be connected, because there is a path in the spanning tree between any two vertices. The converse is also true; that is, every connected simple graph has a spanning tree.

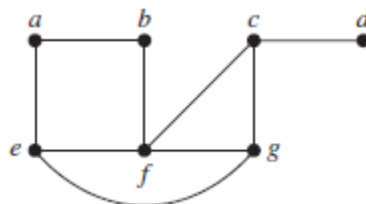
Example 1: Find two different spanning trees of the graph



Solution. Here are two spanning trees.

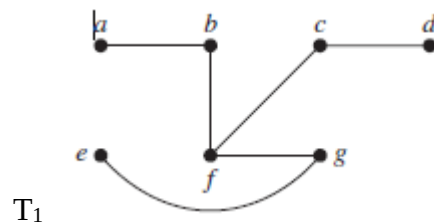


Example 2: Find a spanning tree of the simple graph G .

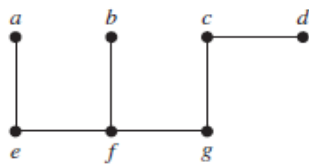


G

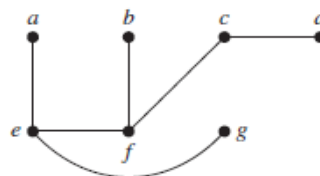
Solution: The graph G is connected, but it is not a tree because it contains simple circuits. Remove the edge $\{a, e\}$. This eliminates one simple circuit, and the resulting subgraph is still connected and still contains every vertex of G . Next remove the edge $\{e, f\}$ to eliminate a second simple circuit. Finally, remove edge $\{c, g\}$ to produce a simple graph with no simple circuits. This subgraph is a spanning tree, because it is a tree that contains every vertex of G .



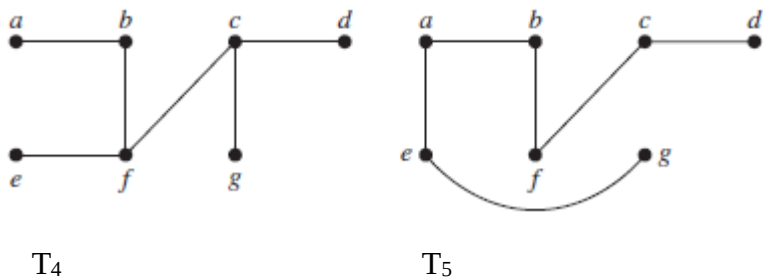
T_1



T_2



T_3



T_1, T_2, T_3, T_4 and T_5 Spanning trees of G

Theorem 7: A simple graph is connected if and only if it has a spanning tree.

3.7 Graph Coloring

Objectives: At the end of this sub topic, students will be able to

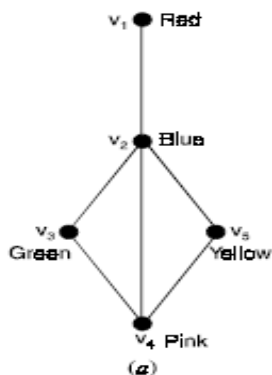
- ✓ Define coloring of graph.
- ✓ Find chromatic number of a graph.

Definition: Properly coloring of a graph

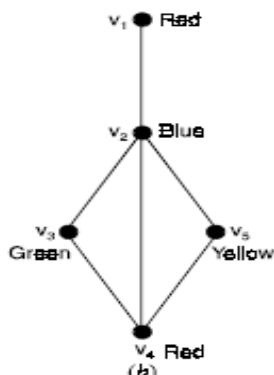
Coloring all the vertices of a graph with colors such that no two adjacent vertices have the same color is called the proper coloring (or simply coloring) of a graph.

The k -colorings of the graph G is a coloring of graph G using k -colors. If the graph G has k -coloring, then the graph G is said to be k -colorable.

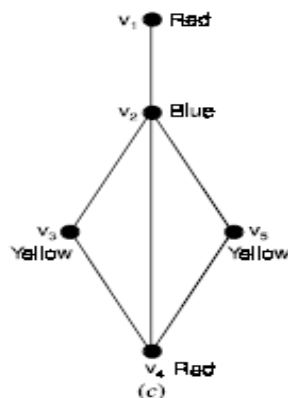
Example: The following are three different coloring of the graph G .



5- Coloring of G



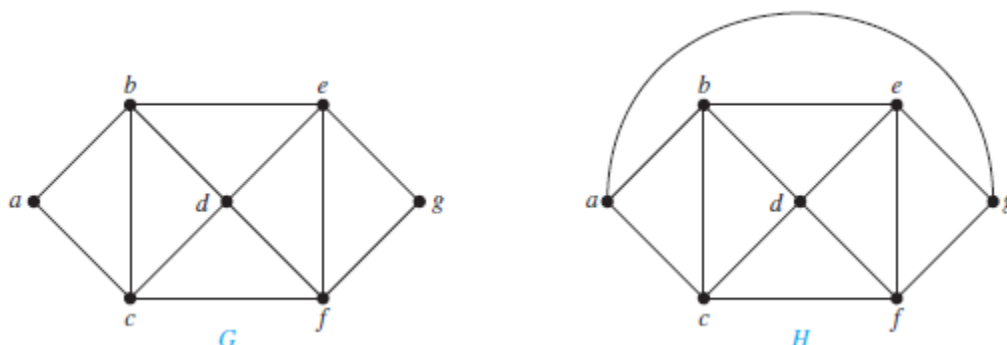
4- coloring of G



3-coloring of G

Definition: The chromatic number of a graph is the least number of colors needed for a coloring of this graph. The chromatic number of a graph G is denoted by $\chi(G)$. (Here χ is the Greek letter *chi*.)

Example 1: What are the chromatic numbers of the graphs G and H given below?

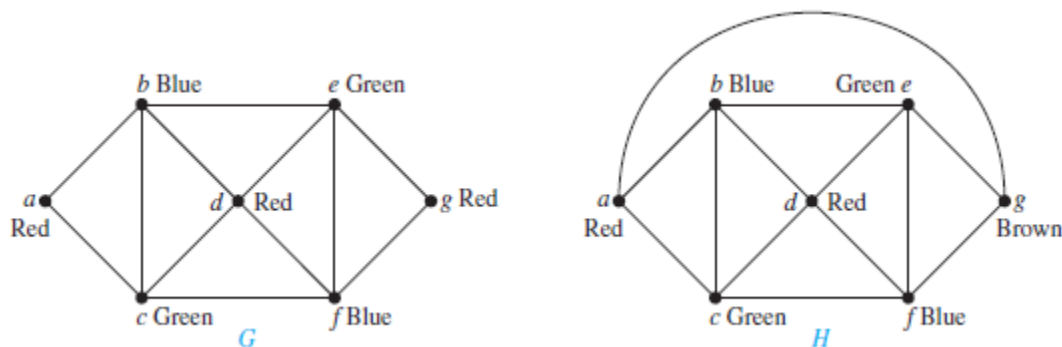


Solution: The chromatic number of G is at least three, because the vertices a , b , and c must be assigned different colors. To see if G can be colored with three colors, assign red to a , blue to b , and green to c . Then, d can (and must) be colored red because it is adjacent to b and c .

Furthermore, e can (and must) be colored green because it is adjacent only to vertices colored red and blue, and f can (and must) be colored blue because it is adjacent only to vertices colored red and green. Finally, g can (and must) be colored red because it is adjacent only to vertices colored blue and green. This produces a coloring of G using exactly three colors. i.e $\chi(G) = 3$.

The graph H is made up of the graph G with an edge connecting a and g . Any attempt to color H using three colors must follow the same reasoning as that used to color G , except at the last stage, when all vertices other than g have been colored. Then, because g is adjacent (in H) to vertices colored red, blue, and green, a fourth color, say brown, needs to be used. Hence, H has a chromatic number equal to 4. i.e $\chi(G) = 4$.

The coloring of G and H using 3 and 4 colors respectively are given below



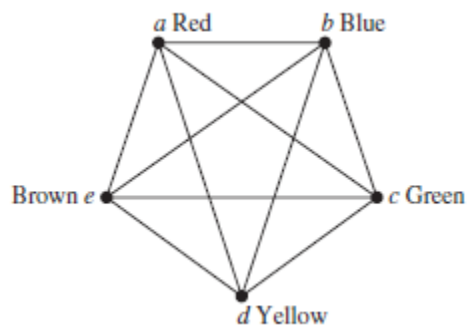
Example 2: What is the chromatic number of K_n ?

Solution: A coloring of K_n can be constructed using n colors by assigning a different color to each vertex. Is there a coloring using fewer colors? The answer is no. No two vertices can be assigned the same color, because every two vertices of this graph are adjacent. Hence, the chromatic number of K_n is n . That is, $\chi(K_n) = n$.

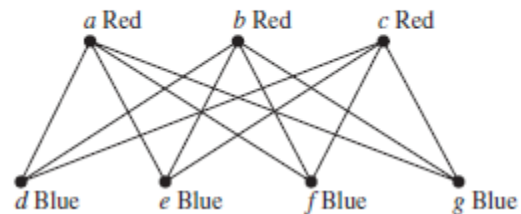
Example 3: What is the chromatic number of the complete bipartite graph $K_{m,n}$, where m and n are positive integers?

Solution: The number of colors needed may seem to depend on m and n . Only two colors are needed, because $K_{m,n}$ is a bipartite graph. Hence, $\chi(K_{m,n}) = 2$. This means that we can color the set of m vertices with one color and the set of n vertices with a second color. Because edges connect only a vertex from the set of m vertices and a vertex from the set of n vertices, no two adjacent vertices have the same color.

The coloring of K_5 and $K_{3,4}$ using 5 and 2 colors respectively are given below.



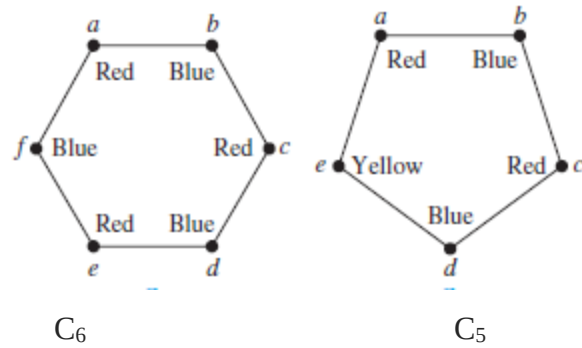
5 coloring of K_5



2 coloring of $K_{3,4}$

Example 4: What is the chromatic number of the graph C_n , where $n \geq 3$?

Solution: We will first consider some individual cases. To begin, let $n = 6$. Pick a vertex and color it red. It is necessary to assign a second color, say blue, to the next vertex reached. Continue in the clockwise direction; the third vertex can be colored red, the fourth vertex blue, and the fifth vertex red. Finally, the sixth vertex, which is adjacent to the first, can be colored blue. Hence, $\chi(C_6) = 2$.



2 coloring and 3 coloring of C_6 and C_5

THEOREM: The four color theorem

The chromatic number of a planar graph is not greater than four.

Note:

- $\chi(G) \leq |V|$, where $|V|$ is the number of vertices of G .
- If some subgraph of G requires k colors then $\chi(G) \geq k$.
- If $\deg(v) = d$, then at most d colors are required to color the vertices adjacent to v .

3.8 Directed Graph

Objectives: At the end of this sub topic, students will be able to

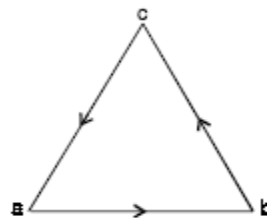
- ✓ Define directed graph.
- ✓ Find adjacency matrix and incidence matrix of a directed graph.

A directed graph or digraph D consists of a set V of vertices and a set E of edges such that $e \in E$ is associated with an ordered pair of vertices.

In other words, if each edge of the graph D has a direction then the graph is called directed graph.

In the diagram of directed graph, each edge $e = (u, v)$ is represented by an arrow or directed curve from initial point u of e to the terminal point v .

The following graph is an example of a directed graph.



Directed graph D

Suppose $e = (u, v)$ is a directed edge in a digraph, then

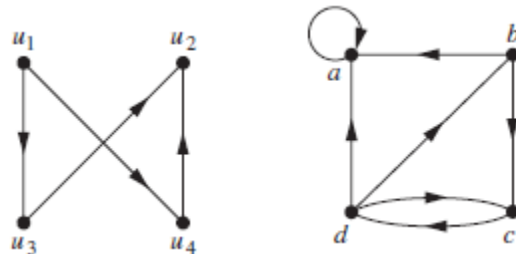
- (i) u is called the initial vertex of e and v is the terminal vertex of e
- (ii) e is said to be incident from u and to be incident to v .
- (iii) u is adjacent to v , and v is adjacent from u .

Definition: A directed graph D that contain a directed self-loop an directed parallel edges is said to be a directed multi graph.

Note: The initial vertex and terminal vertex of loop are the same.

Definition: In a directed graph D , if there is no more than one directed edge in a particular direction between a pair of vertices, then it is called simple directed graph.

Note: A directed graph which is not simple is called directed multi graph.



D_1

D_2

D_1 is simple directed graph

D_2 is directed multi graph

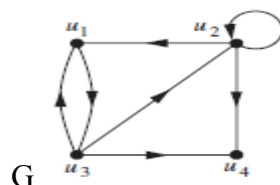
Degree of a vertex on directed graph

Definition: Let D be a directed graph the out degree of a vertex v of D is the number of edge beginning at v , it is denoted by $deg^+(v)$.

Definition: Let D be a directed graph the in degree of vertex v of D is the number of edge $deg^-(v)$.

Note: a directed loop contributes one for out-degree and one for in degree.

Example: Consider the directed graph G .



$$\deg^+(u_1)=1 \quad \deg^-(u_1)=2 \quad \deg^+(u_3)=3 \quad \deg^-(u_3)=1$$

$$\deg^+(u_2)=3 \quad \deg^-(u_2)=2 \quad \deg^+(u_4)=0 \quad \deg^-(u_4)=2$$

Theorem 7: Let $G=(V,E)$ be a graph with directed edges then the sum of the in- degrees and the sum of out- degrees of all vertices in a graph are equal. Both of these sums are equal to number of edges in the graph.

That is, $\sum \deg^-(v) = \sum \deg^+(v) = |E|$

In the above example, $\sum \deg^+(v) = \sum \deg^-(v)$

$$\Rightarrow \deg^+(u_1) + \deg^+(u_2) + \deg^+(u_3) + \deg^+(u_4) = \deg^-(u_1) + \deg^-(u_2) + \deg^-(u_3) + \deg^-(u_4)$$

$$1+3+3+0=2+2+1+2$$

$$\Rightarrow 7 = 7 = |E|$$

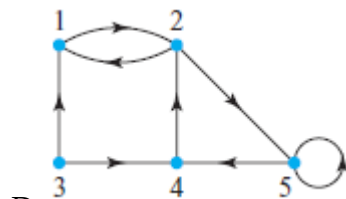
Definition: A vertex with zero in- degree is called a source where as a vertex without zero out- degree is called a sink.

Matrix Representation of Directed Graph

Adjacency Matrix Representation of Directed Graph

Definition: The adjacency matrix for a directed graph G on the ordered list of vertices $v_1, v_2, \dots, v_n$ is the $n \times n$ matrix $A = [a_{i,j}]$ such that $a_{i,j}$ is the number of edges from v_i to v_j .

Example 1: Find the adjacency matrix of the directed graph D.



Solution: D

we take the ordering of the vertices to be 1, 2, 3, 4, 5, then the adjacency matrix obtained is

$$A =$$

$$\begin{matrix} & 1 & 2 & 3 & 4 & 5 \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix} \end{matrix}.$$

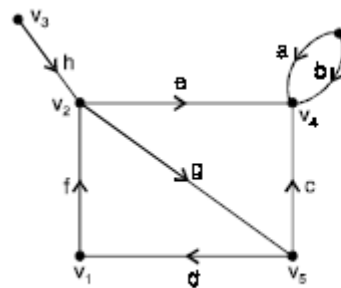
Remark:- The adjacency matrices can also be used to represent directed multi graphs. Again such matrices are not zero one matrices. When there are multiple edges in the same direction connecting two vertices in the adjacency matrix for a directed multi graph a_{ij} equals the number of edges that are associated to (v_i, v_j) .

Incidence Matrix Representation of Directed Graph

Definition: The incidence matrix $B = \{b_{ij}\}$ of a di graph D with n vertices and m edges is the $n \times m$ matrix in which

$$b_{ij} = \begin{cases} 1, & \text{if arc } j \text{ is directed away from a vertex } v_i \\ -1, & \text{if arc } j \text{ is directed towards vertex } v_i \\ 0, & \text{otherwise} \end{cases}$$

Example: A directed graph G and its incidence matrix are shown



G

$$\begin{matrix} & a & b & c & d & e & f & g & h \\ \begin{matrix} v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \\ v_6 \end{matrix} & \begin{bmatrix} 0 & 0 & 0 & -1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & -1 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ -1 & -1 & -1 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & -1 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \end{matrix}$$

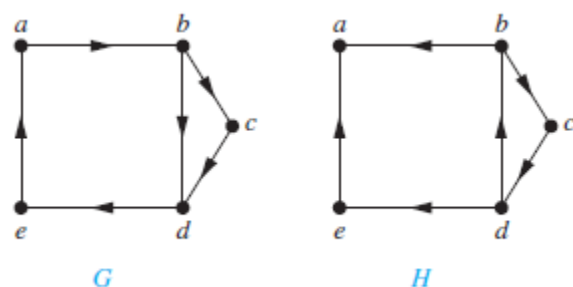
Connectivity of Directed Graph

Definition: A directed graph is strongly connected if there is a path from a to b and from b to a whenever a and b are vertices in the graph. In other word, a directed graph to be strongly connected there must be a sequence of directed edges from any vertex in the graph to any other vertex.

Definition: A directed graph is weakly connected if there is a path between any two vertices in the underlying undirected graph. That is a directed graph is weakly connected if and only if there is always a path between two vertices when the direction of the edges are disregarded.

Remark: Any strongly connected directed graph is also weakly connected.

Example:



Solution: G is strongly connected because there is a path between any two vertices in this directed graph and G is also can be weakly connected. H is not strongly connected because there is no directed path from a to b in this graph , hence H is weakly connected graph.

Check list

To check your mastery the topic put (✓) if you can perform it.; otherwise put (X) in front of each statement.

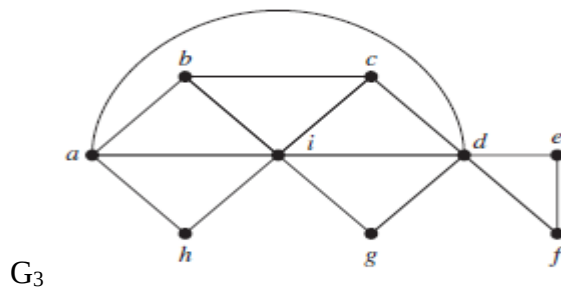
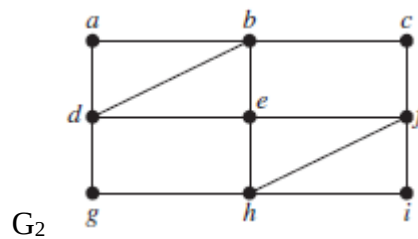
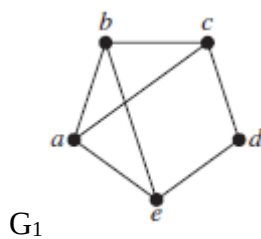
Important points	I Can	I Can't
1. Define Eulerian graph and Hamiltonian graph.		
2. Find Eulerian circuit in a graph if it exist.		
3. Find Hamiltonian cycle in a graph if it exist.		
4. Determine whether a graph is Eulerian or not.		
5. Determine whether a graph is Hamiltonian or not.		
6. Define planar graph.		
7. Determine whether a graph is planar or nonplanar.		
8. State Euler's formula.		
9. Verify Euler's formula for a connected planar graph.		
10. Define tree graph.		
11. Determine whether a graph is a tree or not.		

12. Define rooted tree and spanning tree of a connected graph.		
13. Define directed graph.		
14. Find adjacency matrix and incidence matrix of a directed graph.		

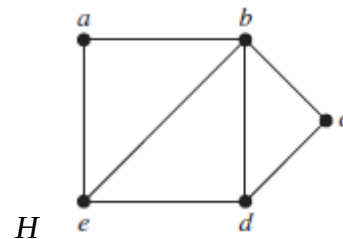
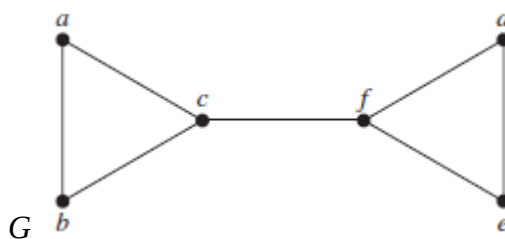
Is your response positive for the above questions? If your answer is “I Can”, go to the next section. If your answer is “I Can’t”, go back and read once again.

Exercise 3.3

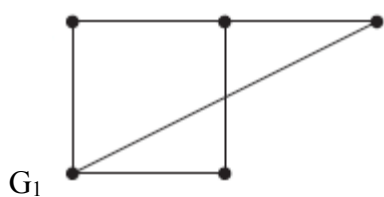
- Determine whether the given graph has an Euler circuit or not. Construct such a circuit when one exists.



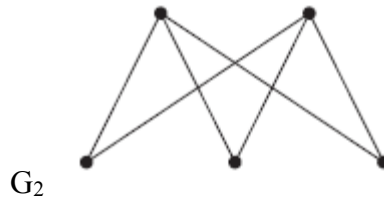
- For which values of n do these graphs have an Euler circuit?
a) K_n b) C_n
- Determine whether the given graph has a Hamilton circuit.



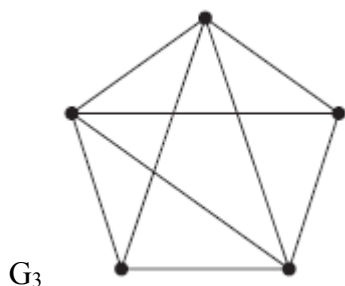
- Draw the given planar graph without any crossings.



G_1

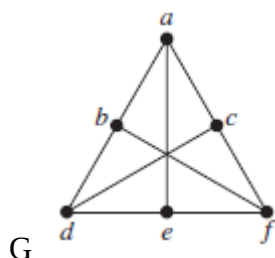


G_2

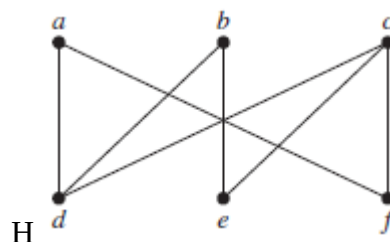


G_3

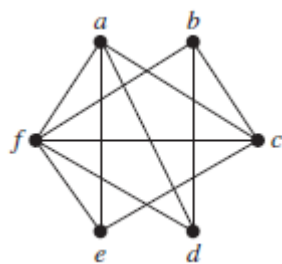
5. Determine whether the given graph is planar. if so, draw it so that no edges cross.



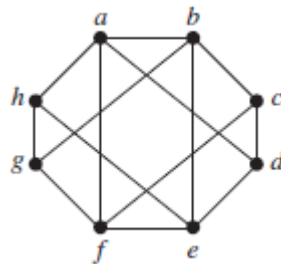
G



H

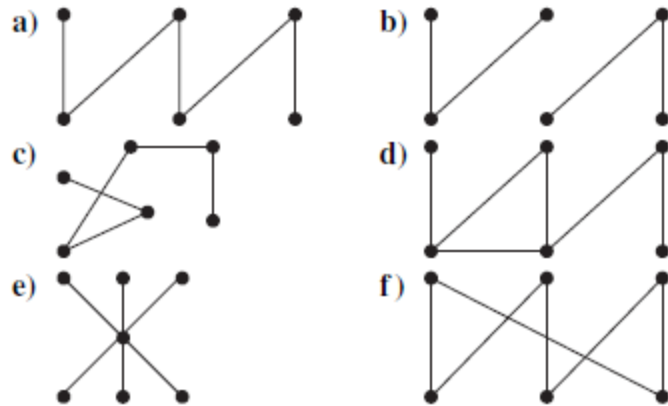


H_1



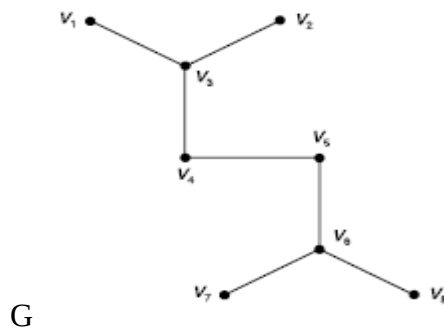
H_2

6. Suppose that a connected planar graph has eight vertices, each of degree three. Into how many regions is the plane divided by a planar representation of this graph?
7. Suppose that a connected planar graph has six vertices, each of degree four. Into how many regions is the plane divided by a planar representation of this graph?
8. Suppose that a connected planar graph has 30 edges. If a planar representation of this graph divides the plane into 20 regions, how many vertices does this graph have?
9. Which of these nonplanar graphs have the property that the removal of any vertex and all edges incident with that vertex produces a planar graph?
 - a) K_5
 - b) K_6
 - c) $K_{3,3}$
 - d) $K_{3,4}$
10. Which of these graphs are trees?



11. In the graph G below , find

- Eccentricity of G.
- Center of G.



12. Which of the following graphs are trees?

- $G = (V, E)$ with $V = \{a, b, c, d, e\}$ and

$$E = \{\{a, b\}, \{a, e\}, \{b, c\}, \{c, d\}, \{d, e\}\}$$

- $G = (V, E)$ with $V = \{a, b, c, d, e\}$ and

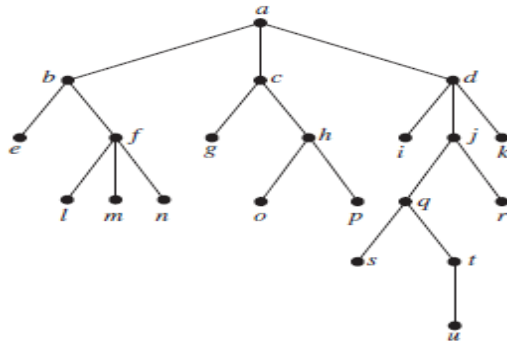
$$E = \{\{a, b\}, \{b, c\}, \{c, d\}, \{d, e\}\}$$

- $G = (V, E)$ with $V = \{a, b, c, d, e\}$ and

$$E = \{\{a, b\}, \{a, c\}, \{a, d\}, \{a, e\}\}$$

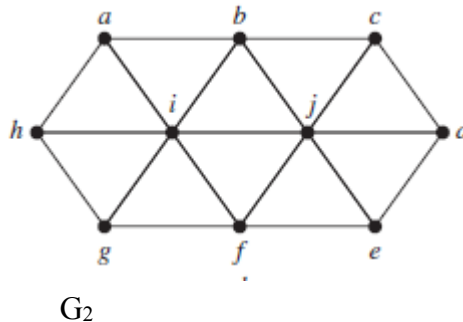
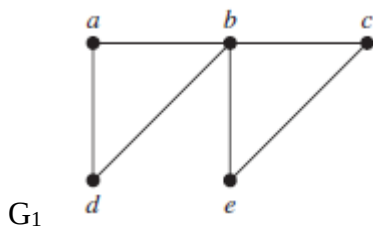
- $G = (V, E)$ with $V = \{a, b, c, d, e\}$ and $E = \{\{a, b\}, \{a, c\}, \{d, e\}\}$

13. Answer these questions about the rooted tree illustrated.



- Which vertex is the root?
- Which vertices are internal?
- Which vertices are leaves?
- Which vertices are children of j ?
- Which vertex is the parent of h ?
- Which vertices are siblings of o ?
- Which vertices are ancestors of m ?
- Which vertices are descendants of b ?

14. Find a spanning tree for the graph shown by removing edges in simple circuits.

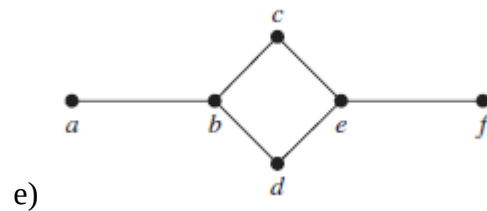
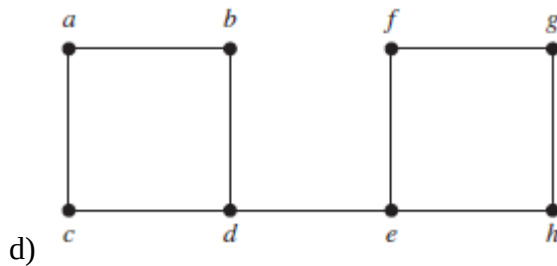


15. Find a spanning tree for each of these graphs.

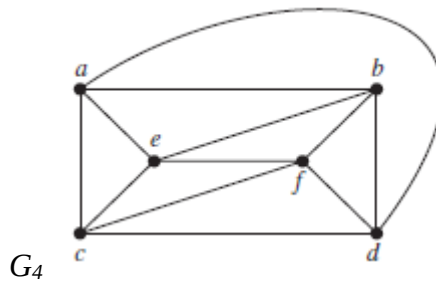
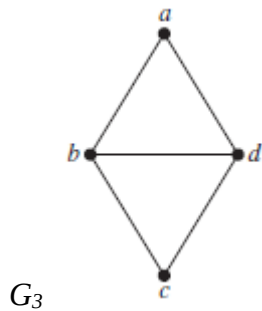
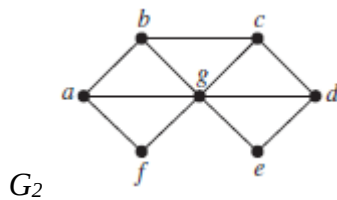
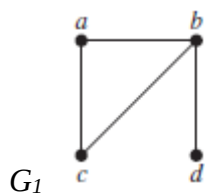
a) K_5

b) $K_{4,4}$

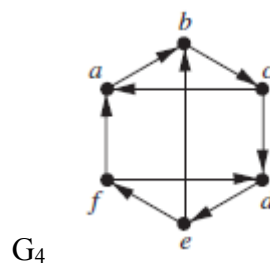
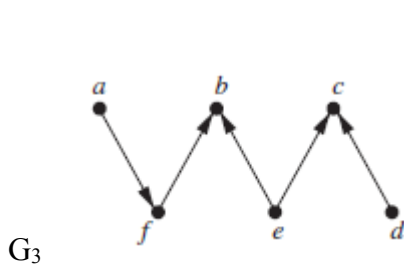
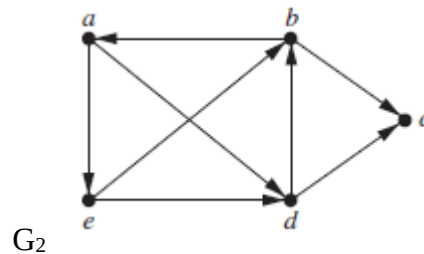
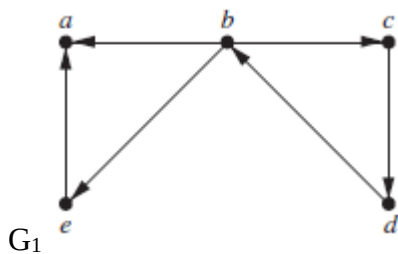
c) C_5



15. Find the chromatic number of the following graphs.



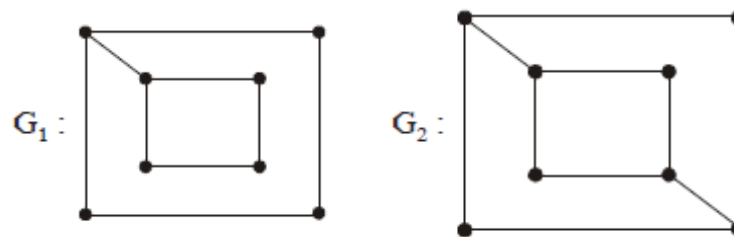
16. Determine whether each of these graphs is strongly connected and if not, whether it is weakly connected.



Review Exercise

- How many vertices do the following graphs have if they contains
 - 16 edges and all vertices of degree 2
 - 21 edges, 3 vertices of degree 4 and others each of degree 3.

2. If $G = (V, E)$ be an undirected graph with e edges then show that $\sum_{v \in V} \deg_G(v) = 2e$.
3. Determine the number of edges in a graph with 6 vertices, 2 of degree 4 and 4 of degree 2. Draw two such graphs.
4. How many vertices are needed to construct a graph with 6 edges in which each vertex is of degree 2.
5. Show that the maximum number of edges in a simple graph with n vertices is $\frac{n(n-1)}{2}$.
6. Does a 3-regular graph on 14 vertices exist? What can you say on 17 vertices?
7. Draw all six graphs with five vertices and five edges.
8. Find all possible non-isomorphic induced subgraphs of the following graph.
9. If a graph G of n vertices is isomorphic to its complement $\overline{G}$, show that n or $(n-1)$ must be multiple of 4.
10. Determine whether the following graphs are isomorphic or not



11. Prove that any party with six people, there are three mutual acquaintances or three mutual nonacquaintances.
12. Prove that a simple graph with n vertices must be connected if it has more than $\frac{(n-1)(n-2)}{2}$ edges.
13. Prove that A simple graph with n vertices and k components cannot have more than $\frac{(n-k)(n-k+1)}{2}$ edges.

Chapter 4

Weighted Graphs and Their Applications

Chapter Objectives

On completion of this chapter, students will be able to

- Understand weighted graph.
- Know the minimal spanning tree of a weighted graph.
- Know the shortest path between two vertices.
- Understand algorithms (kruskal's , Prim's) algorithms to find the minimal spanning tree of a graph and the shortest path between two vertices.
- Understand algorithms Dijkstra's algorithm to find the shortest path between two vertices.
- Understand travelling sales man problem.

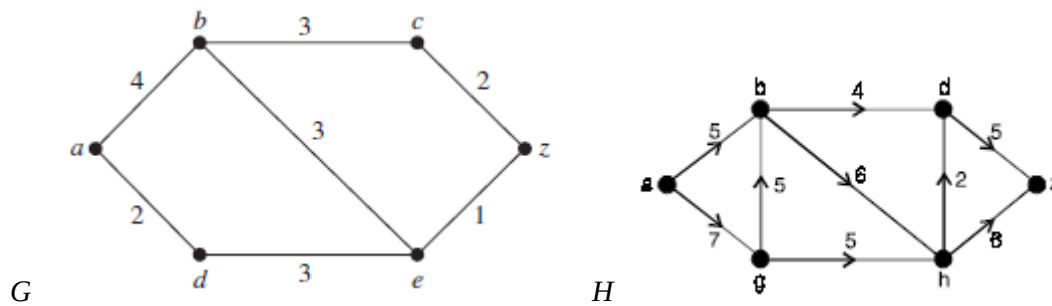
4.1 Definition and Examples of Weighted Graph

Objectives: After completion of this topic, students will be able to

- ✓ Define weighted graph.
- ✓ Give examples of weighted graph.
- ✓ Define the minimal spanning tree of a weighted graph.
- ✓ Find the minimal spanning tree of a weighted graph.
- ✓ Define the shortest path between two vertices.
- ✓ Find the shortest path between two vertices.
- ✓ Use kruskal's algorithm and Prim's algorithm to find the minimal spanning tree of a graph and the shortest path between two vertices.
- ✓ Use Dijkstra's algorithm to find the shortest path between two vertices.

Definition: A graph $G = (V, E)$ is called weighted graph if each edge of graph G is assigned positive number we called the weight of the edge. Such weights might represent for example costs, lengths or capacities, depending on the problem at hand. Some authors call such a graph a network.

Example 1: The following two graphs are weighted graphs.



G is weighted undirected graph and H is directed weighted graph

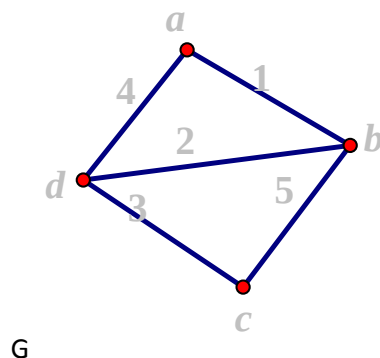
4.2 Minimum Spanning Trees

Objectives: After completion of this topic, students will be able to

- ✓ Define the minimal spanning tree of a weighted graph.
- ✓ Find the minimal spanning tree of a weighted graph.
- ✓ Use kruskal's algorithm and Prim's algorithm to find the minimal spanning tree of a graph.

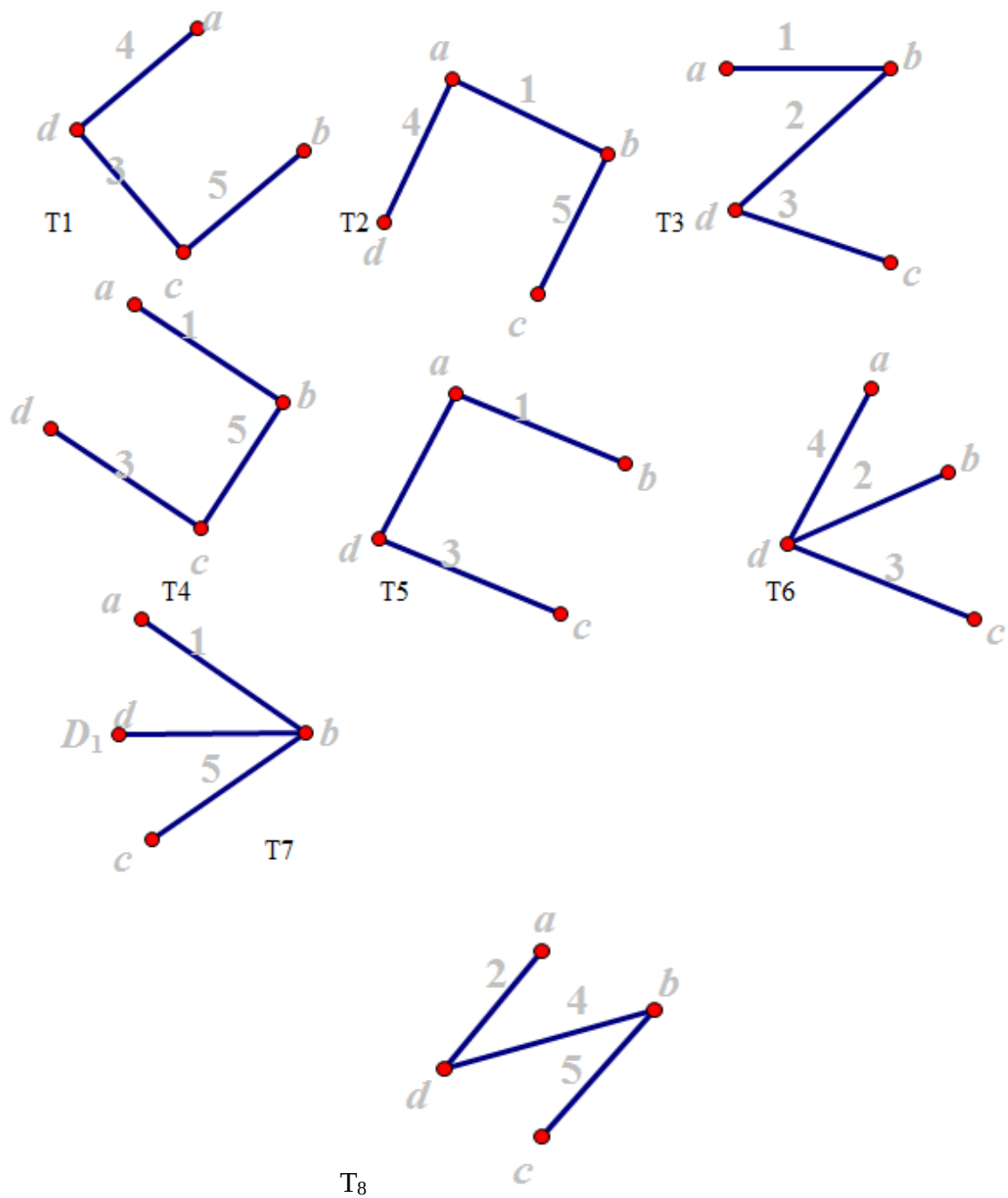
Definition: Let $G = (V, E)$ be a connected weighted graph. A minimum spanning trees T of the graph G is a tree whose total weight is smallest among all the spanning trees of the graph G . Total weight of a spanning tree is the sum of the weights of each edges on the spanning tree.

Example: Consider the given graph G ,



Find the minimum spanning tree of this graph.

Solution: First find all spanning tree of this graph.



Let T_w is the weight of the tree, then

$$T_w(T_1) = 4+3+5 = 12, T_w(T_2) = 4+1+5 = 10, T_w(T_3) = 1+2+5 = 8, T_w(T_4) = 4+3+5 = 12, T_w(T_5) = 4+2+5 = 11, T_w(T_6) = 4+2+5 = 9, T_w(T_7) = 1+2+3 = 6, T_w(T_8) = 4+3+2 = 9$$

∴ The minimum spanning tree of above graph is

$$T_m = \min(T_w(T_1), T_w(T_2), T_w(T_3), T_w(T_4), T_w(T_5), T_w(T_6), T_w(T_7), T_w(T_8)) \\ = \min(8, 10, 8, 12, 11, 9, 6, 9)$$

$T_w(T_7) = 6$ is the minimum spanning tree of this graph.

There are two famous algorithms finding the minimum spanning tree:

i. Kruskal's Algorithm

Kruskal's Algorithm builds the spanning tree by adding edges one by one into a growing spanning tree. This algorithm finds an edge which has least weight and add it to the growing Kruskal's Algorithm

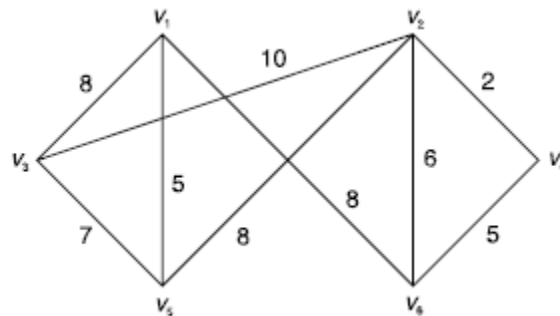
Input: A connected weighted graph G with n vertices.

Step 1: Arrange the edges of in order of increasing weights and select the edge with minimum weight.

Step 2: Proceed sequentially, add each edge which does not result in a cycle until $n - 1$, edges are selected.

Step 3: Exit.

Example 1: Consider the graph G .



G

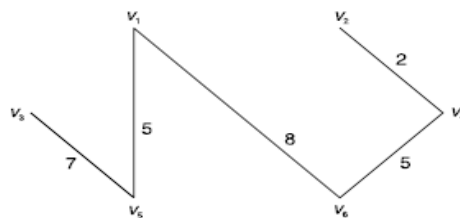
Solution: We have $n = 6$

We order the edges by increasing weights (v_2, v_4) is edge with minimum weight. Select the edge (v_2, v_4) we successively add edges to (v_2, v_4) , without forming cycles until $6 - 1 = 5$ edges

are selected. This yields:

Edges	(v_2, v_4)	(v_1, v_5)	(v_4, v_6)	(v_2, v_6)	(v_3, v_5)	(v_1, v_3)	(v_1, v_6)	(v_2, v_5)	(v_2, v_3)
Weight	2	5	5	6	7	8	8	8	10
Add?	Yes	Yes	Yes	No	Yes	No	Yes	No	No

Edges in the minimum spanning tree are (v_2, v_4) , (v_1, v_5) , (v_4, v_6) , (v_3, v_5) , (v_1, v_6) . The resulting minimal (optimal) spanning tree is



We apply the steps of Kruskal's algorithm to the graph G as follows:

(v_2, v_4) is the edge with minimum weight, therefore we select the edge (v_2, v_4) . The next edge with minimum weight is (v_1, v_5) , selection of (v_1, v_5) does not result in a cycle.

edge (v_1, v_5) is selected.

ii. Prim's Algorithm

In Prim's Algorithm we grow the spanning tree from starting position. Unlike an edge in Kruskal's, we add vertex to the growing spanning tree in Prim's

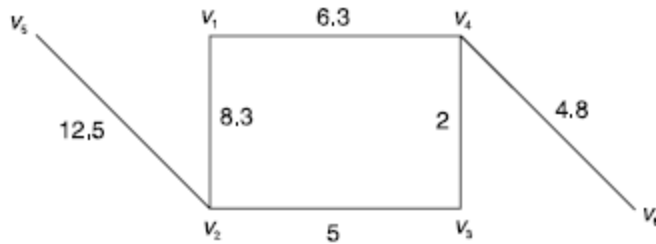
Step1. Choose any vertex v_1 of G

Step2. Choose an edge $e_1 = (v_1, v_2)$ of G such that $v_2 \neq v_1$ and e_1 has smallest weight among the edges of G incident with v_1 .

Step3. If edges $e_1, e_2, \dots, e_i$ have been chosen involving end points $v_1, v_2, \dots, v_{i+1}$. Choose an edge $e_{i+1} = (v_j, v_k)$ with $v_j \in (v_1, v_2, \dots, v_{i+1})$ and $v_k \notin (v_1, v_2, \dots, v_{i+1})$ such that e_{i+1} has smallest weight among the of G with precisely one end in $(v_1, v_2, \dots, v_{i+1})$.

Step4. Stop after $n-1$ edges $n-1$ edges have been chosen

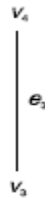
Example: Find the minimal spanning tree of the weighted graph in figure below by Prim's algorithm.



Solution: Let denote the edge of G ; $e_1 = (v_1, v_2)$, $e_2 = (v_2, v_3)$, $e_3 = (v_3, v_4)$, $e_4 = (v_4, v_1)$, $e_5 = (v_2, v_5)$ and $e_6 = (v_4, v_6)$.

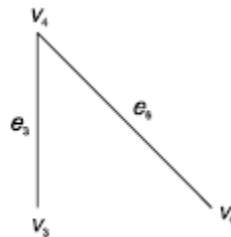
We apply Prim's algorithm to the graph as follows:

The edge $e_3 = (v_3, v_4)$ is an edge with minimum weight. Hence, we start with the vertex v_3 and select the edge e_3 incident with v_3 .

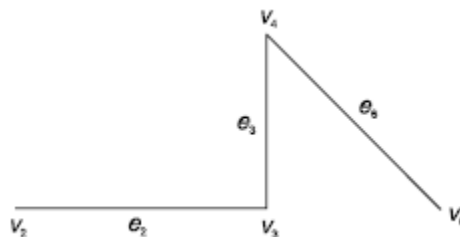


We next consider the edges connecting a vertex $\{v_3, v_4\}$ with the vertex of the set $V - \{v_3, v_4\}$.

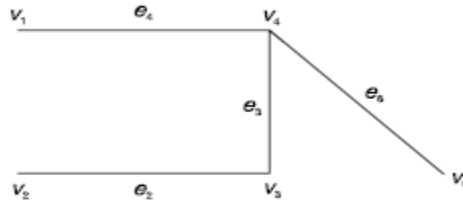
We observe that e_6 the edge with minimum weight.



Consider the edges connecting the vertices of the set $\{v_3, v_4, v_6\}$ with the vertices of $V - \{v_3, v_4, v_6\}$. The edge e_2 has the minimum weight. The edge e_2 is selected.



of the connecting the vertices of $\{v_2, v_3, v_4, v_6\}$; with the vertex set $V - \{v_2, v_3, v_4, v_6\}$, e_4 has minimum weight, therefore e_4 is selected.



e_1 , e_5 are the edges remaining. e_5 is the only edge connecting $\{v_1, v_2, v_3, v_4, v_5, v_6\}$ and $\{v_5\}$ such that the inclusion of e_5 does not result in a cycle. Hence e_5 is selected. Since number of edges selected is 5 we stop.

The minimal spanning tree obtained is shown below



Weight of the minimal spanning tree = $2 + 4.8 + 5 + 6.3 + 12.5 = 30.6$

4.3 Shortest path problem

Objectives: After completion of this topic, students will be able to

- ✓ Define the shortest path between two vertices.
- ✓ Find the shortest path between two vertices.
- ✓ Use Dijkstra's algorithm to find the shortest path between two vertices.

In many areas like transportation, cartoon motion planning, communication network topology design etc, problems related to finding shortest path algorithms.

The shortest path problem is concerned with finding the least cost (that costs minimum) path from an originating node in a weighted graph to a destination node in that graph.

Shortest path algorithms

Dijkstra's algorithm

To find the length of a shortest path from vertex A to vertex E in a weighted graph, proceed as follows:

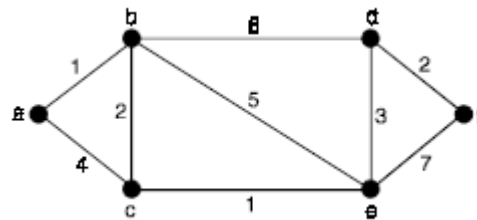
Step 1: Set $v_1 = A$ and assign to this vertex the permanent label 0.

Assign every other vertex a temporary label of ∞ , where ∞ is a symbol which, by definition, is deemed to be larger than any real number.

Step 2: Until E has been assigned a permanent label or no temporary labels are changed in (a) or (b), do the following.

- Take the vertex v_i which most recently acquired a permanent label, say d . For each vertex v which is adjacent to v_i and has not yet received a permanent label, if $d + w(viv) < t$ the current temporary label of v to $d + w(viv)$.
- Take a vertex v which has a temporary smallest among all temporary labels in the graph. Set $v_{i+1} = v$ and make its temporary label permanent. If there are several vertices v which tie for smallest temporary label, make any choice.

Example: Apply Dijkstra's algorithm to the graph given below and find the shortest path from a to f.



Solution: The initial labelling is given by

Vertex V	a	b	c	d	e	f
$L(v)$	0	α	α	α	α	α
T	{a,}	b,}	c,}	d,}	e,}	f}

Iteration 1 : $u = a$ has $L(u) = 0$. T becomes $T - \{a\}$. There are two edges incident a, i.e., ab and ac where b and c are in T.

$$L(b) = \min \{ \text{old } L(b), L(a) + w(ab) \}$$

$$= \min \{ \alpha, 0 + 1.0 \} = 1.0$$

$$L(c) = \min \{ \text{old } L(c), L(a) + w(ac) \}$$

$$= \min \{ \alpha, 0 + 4.0 \} = 4.0$$

Hence minimum label is $L(b) = 1.0$

Vertex V	a	b	c	d	e	f
L(v)	0	1.0	4.0	α	α	α
T	{	b,	c,	d,	e,	f}

Iteration 2 : $u = b$, the permanent label of b is 1.0 T becomes $T - \{b\}$ there are three edges incident with b , i.e., bc , bd and be where $c, d, e \in T$.

$$L(c) = \min \{\text{old } L(c), L(b) + w(bc)\}$$

$$= \min \{4.0, 1.0 + 2.0\} = 3.0$$

$$L(d) = \min \{\text{old } L(d), L(b) + w(bd)\}$$

$$= \min \{\alpha, 1.0 + 6.0\} = 7.0$$

$$L(e) = \min \{\text{old } L(e), L(b) + w(be)\}$$

$$= \min \{\alpha, 1.0 + 5.0\} = 6.0$$

Vertex V	a	b	c	d	e	f
L(v)	0	1.0	3.0	7.0	6.0	α
T	{		c,	d,	e,	f}

Iteration 3 : $u = c$, the permanent label of c is 3.0, T becomes $T - \{c\}$. There is one edge incident with c , i.e., ce where $e \in T$.

$$L(e) = \min \{\text{old } L(e), L(c) + w(ce)\}$$

$$= \min \{6.0, 3.0 + 1.0\} = 4.0$$

Thus minimum label is $L(c) = 4.0$

Vertex V	a	b	c	d	e	f
L(v)	0	1.0	3.0	7.0	4.0	α
T	{			d,	e,	f}

Iteration 4 : $u = e$, the permanent label of e is 4.0, T becomes $T - \{e\}$. There are two edges incident with e , i.e., ed and ef where $d, f \in T$

$$L(d) = \min \{\text{old } L(d), L(e) + w(ed)\}$$

$$= \min \{7.0, 4.0 + 3.0\} = 7.0$$

$$L(f) = \min \{\text{old } L(f), L(e) + w(ef)\}$$

$$= \min \{\alpha, 4.0 + 7.0\} = 11.0$$

Vertex V	a	b	c	d	e	f
L(v)	0	1.0	3.0	7.0	4.0	11.0
T	{			d,		f}

Thus minimum label is $L(d) = 7.0$

Iteration 5 : $u = d$, the permanent label of d is 7.0. T becomes $T - \{d\}$. There is one edge incident with d , i.e., d, f where $f \in T$

$$L(f) = \min \{ \text{old } L(f), L(d) + w(df) \}$$

$$= \min \{ 11.0, 7.0 + 2.0 \} = 9.0$$

Vertex V	a	b	c	d	e	f
$L(v)$	0	1.0	3.0	7.0	4.0	9.0
T	{					f }

The minimum label is $L(f) = 9.0$

Since $u = f$, the only choice. Iteration stops. Thus the shortest distance between a and f is 9, and moreover, the shortest paths is $\{a, b, c, e, d, f\}$.

Applications of weighted graphs

Traveling Salesman Problem

A salesman required to visit a number of cities going through each city exactly once and return to the starting point with the minimum kilometers travelled, given the distance and the routes.

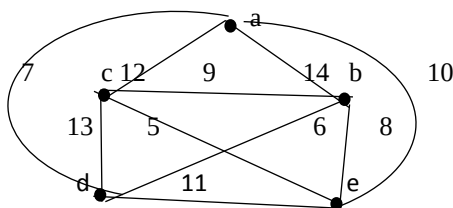
However, currently there is no algorithm to solve this problem with in fewer steps. If the graph is complete with a n vertices then there are $(n-1)!$ different circuits. For n larger we have closest-neighbour algorithm, which gives near optimal solution.

Closest Neighbour Algorithm

Let $G = (V, E, W)$ be a complete graph with n vertices. The closet- neighbour circuit of G starting at v_1 is $(v_1, v_2, v_3, \dots, v_n)$ got by the following steps.

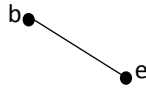
1. $v_1 = v_1 - \{v_1\}$
2. For $i=2$ to $n-1$ do;
 - 2.1. v_i ; the closest vertex in v_{i-1} to v_{i-1} ; $w(v_{i-1}, v_i) = \min[w(v_{i-1}, v) : v \in v_{i-1}]$
 - a. $v_i = v_{i-1} - \{v_i\}$
 - iii. u_n = the only element in v_n .

Example 1: solve the sales man problem of the following graph.

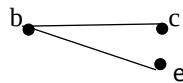


Solution:

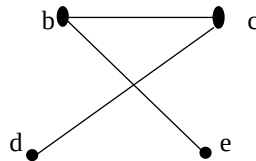
Step1: Take one vertex that has small weight to another vertex, i.e. e. Now edge with smallest weight incident on e is $(e, d) = 11$, $(e, c) = 8$, $(e, b) = 5$, $(e, a) = 10$. Now choose (e, b) because it has minimum weight



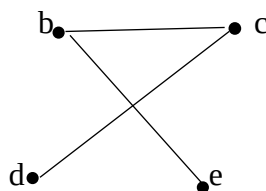
Step 2: Now $w(b, a) = 14$, $w(b, c) = 9$, $w(b, d) = 13$, then we choose (b, c) because it has minimum weight to each other. i.e



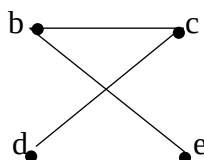
Step 3: Now, $w(c, e) = 8$, $w(c, d) = 6$, $w(c, a) = 12$, we choose (c, d) because it has minimum weight when compare to each other. i.e



Step 4: Now $w(d, a) = 7$, $w(d, b) = 13$, $w(d, e) = 11$, we choose (d, a) because it has minimum weight when we compare each other. i.e

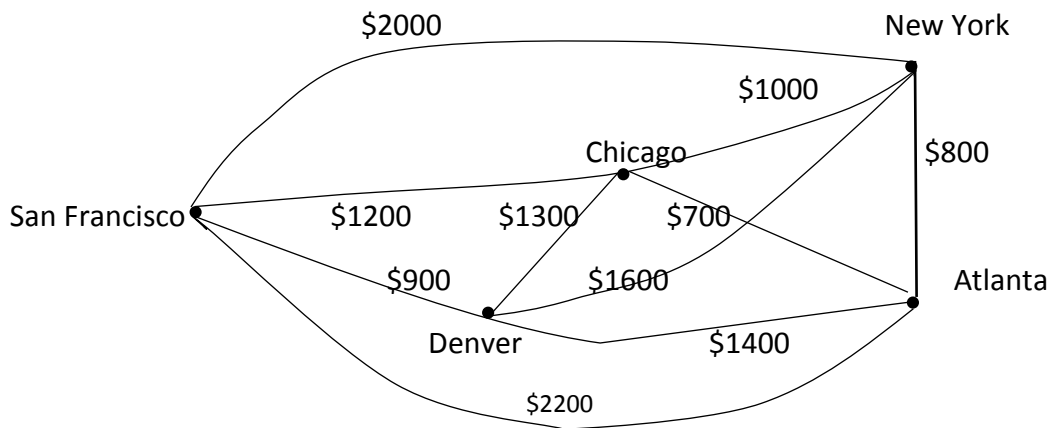


Step 5: Now $w(a, b) = 14$, $w(a, c) = 12$, $w(a, e) = 10$, we choose (a, e) because it has minimum weight when we compare to each other. i.e



Now we stop to draw the edge because it has form a circle, then the sales man problem of the above is this

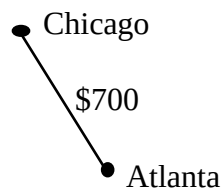
Example 2: Look the following a weighted graph showing monthly lease cost for line in a computer network.



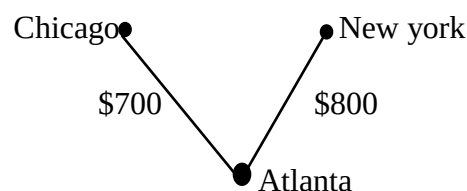
Use prim's algorithm to design a minimum cost communications network connecting all the computers represented by the graph below.

Solution: we solve this problem by finding spanning tree in the graph in figure above. Prim's algorithm is carried out by choosing an initial edge of minimum weight and successively adding edges of minimum weight that are incident to a vertex in the tree and that do not form simple circuits.

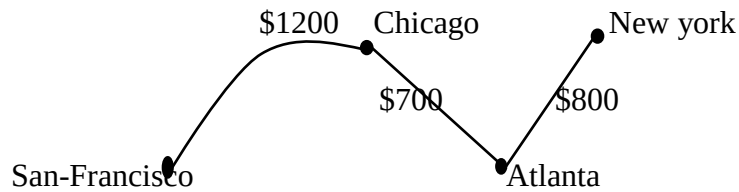
	Choice	Edge	cost
Step1	1	{Chicago, Atlanta}	\$700



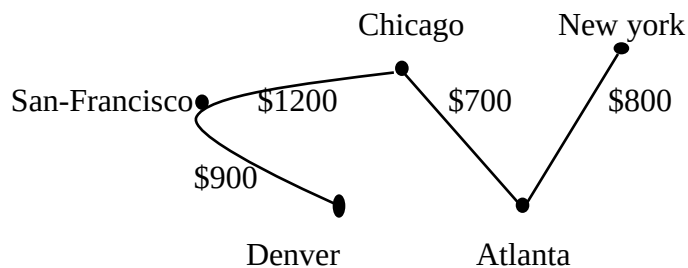
Step2	2	{Atlanta, New york}	\$800
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Step3 3 {Chicago, San-Francisco} \$1200



Step4 4 {San-Francisco, Denver} \$900



This graph shows minimum spanning tree produced by Prim's algorithm with choice made at each step displayed.

Therefore, the minimum cost communications network connecting all the computers represented by above graph,

$$\$800 + \$700 + \$1200 + \$900 = \underline{\underline{\$3600}}$$

Check list

To check your mastery the topic put (✓) if you can perform it.; otherwise put (X) in front of each statement.

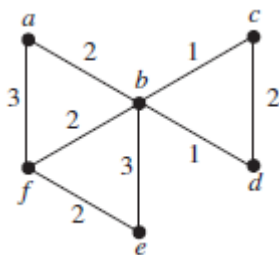
Important points	I Can	I Can't
1. Define weighted graph.		
2. Give examples of weighted graph.		
3. Define the minimal spanning tree of a weighted graph.		
4. Give examples of minimal spanning trees.		
5. Find the minimal spanning tree of a weighted graph.		

6. Define the shortest path between two vertices.		
7. Find the shortest path between two vertices.		
8. Use kruskal's algorithm and Prim's algorithm to find the minimal spanning tree of a graph and the shortest path between two vertices.		
9. Use Dijkstra's algorithm to find the shortest path between two vertices.		

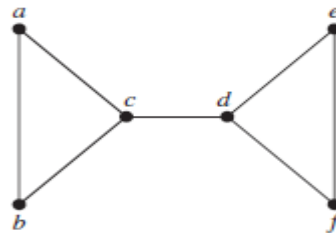
Is your response positive for the above questions? If your answer is “I Can”, go to the next section. If your answer is “I Can’t”, go back and read once again.

Exercise

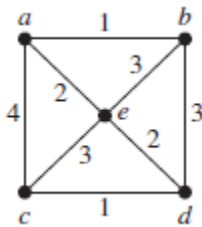
- Determine whether the following graphs are weighted graphs or not.



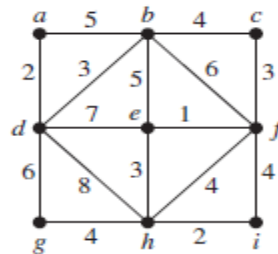
G



- Prim's algorithm to find a minimum spanning tree for the given weighted graphs.

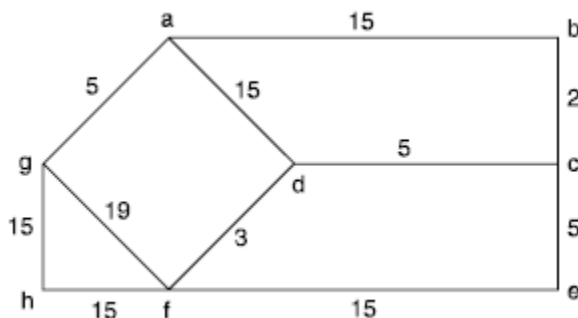


G₁

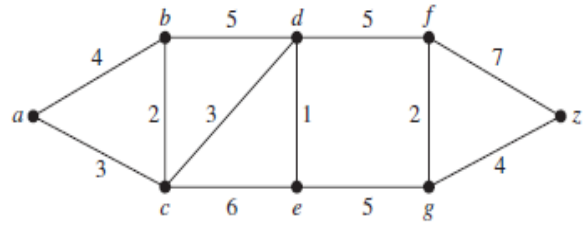
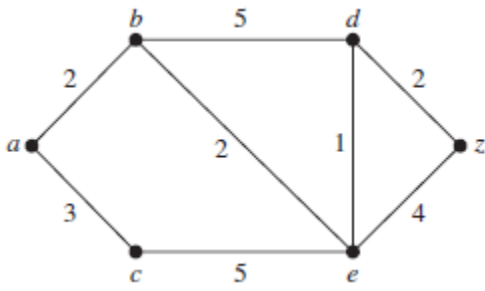


G₂

- Determine a railway network of minimal cost for the cities.



4. Find the length of a shortest path between a and z in the given weighted graph.



Chapter 5

Network and Flow

Chapter Objectives

On completion of this chapter, students will be able to

- Understand the concept of network and network flow.
- Know the maximum flow of a given network flow.
- Understand max- flow and min- cut theorem and apply the theorem.

5.1 Definition and Examples of Network and Flow

Objectives: After completion of this topic, students will be able to

- ✓ Define network on a weighted directed graph.
- ✓ Define network flow.

Network

Let $N = (V, E)$ be a loop-free connected directed graph. Then N is called a network, or transport network, if the following conditions are satisfied:

- i) There exists a unique vertex $a \in V$ with $id(a)$, the in degree of a , equal to 0. This vertex a is called the source.
- ii) There is a unique vertex $z \in V$, called the sink, where $od(z)$, the out degree of z , equals 0.
- iii) The graph N is weighted, so there is a function from E to the set of non negative integers that assigns to each edge $e = (v, w) \in E$ a capacity, denoted by $c(e) = c(v, w)$.

If $N = (V, E)$ is a transport network, a function f from E to the non-negative integers is called a flow for N .

(i) $f(e) \leq c(e)$ for each edge $e \in E$, and

(ii) for each $v \in V$, other than the source a or the sink z , $\sum_{w \in V} f(w, v) = \sum_{w \in V} f(v, w)$

If there is no edge (v, w) , then $f(v, w) = 0$.

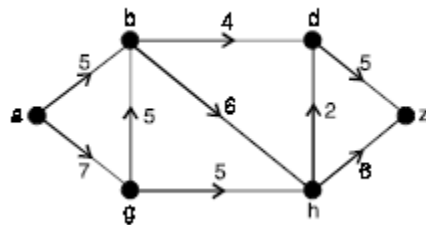
Let f be a flow for a transport network $N = (V, E)$

(i) An edge e of the network is called **saturated** if $f(e) = c(e)$. When $f(e) < c(e)$, the edge is called **unsaturated**.

(ii) If a is the source of N , then $\text{val}(f) = \sum_{v \in V} f(a, v)$ is called the **value of the flow**.

If $N = (V, E)$ is a transport network and C is a cut-set for the undirected graph associated with N , then C is called a **cut**, or an a - z cut, if the removal of the edges in C from the network results in the separation of a and z .

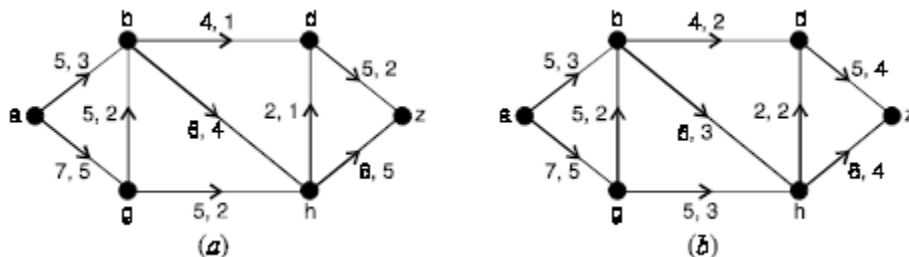
Example: The graph N below is network. Here vertex a is the source, the sink is at vertex z , and capacities are shown beside each edge. Since $c(a, b) + c(a, g) = 5 + 7 = 12$, the amount of the commodity being transported from a to z cannot exceed 12. With $c(d, z) + c(h, z) = 5 + 6 = 11$, the amount is further restricted to be no greater than 11.



For the network, the label x, y on each edge e is determined so that $x = c(e)$ and y is the value assigned for a possible flow f . The label on each edge e satisfies $f(e) \leq c(e)$.

In part (a) of the network flow, the flow into vertex g is 5, but the flow out from that vertex is $2 + 2 = 4$.

Hence the function f is not a flow in this case.



For the network flow in above (b), only the edge (h, d) is saturated. All other edges are unsaturated.

The value of the flow in this network is

$$\text{val}(f) = \sum_{v \in V} f(a, v) = f(a, b) + f(a, g) = 3 + 5 = 8.$$

We observe that in the network flow (b)

$$\sum_{v \in V} f(a, v) = 3 + 5 = 8 = 4 + 4 = f(d, z) + f(h, z) = \sum_{v \in V} f(v, z).$$

Consequently, the total flow leaving the source a equals the total flow into the sink z .

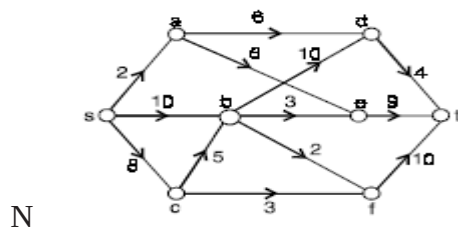
5.2 Max-flow and Min-cut

Objectives: After completion of this topic, students will be able to

- ✓ Find the maximum flow of a given network flow.
- ✓ State max- flow and min- cut theorem and apply the theorem.

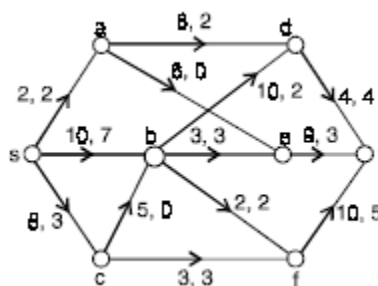
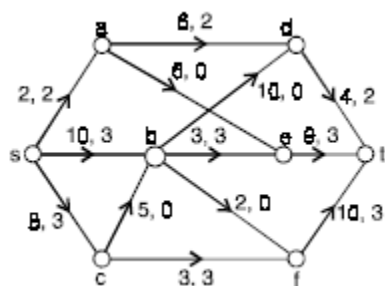
In any network, the value of any maximum flow is equal to the capacity of any minimum cut.

Example: Find a maximum flow in the network shown below and prove that it is a maximum.



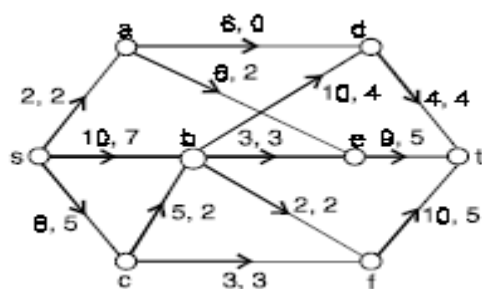
Solution: We start by sending a flow of 2 units through the path $sadt$, a flow of 3 units through $sbet$, and a flow of 3 units through $scft$, obtaining the flow shown on graph A below.

Continue by sending flows of 2 units through $sbdet$ and 2 units through $sbft$, obtaining the flow shown on graph B below.



At this point, there are no further flow-augmenting chains from s to t involving only forward arcs.

However, we can use the backward arc da to obtain a flow-augmenting chain $scbdaet$. Since the slack of this chain is 2, we add a flow of 2 to sc , cb , bd , ae , and et , and subtract 2 from ad . The result is shown in graph C below.



C

A search for further flow-augmenting chains takes us from s to c or b and on to d , where we are stuck.

This tells us that the current flow (of value 14) is maximum.

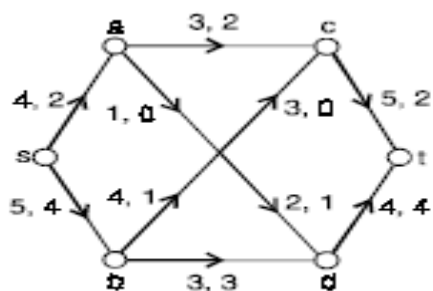
It also presents us with a cut verifying maximality, namely, $S = \{s, b, c, d\}$ (those vertices reachable from s by flow-augmenting chains) and $T = \{a, e, f, t\}$ (the complement of S).

The capacity of this cut is

$$C_{sa} + C_{be} + C_{bf} + C_{cf} + C_{dt} = 2 + 3 + 2 + 3 + 4 = 14.$$

Since this is the same as the value of the flow, we have verified that our flow is maximum.

Example 2: In the network given below



- i) Verify the law of conservation of flow at a , e , and d .
- ii) Find the value of the indicated flow.
- iii) Find the capacity of the (s, t) -cut defined by $S = \{s, a, b\}$ and $T = \{c, d, e, t\}$
- iv) Can the flow be increased along the path $sbedt$? If so, by how much?
- v) Is the given flow maximum? Explain.

Solution: The law of conservation holds at a because

$$\sum_v f_{va} = f_{za} = 2 \text{ and } \sum_v f_{av} = f_{ac} + f_{ae} = 2 + 0 = 2.$$

It holds at e because

$$\sum_v f_{ve} = f_{ae} + f_{be} = 0 + 1 = 1$$

$$\text{and } \sum_v f_{ev} = f_{ec} + f_{ed} = 0 + 1 = 1$$

It holds at d because

$$\sum_v f_{vd} = f_{bd} + f_{ed} = 3 + 1 = 4$$

$$\text{and } \sum_v f_{dv} = f_{dt} = 4.$$

ii) the value of the flow is 6.

iii) The capacity of the cut is $C_{ac} + C_{ae} + C_{be} + C_{bd} = 3 + 1 + 4 + 3 = 11$.

iv) No. Arc dt is saturated.

v) The flow is not maximum. For instance, it can be increased by adding 1 to the flow in the arcs along $sact$.

Check list

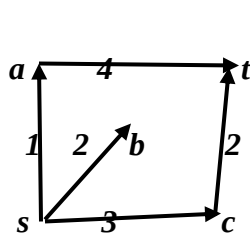
To check your mastery the topic put ($\checkmark$) if you can perform it.; otherwise put (X) in front of each statement.

Important points	I Can	I Can't
1. Define network on a weighted directed graph.		
2. Define network flow.		
3. Find the maximum flow of a given network flow.		
4. State max- flow and min- cut theorem and apply the theorem.		

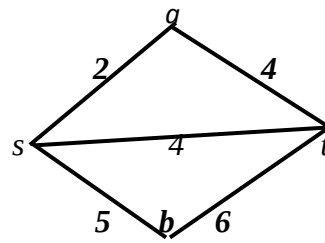
Is your response positive for the above questions? If your answer is “I Can”, go to the next section. If your answer is “I Can’t”, go back and read once again.

Exercise

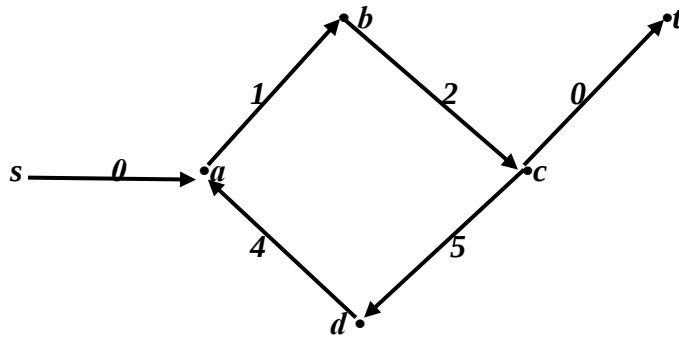
- Determine which of the graphs are networks.



G

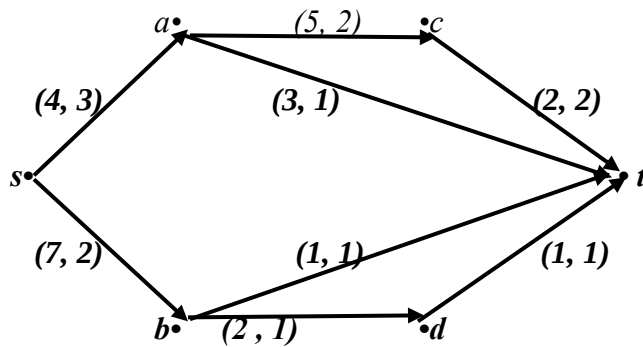


H



K

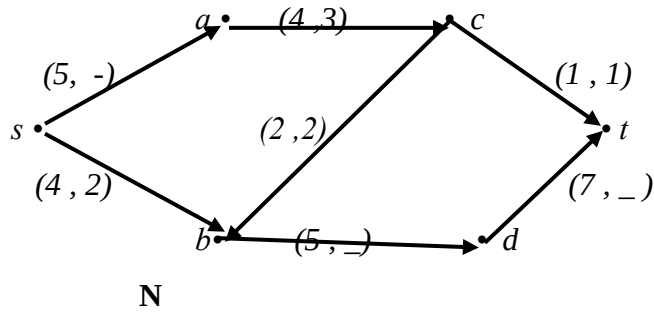
- Consider the flow f in the network N shown below.



N

- Find the capacity of the edges (s, a) and (a, c) .
- Find $f(s, a)$, $f(a, c)$, $f(a, d)$ and $f(b, d)$.
- Determine which edges are saturated.
- Find the value of the flow $(|F|)$.

3. Consider the network **N** on Q. No. 2 above.
 - a) Find two different maximal flows.
 - b) Find the value of each of these flows.
4. In the following network **N**, fill in the missing edge flows so that the result is a flow in the given network. Determine the value of each of the flows.



Reference:

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- Thomas K. Discrete Mathematics with Applications.
- Oscar L. Discrete mathematics. An open introduction. Third Edition.
- Kevin F. Discrete Mathematics. An introduction to proofs and combinatorics.